

Approval Page

A Case Study of Process Maturity and Improvement: Quantitatively Examining the Capability Maturity Model Integrated (CMMI) as a Maturity Model for Implementing an Innovative Telehealth Maturity Model at the Louis Stokes Cleveland Medical Center, United States (U.S.) Department of Veterans Affairs

By

James Carlton Jamison

Approved by the Doctoral Committee:

Dissertation Chair:

Degree Held

Date

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Daniel Adrian Doss

[Daniel Adrian Doss \(Jul 30, 2025 15:04:10 CDT\)](#)

Committee Member:

Degree Held

Date

Ph.D.

Jul 30, 2025

John Enamait

[John Enamait \(Jul 30, 2025 17:53:18 EDT\)](#)

Committee Member:

Degree Held

Date

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JAMES CARLTON JAMISON

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Abstract

The rapid expansion of telehealth technology had created both opportunities and challenges in the healthcare industry, particularly within the Veterans Health Administration (VHA). Despite improvements in accessibility, the lack of a standardized maturity model had hindered the consistent implementation of telehealth technologies across various healthcare settings. This study addressed this gap by developing a Telehealth Maturity Model (TMM) based on the Capability Maturity Model Integration (CMMI) framework, offering a structured approach for telehealth adoption and process enhancement.

The purpose of this quantitative study was to examine the applicability of the CMMI framework in telehealth environments for assessing process maturity and standardization. The study analyzed the perceptions of 285 participants from VHA and other healthcare organizations, comparing managerial and non-managerial views on telehealth process maturity. The research investigated five maturity levels: Initial, Managed, Defined, Quantitatively Managed, and Optimized, focusing on process standardization, efficiency, and scalability.

Statistical analyses, including ANOVA and Chi-Square tests, revealed significant discrepancies between managerial and non-managerial perspectives, particularly concerning process predictability and scalability. The findings indicated that managers exhibited a greater agreement with maturity model principles, while variability in early-stage process standardization resulted in inconsistencies in telehealth deployment. The study identified seventeen statistically significant results related to process definition, standardization, scalability, and adherence to best practices. Three hypotheses regarding early-stage process predictability and effectiveness were not statistically significant, underscoring the variability in initial process maturity across different telehealth implementations.

The findings highlight the necessity for a structured CMMI-based maturity model to enhance telehealth deployment, mitigate risks, and optimize processes. Practical recommendations involve adopting a sector-specific maturity model to address the gaps between managerial and non-managerial perspectives, along with implementing standardized telehealth protocols within the VHA. Future research should investigate qualitative insights and longitudinal studies to further refine the maturity model and assess its adaptability to various healthcare systems in both the public and private sectors. This study adds to the existing literature by integrating theoretical maturity frameworks with real-world telehealth implementation and offering actionable recommendations for policymakers, healthcare administrators, and technology developers.

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Chapter 1: Introduction

Telehealth, or telemedicine, was a developing area of medicine that used medical analysis data by various technologies (Lunney et al., 2018). It communicates healthcare diagnoses to assist or improve needed medical service delivery and to build a successful liaison between the provider and patient (Hu et al., 2002). Tele-healthcare progressions cover many areas aside from technical, creating challenges in developing processes (Valenta et al., 2021). Telehealth technology had historically established a role in offering solutions to divert a crisis (Kumar, 2022). Historically, the crisis's needs outweighed the process's necessity (Wright et al., 2021), and telehealth established a role in offering solutions to divert disasters (Kumar, 2022).

In 1879, medical communication using the telephone was introduced three years after the invention's debut, and no process was written about using the telephone as a medical communication medium (Soet & Basch, 1997). During the 1920s, a reference about Norwegian doctors assisting sick ship crew members at sea via a radio link was made, but no process was described for practice (Strehle & Shabde, 2006). A 1925 Science and Invention magazine cover illustrated a doctor diagnosing a patient by radio. It proposed a device allowing video examination over a distance, describing presented-day telehealth (Lustig, 2012, p. 11). In the 1930s, Joseph Schumpeter's theory of Creative Destruction described any innovation, including telehealth, as a theory of economic innovation and the business cycle, where industries adopting innovative technologies benefited from opportunities that reputable companies might overlook (Grady, 2014). In the 1940s, the telehealth process was redistributed after phone lines were used to convey medical data (Rosner et al., 2020). In the succeeding decades, computing teleradiology was used as film passed through a digitizer, producing direct digital captures and allowing for online reading of those digital images (Lustig, 2012). By the 1950s, the earliest stages of modern telehealth were initially revealed in the United States and Canada for use (Fong & Usman, 2020).

In the 1960s, X-rays and electrocardiograms were the initial caseloads of asynchronous or store-and-forward telehealth. There were four categories of telehealth: Live Video Conferencing, Asynchronous Video (Store-and-Forward), Remote Patient Monitoring (RPM), and Mobile Health (mHealth, 2022). In 1964, the first collaborative video link was established between the Psychiatric Institute in Omaha and the Norfolk State Hospital in Nebraska, 112 miles away (Zundel, 1996). It was the first recorded event of implementation of live synchronous technology used by telehealth (Ford et al., 2021). In 1999, a telehealth link was established aboard the USS George Washington aircraft carrier as part of a crisis evaluation, allowing medical assistance to be provided remotely and avoiding the extensive costs and logistical challenges of air-evacuating a patient over 5,000 miles (James & Folen, 1999, p. 16). The evaluated literature indicated that no telehealth process maturity model stemmed from these incidents.

Telehealth compels intervention, provides consultation, involves pre-post-care management, and offers healthcare education (Nickelson, 1998). The World Health Organization (WHO), on March 12, 2020, declared COVID-19 a pandemic, and there were 134,576 reported cases and 4,981 deaths worldwide (Fisk et al., 2020). In just two weeks, reported cases increased to 531,865, and casualties rose to 24,073 (Fisk et al., 2020). At the pandemic's peak, in-person care incorporated social distancing and quarantines, while virtual care deployment doubled (Mann et al., 2020). The pandemic produced a large-scale adoption of telehealth tools (Golinelli et al., 2020). These tools enabled healthcare providers to reach severely needed treatment processes for various medical programs (Golinelli et al., 2020). As a preventive measure, screening was held using leased office space, tents, or in-car testing for remote patients to establish and combine into telemedicine plans (Hollander & Carr, 2020). The COVID-19 pandemic altered the economy, society, and healthcare system (Wosik et al., 2020). The

innovation and implementation of telehealth networks allowed providers to network with patients remotely (Cho, 2007).

With the addition of big data and artificial intelligence (Whitelaw et al., 2020), the COVID-19 pandemic facilitated the distribution of telehealth technology in an adverse social setting (Ebersberger & Kuckertz, 2021). There was a need for process maturity and a policy-relevant framework (Edmunds et al., 2017). Applying technology during a crisis without adhering to an organized, defined process, unresearched hardware devices, and software applications created uncomfortable associations and economic consequences (Ebersberger & Kuckertz, 2021). With the upsurge in telehealth, users needed to be oriented to understand what was being used before its implementation (Naik et al., 2021). Unexpected performance strained the user relationship in terms of functionality, the quality of the rapport, examination, and the prominent social issues regarding the digital divide, mainly affecting the aged (van Dijk, 2006). There was a need to modernize telehealth implementation processes, but no significant process existed between using telehealth and its implementation (Ross et al., 2016). The perception of maturity models gradually made headway in hospital information systems as their use had been delayed during telehealth implementation (Carvalho et al., 2016). As innovative technology in telehealth was introduced, a process for maturity integrated within a normalized process was required (Greenhalgh et al., 2017).

Statement of the Problem

The problem to be addressed in this study was to develop an effective and customizable maturity model to be used in healthcare for telehealth for implementing innovative technology sustainably (Gomes & Romão, 2018). Although there were various Capability Maturity Model Integration (CMMI) frameworks accessible to incorporate maturity models effectively, there were still many shortcomings in the process of developing these models to implement innovative

technology in the telehealth industry (Scanlon et al., 2020). Several telehealth technologies had been introduced without undergoing proper training or field testing, following a well-constructed maturity model procedure, or implementing measures for controlling and policies for a successful roll-out (Wisniewski et al., 2020). When a maturity model isn't followed during the implementation of new technology, it could negatively affect both the provider and patient's safety and the effectiveness of the telehealth communications platform (Karsh, 2004). It was unclear which factors and external pressures hinder the ability to make significant advancements (Faiola et al., 2019). Several obstacles related to telehealth technology were.

- Patient/provider motivation for learning the technology (Bauer et al., 2018).
- Accurate evidence on both clinical outcomes and cost-effectiveness (Gomes & Romão, 2018).
- A maturity model framework which presented less risk in implementation (Bak & Jedynek, 2022).

Maturity models assess and align a specific set of skills that healthcare organizations need to achieve their desired level of proficiency (Dikhanbayeva et al., 2020). In healthcare, maturity models had been utilized to implement innovative technology, and the telehealth sector needs to demonstrate a more precise adoption of a written Capability Maturity Model integration process (Bashshur et al., 2005). Implementing technology often comes with its fair share of challenges. However, an organization's ability to mitigate risks largely depends on its maturity model process and its competence in handling such situations. This could ultimately add value to the organization (Scremin et al., 2021).

Purpose of the Study

The purpose of this study was to examine a derivation of the CMM as a telehealth maturity model. Within this study, the researcher aimed to address shortcomings in the

functional processes needed to successfully integrate new technology and minimize potential risks to services (Watts, 2020). Within this study, the researcher aimed to demonstrated process improvement within quantitative research for implementing innovative technology for telehealth (Nematovich & Karimberdiyevich, 2022). The quantitative research process would display a logical path as it explores the benefits of implementing a maturity model design (Mishra & Sharma, 2022). The process combines the characteristics of various models into a single framework to address shortcomings in development procedures (White, 2021).

A maturity model was a tool which assesses the current effectiveness of a process (Carvalho et al., 2016). Relative to this study, the maturity model would involve five levels of activities and would conduct an IT Service Management (ITSM) maturity assessment (Santos & Martinho, 2019). Project management, technology hardware, software, providers and clients were considered variables for analysis. The target population and sample size to ensure the representativeness of this study was 218 responses calculated out of a population size of 500, a confidence level of 98% with a 5% margin of error, and a population proportion of 50% (Andrade, 2020). A healthcare environment involving telehealth technology would be the site of this study. This study would initiate a quantitative survey to determine whether the CMMI might be adapted within the telehealth domain. The survey would also probe the current use of process improvement and quality paradigms within the work setting (Doss et al., 2017).

Introduction to Theoretical Framework

This study was based on the CMMI and CMM theoretical frameworks. The CMMI framework consists of five maturity levels that establish a hierarchy for assessing process maturity and determining the capability of these processes to prioritize development efforts (Watts, 2020). Each level represented a step towards continuous improvement in process development, built on established augmentations. The process starts with fundamental

management practices and advances through a predetermined and validated sequence of levels (Poeppelbuss et al., 2011). The methodology of a maturity model focused on a problem encountered which guides this study in a specific set of interventions, evaluation, and fabrication of an item adopting the class of difficulties (Blondiau et al., 2016).

1. Initial. The first stage of CMM determines an unpredictable environment, leading to increased risks and wastefulness. At this level, the processes were seen as irregular, poorly controlled, and reactive (White, 2021).
2. Managed. This stage was considered a progression for projects. It was often reactive to ensure requirements were addressed and processes were planned, performed, and measured to consider the most relevant aspects related to sustainability (David Patón-Romero et al., 2019).
3. Defined. A process used in development as the organization had achieved all aspects of goals in areas assigned to maturity levels and must implement engineering process from a usability viewpoint, achieving all requirements developments, technical provide solutions, provide verification, and validation (Jokela & Lalli, 2003).
4. Quantitatively Managed. Obtains chronological results for level 3 projects to make trade-offs, with anticipated outcomes, among rival dimensions of business performance (cost, quality, timeliness). A process maturity level was depicted by refining organizational performance (Royce, 2002).
5. Optimizing. Examine and control process performance proactively by analytically using process performance data to improve the processes (Lee et al., n.d.).

This study would incorporate a unique CMM/CMMI framework perspective within telehealth administrative processes. The proposed framework would address the progressive maturing of technical and administrative processes aligned with the typical stages of the

established CMM/CMMi constructs by reviewing the literature (Blondiau et al., 2016). The CMMi model was a comprehensive process that could be utilized to assess and improve existing systems to measure progress. Once created, The telehealth maturity model would include different maturity levels and their respective characteristics (Goldenson & Gibson, 2003).

Introduction to Research Methodology and Design

This quantitative study assertively assesses the efficacy of utilizing a quantitative methodology approach to implement innovative technology in telehealth effectively, applying the CMMI framework. The quantitative research approach was grounded in proven scientific principles that could be measured and statistically analyzed precisely (Tuli, 2011, p. 103). To assess the effectiveness of the CMMI framework in telehealth, a survey would be conducted using a quantitative methodology approach (Overby et al., 2010). The quantitative research plans to use a five-point Likert-scale survey to conduct a cross-sectional study while also utilizing an integrated process model (Ning et al., 2020). The Likert scale of questions collects and analyzed responses that reflect both the complete maturity framework and each individual maturity level (Doss et al., 2019). The respective Likert categories were:

1. Strongly Disagree
2. Disagree
3. Indifference
4. Agree
5. Strongly Agree

Within the survey, questions 1 through 5 would examine the overall CMMI framework, questions 6 through 8 examine CMMi Level 1, questions 9 and 10 examine Level 2, questions 11 through 13 read Level 3, questions 14 through 16 analyzed Level 4, and questions 17 through 19

examine Level 5 (Doss et al., 2019). The corresponding levels and queried process maturity concepts were:

- Level 1: Most processes were theoretically erratic, chaotic, and ad hoc.
- Level 2: The processes were reoccurring and reactive.
- Level 3: The processes were characterized and expressed.
- Level 4: The processes were measured, contained and investigated quantitatively.
- Level 5: All processes were optimized.

The means analysis determines the response's overall agreement, neutrality, or disagreement regarding response directionality. Means analysis would examine the directionality of responses between managerial and non-managerial personnel. This study's mean ranges were as follows (Doss et al., 2019):

- Less than 2.5 produced an overall view of disagreement
- Between 2.5 and 3.5 produced an overall view of neutrality
- Larger than 3.5 produced an overall view of the agreement

Finally, the minimum necessary sample size to ensure the representativeness of this study was 218 responses calculated out of a population size of 500, a confidence level of 98% with a 5% margin of error, and a population proportion of 50%. Response bias would be examined using the Chi-Square method. Respondents would be classified as either non-managerial or managerial personnel for hypothesis testing. The hypothesis testing method would incorporate ANOVA using a 0.05 significance level. The Omega squared method would showed the effect size of statistically significant findings. The Cronbach method would examine the reliability of the study.

Research Questions

Within this study, the researcher used a quantitative research method to appraise how effective it was to apply the CMMi/CMM maturity framework to administrative telehealth processes. The main question for this research was whether the CMMi/CMM framework could be adapted as a process improvement framework for telehealth. To examine this issue, the study investigated questions related to each level of the maturity model framework (Doss et al., 2019).

RQ1

What was the difference in the opinions of managers versus non-managers regarding the notion telehealth processes might be defined as being ad-hoc, random, or chaotic?

RQ2

What was the difference in the opinions of managers versus non-managers regarding the notion telehealth processes were managed, disciplined, and repeatable?

RQ3

What was the difference in the opinions of managers versus non-managers concerning the notion telehealth processes were defined and specific?

RQ4

What was the difference in the opinions of managers versus non-managers relating to the notion telehealth processes were quantitatively managed?

RQ5

What was the difference in the opinions of managers versus non-managers concerning the notion telehealth processes were optimized?

Hypotheses

The hypotheses for the research questions were listed below in corresponding order. The main inquiry for this study aligns with the generic null hypothesis.

H1₀

No statistically significant difference exists in the opinions of managers versus non-managers regarding the notion telehealth processes might be defined as being ad-hoc, random, or chaotic.

H1_a

A statistically significant difference exists in the opinions of managers versus non-managers regarding the notion telehealth processes might be defined as being ad-hoc, random, or chaotic.

H2₀

No statistically significant difference exists in the opinions of managers versus non-managers regarding the notion telehealth processes were managed, disciplined, and repeatable.

H2_a

A statistically significant difference exists in the opinions of managers versus non-managers regarding the notion telehealth processes were managed, disciplined, and repeatable.

H3₀

No statistically significant difference exists in the opinions of managers versus non-managers concerning the notion telehealth processes were defined and specific.

H3_a

A statistically significant difference exists in the opinions of managers versus non-managers concerning the notion telehealth processes were defined and specific.

H4₀

No statistically significant difference exists in the opinions of managers versus non-managers relating to the notion telehealth processes were quantitatively managed.

H4_a

A statistically significant difference exists in the opinions of managers versus non-managers relating to the notion telehealth processes were quantitatively managed.

H5₀

No statistically significant difference exists in the opinions of managers versus non-managers concerning the notion telehealth processes were optimized.

H5_a

A statistically significant difference exists in the opinions of managers versus non-managers concerning the notion telehealth processes were optimized.

Significance of the Study

The researcher used a quantitative research methodology to presented a theoretical framework based on the Capability Maturity Model Integration (Ghazal et al., 2013). The findings of this study could aid healthcare providers and patients in implementing innovative Telehealth technology using the maturity model. It would contribute to the existing literature and help improve medical service delivery by communicating telehealth diagnoses and building a successful relationship between the provider and patient (Hu et al., 2002). This research would help fill gaps in the use of telehealth technology that frequently arises at the beginning of its deployment (Van Cleave et al., 2022). Telehealth had developed into a technology-driven platform that offered a service delivery model, which provided exclusive opportunities to expand

access to treatment and reduce barriers to healthcare (Butzner & Cuffee, 2021). Implementing the Capability Maturity Model Integration (CMMI) would preserve the implementation and development of Telehealth technology. A maturity model would improve deployment by integrating a framework that covers competencies in areas such as task-sharing, training clients and providers to use technology to improve access, reducing costs, and eliminating stigma (Doss et al., 2017). This study would introduce a viable maturity model facilitating process improvement and augment process maturity standards within the telehealth domain, benefiting from finding a proactive approach to producing a viable maturity model (Otto et al., 2019). This study presented a useful maturity model to enhance process improvement and maturity standards in the telehealth domain. The goal was to take a proactive approach and provide a viable model to benefit the healthcare industry (Otto et al., 2019).

Definitions of Key Terms

CMMI

Capability Maturity Model Integration - A process and interactive model helping modernize process improvement and support productive, efficient performances to decrease risks in implementing products and service development (White, 2021).

IT Service Management (ITSM)

A methodology to IT operations was categorized by its emphasis on IT services, customers, service level agreements, and IT roles in handling its day-to-day activities and processes (Iden & Eikebrokk, 2013).

Maturity Model

A commonly used technique valuable to assess processes used to measure the current process level (Proença & Borbinha, 2016).

Process

Phases and assessments to complete work (Baird, 2023).

Process Efficiency

Suggests the fastest or the cheapest way of operating processes and means executing the right strategies and attaining the desired goals (Wilson et al., 2007).

Process Effectiveness

Providing the desired output at the right time, the right way, and at the right place at the right cost (Wilson et al., 2007).

Process Improvement

Preparing to identify, analyzed and improve current processes to optimize performance for best practice standards and improve customer relationship quality and experience (White, 2019).

Process Maturity

A method of measuring progress, continuously improving processes, and a specific dynamic mechanism facilitating improvement (Kosieradzka & Rostek, 2021).

Process Optimization

Regulating continuously adapting processes to advance them (Sydle, 2022).

Process Quality

The point to a suitable process, comprising measurements and criteria for quality, had been implemented and followed to produce the product (Sceweb, 2001).

Process Variation

The average performance per se or somewhat dependability in a process following a precise pattern for total quality management of implementing variations to increase or reduce items in the process improves quality (Tsikriktsis & Heineke, 2004).

Summary

Telehealth was a technology-based platform that provided an alternative to traditional healthcare services. It offered distinct possibilities to enhance access to treatment and remove obstacles to healthcare (Butzner & Cuffee, 2021). This study was needed because the existing literature revealed a lack of telehealth maturity models that could aid in improving processes and enhancing the process maturity paradigm within the information technology telehealth field.

While some CMM and CMMI derivative frameworks were mentioned in the literature reviewed, none were explicitly designed for telehealth. This study would examine the latest telehealth technologies and use the CMMI model to improve the maturity of managerial processes (Gresse von Wangenheim et al., 2013). Specifically focusing on enhancing vendor and supplier interactions, properly classifying and resolving process issues, and mitigating risks. The goal was to create a telehealth-specific maturity framework using the CMMI process maturity framework to advance and improve processes (Picket, 2016). The CMMI framework focused on enhancing processes, minimizing service delivery risks (Watts, 2020), and combining multiple business maturity models into one framework (White, 2021). It would add to the body of knowledge, given that the reviewed literature did not showed a derivative framework uniquely suited to the needs of telehealth processes.

Chapter 2: Literature Review

This literature review supports the purpose and problem statement of the study, which aims to examine how the Capability Maturity Models (CMM) of various industries compare to processes that cannot be used in a telehealth maturity model. The purpose of this quantitative research would be to investigate the potential of adapting a CMM as a telehealth framework for implementing innovative telehealth technologies designed to follow a Capability Maturity Model Integration (CMMI). The CMMi consists of process improvement, decreasing risks in service (Watts, 2020), and combining multiple business maturity models into one framework (White, 2021). The model was tailored to software engineering in its first instance as the Software CMM and was established between 1987 and 1997. It then became more conceptually generalized, allowing it to be applied to hardware, software, and service development (White, 2021) and given its adapting and implementation among unrelated domains, formulating the basic maturity framework from unrelated domains to telehealth. The problem to address was the need for an effective and customizable maturity model to implement innovative technology sustainably in healthcare for telehealth (Gomes & Romão, 2018).

This literature review contains different maturity models and their applications across various industry sectors, including the Blockchain Maturity Model (Richter et al. (2018), Business Intelligence Maturity Model (Najmi et al., 2010), Business Process Maturity Models (Klun & Trkman, 2018), Capability Maturity Model Integration (Doss et al., 2017) and Data Science Capability Maturity Model (Carvalho et al., 2016). Energy and Utility Management Maturity Model (Finnerty et al., 2011), Environmental Maturity Model (Doss et al., 2017), Healthcare Support Maturity Model (Gomes & Romão, 2018), and Maturity Model for Data-Driven Manufacturing (Gökalp et al., 2021). Maturity Model for Information Technology

Management (Becker et al., 2009), Maturity Model for Manufacturing Enterprises (Mrugalska & Stasiuk-Piekarska, 2020), and Project Management Process Maturity Model (Curtis et al., 2009).

Each maturity model was presented through different viewpoints, offering a comprehensive perspective. The review examined 'maturity' in models, defining it as complete, perfect, or ready. It critiques the existing research, especially in implementing a telehealth maturity model. The review focused on the general features of these models and their use in the telehealth industry, emphasizing the importance of understanding digital health technologies, patient engagement, data privacy, and the interoperability of health information systems. The review also highlights the potential limitations of these models, such as the focus on quantitative evaluation indices, which might overlook qualitative aspects like patient satisfaction and quality of care (Lianying et al., 2012). It further examined the portability of the maturity framework across domains, with findings suggesting that this might only sometimes be feasible (Doss et al., 2017). This review provided valuable insights into maturity models, their applications, and their limitations. It was helpful for those interested in telehealth and developing more effective and integrated models.

This review delves into the intricacies of implementing telehealth technology and leverages a Capability Maturity Model Integration (CMMI) maturity model to guide its integration. The primary objective was to discern the adaptability of CMMI within telehealth. The literature review begins with an outline that underscores the importance of a structured approach to implementing telehealth technology. By leveraging a maturity model, the research provided invaluable insights into the adaptability of CMMI within telehealth, offering a novel framework that the existing literature had overlooked. The meticulous methodology, comprehensive data analysis, and the study's significant contribution accentuate its relevance and

potential in shaping the future of telehealth technology. The review continues by offering a novel framework that the existing literature had overlooked.

The focus of this literature review was to delineate a structured approach, beginning with formulating research questions and culminating in navigating dynamic research landscapes. A search strategy for this research utilized various multidisciplinary platforms like the National University Library, Google Scholar, PubMed, IEEE Xplore, Scopus, ScienceDirect and Web of Science. Scholarly journals were sourced from JSTOR and niche academic interests in psychology and platforms like PsycINFO and Agricola catered to agriculture. The review's commitment to quality was also emphasized by sourcing articles from esteemed publishers like Taylor and Francis resulting in a diverse and relevant set of articles.

Keywords used in the search were Synonyms, Boolean Operators "AND" to combine keywords, "OR" for synonyms, and "NOT" to exclude terms. Additional search filters like date range focused on recent publications or include older seminal works. Publication type filtered by journal articles, conference papers, theses, etc., as needed. Access type that prioritizes open-access sources if subscription-based sources were not accessible.

Theoretical Framework

Lee et al., (2019) proposed using a maturity model framework to improve telehealth processes progressively and incrementally. The proposed model would assess and enhance managerial procedures in telehealth organizations (Bashshur et al., 2011). The CMMi framework could improve vendor and supplier interactions and process issues and mitigate risks in telehealth implementation (van Dky & Schutte, 2012). The CMMi framework offered a structured approach to assess process maturity and determine process capability (Gökalp et al., 2021). It includes five maturity levels, establishing a hierarchy for assessing process maturity and continuous improvement (Paulk, 2002). Each level represented a progressive step in process

development, building upon established augmentations (Poepelbuss et al., 2011). The journey starts with fundamental management practices and advances through a predetermined sequence of levels.

1. During the initial stage, telehealth processes were often unstructured, unregulated, and reactive, leading to potential risks and inefficiencies (White, 2021).
2. The managed stage of telehealth projects focused on addressing requirements and planning, performing, and measuring processes to ensure sustainability (David Patón-Romero et al., 2019).
3. In the defined stage, telehealth organizations achieve maturity-level goals by implementing engineering processes focusing on usability. This ensures that all development requirements were met, technical solutions were provided, and verification and validation were performed (Jokela & Lalli, 2003).
4. Quantitatively Managed was a project management stage that allows for trade-offs among cost, quality, and timeliness by refining organizational performance through obtaining chronological results for level 3 projects (Royce, 2002).
5. The optimizing stage focused on analyzing telehealth process performance data to drive improvements proactively (Lee et al., n.d.).

In the sphere of healthcare innovation, the prospect of a forthcoming telehealth maturity model arises (Bashshur et al., 2011). It might guide healthcare organizations toward procedural enhancement and attaining heightened maturity levels (Carvalho et al., 2016). This model proposes a perspective based on the Capability Maturity Model Integration (CMMI) framework (Soydan & Kokar, 2006). This integration aims to align technical and administrative procedures within the context of telehealth, thereby fostering improved interactions with vendors and

suppliers, offering resolutions to pertinent issues, and proactively mitigating risks (Gibson et al., 2006).

CMMI Framework and Telehealth Process Maturity

The CMMI framework, as discussed by van Dky & Schutte (2012), provided a structured approach to assess process maturity and determine the capability of processes in prioritizing development efforts. The progressive steps were delineated as follows (Watts, 2020):

1. Initial: At the initial stage, telehealth processes were considered irregular, poorly controlled, and reactive. This level signifies an unpredictable environment that leads to increased risks and inefficiencies (White, 2021).
2. Managed: The managed stage represented a progression for telehealth projects. It focused on ensuring requirements were addressed and processes were planned, performed, and measured to consider relevant aspects related to sustainability (David Patón-Romero et al., 2019).
3. Defined: In the defined stage, telehealth organizations had achieved the goals assigned to maturity levels. They implement engineering processes from a usability viewpoint, ensuring all development requirements were met, technical solutions were provided, and verification and validation were performed (Jokela & Lalli, 2003).
4. Quantitatively Managed: The stage involves obtaining chronological results for level 3 projects. It enables trade-offs among rival dimensions of business performance (cost, quality, timeliness) by refining organizational performance (Royce, 2002).
5. Optimizing: The optimizing stage focused on proactively examining and controlling telehealth process performance using analytical methods and process performance data to drive improvements (Lee et al., n.d.).

Telehealth offered significant potential to enhance access to healthcare services and overcome obstacles in traditional healthcare delivery (Howard & Kaufman, 2018). By incorporating a unique CMM/CMMI framework perspective within telehealth administrative procedures, this proposed telehealth model aims to address the literature gap reflecting an absence of discussions involving telehealth process improvement via maturity modeling (Gibson et al., 2006). The model's development would align technical and administrative processes with the established stages of the CMM/CMMI constructs by reviewing relevant literature (Blondiau et al., 2016). Once created, the telehealth maturity model would encompass different maturity levels and their corresponding characteristics (Goldenson & Gibson, 2003). By leveraging the CMMi process maturity framework, the proposed telehealth model seeks to advance and improve telehealth processes, enhancing vendor and supplier interactions, resolving process issues, and mitigating risks (Gibson et al., 2006). It would contribute to the body of knowledge by providing a telehealth-specific maturity framework tailored to the unique needs of the telehealth domain (Gibson et al., 2006). Applying the CMMI framework to the telehealth industry might elevate the maturity of managerial processes and promote the effective implementation of innovative technologies (Gibson et al., 2006). The proposed telehealth model, emphasizing process maturity, could enable organizations to enhance their telehealth practices, improve service delivery, and achieve desired outcomes in the rapidly evolving healthcare landscape (Gibson et al., 2006).

The realm of maturity models and their derivatives offered a rich tapestry of frameworks designed to assess and improve organizational capabilities across various industries. These models provide organizations with structured approaches to evaluate their maturity levels, identify areas for improvement, and guide their growth and development. By leveraging these maturity models, organizations could embark on transformative journeys, enhance their

processes, optimize efficiencies, and achieve higher levels of success within their respective domains.

Maturity Model Derivatives

The evaluation of telehealth maturity models offered a composite outlook thanks to the diversity of scholarly perspectives. One salient point of focus was the 'maturity' metric, which evaluated models based on their level of completion, readiness, and efficacy. Research in this domain primarily examined the attributes that make these models functional and relevant within telehealth. Elements such as integrating digital health technologies, patient engagement, data security, and interoperability among health information systems were often cited as vital components (Lianying et al., 2012).

However, the literature also indicated potential areas for improvement. One such limitation identified was an overreliance on quantitative measures for evaluation. While these offer valuable insights, they might need to pay more attention to qualitative factors such as patient satisfaction and the overall quality of care (Lianying et al., 2012). The scope of applicability for these maturity models across various domains was another critical subject of investigation. Research findings propose that the portability of these frameworks was not universally guaranteed and might be context-specific (Doss et al., 2017).

The current body of research provided a multifaceted understanding of telehealth maturity models, emphasizing the importance of a comprehensive approach to digital health technologies, patient involvement, data confidentiality, and system interoperability. Nevertheless, attention must also be devoted to qualitative aspects and the generalizability of these models across different contexts.

Blockchain Maturity Model

Navigating through the lattice of blockchain maturity models presented a complex but intriguing scenario. Richter et al. (2018) were among the pioneers in crafting the Blockchain Maturity Model for Local Electricity Markets (BMM-LEM), addressing the sluggish pace of evolution and stringent regulatory practices in the energy sector (Zhu et al., 2020). However, it was essential to note that while this model was the first to concentrate exclusively on blockchain-based LEMs (Akbar et al., 2022), its youthful stage of development poses challenges for establishing benchmarks, particularly for its latter stages (Gajdzik, 2022).

The impact of blockchain extends beyond LEMs. For example, it could significantly overhaul supply chain systems (Federgruen et al., 2000), offering a bulwark against the vulnerabilities of traditional centralized systems like Enterprise Resource Planning, which was prone to errors and corruptibility (Hedman & Borell, 2002). Its promise spans multiple sectors, including agriculture, finance, and healthcare, with features like traceability and smart contracts bringing in new levels of efficiency and security (Ronaghi, 2021; Javaid et al., 2021). However, to superimpose BMM-LEM onto telehealth was to flirt with a square-peg-round-hole situation. The telehealth landscape was complex, involving diverse variables, such as patient data privacy, interoperability, and health-specific regulatory compliances (Kang et al., 2010). Even if blockchain technology could offer advancements to telehealth, these sector-specific considerations render BMM-LEM an unsuitable model for direct application (Lundon, 2023; Higdon, 2018). Moreover, the agricultural sector's distinct facets, like smart contracts, IoT, and transaction records, demand their blockchain maturity model (Musah et al., 2019; Cao et al., 2020). Despite the sectoral disparities, blockchain technology remains a potentially transformative agent, demanding specific assessment metrics for each field it enters (Su et al., 2023; Ronaghi, 2021).

In summary, while the BMM-LEM model propounds a structured way to assess blockchain maturity within LEMs, its direct application to the telehealth sector must be matched (Kahn et al., 2016; Harahap et al., 2020). Given the complexities and unique requirements of the telehealth field, a new maturity model tailored explicitly to its attributes would be more applicable. Therefore, while the existing blockchain maturity models contribute to a broader understanding of technology applications across different domains, the telehealth sector necessitates its specialized model for practical evaluation.

Business Intelligence Maturity Models

In an empirical investigation conducted within the Iranian banking sector, the maturity level of Business Intelligence (BI) processes was put under the microscope. Utilizing questionnaires, the study “The evaluation of business intelligence maturity level in the Iranian banking industry” by Najmi, Sepehri, and Hashemi (2010) concluded that the industry's BI maturity resides at level three, indicating that processes like data acquisition, analysis, and action-taking were well-established (Najmi et al., 2010). While the research conducted by Najmi, Sepehri, and Hashemi in 2010 had advanced the understanding of the Business Intelligence (BI) process, its various sub-processes, such as data extraction and warehousing, including aspects like lag, warrant further exploration and development (Raghupathi & Raghupathi, 2021). Additionally, Najmi, Sepehri, and Hashemi's (2010) study sheds light on the higher maturity level of private banks compared to their public counterparts. Furthermore, the research provided specific recommendations for improving data warehousing and Online Analytical Processing (OLAP) areas.

Given the complexity of industries like healthcare, domain-specific maturity models offer more nuanced insights, as underscored by works from Brooks et al. (2015) and Zahrullaili & Noordin (2018). However, applying these banking-specific findings to a Telehealth Maturity

Model was akin to comparing apples to oranges. The metrics for success and constraints in telehealth, from patient care to regulatory compliance, differ vastly from those in the banking sector (Kang et al., 2010; Pool et al., 2022).

Turning towards churn management, Singh and Samalia in 2014 suggested that BI was a pivotal tool in mitigating customer and employee attrition, a vital aspect of the long-term success of any organization (Kozak et al., 2021). BI provided the bedrock for decision-making processes that were increasingly intricate in today's competitive landscape. It was not merely about reactive responses; instead, BI enables organizations to adopt a more proactive, long-term stance toward challenges and opportunities (Gupta et al., 2015; Nguyen et al., 2005). Nevertheless, BI's definition and utility still generate considerable debate, as its interpretation varies depending on whether you ask software vendors or consultancy agencies (Felden et al., 2015). This influences how BI was employed across different sectors, such as marketing and human resources, to consolidate and analyzed pertinent data (Ahmad et al., 2020). The broad scope of BI's impact was acknowledged in various studies, emphasizing its role in enhancing both tangible and intangible assets, thereby recalibrating the value chain within organizations (Elbashir et al., 2008; Singh & Samalia, 2014; Gibson et al., 1970).

However, one should exercise caution when transplanting these insights onto a Telehealth Maturity Model. The telehealth domain grapples with specialized issues like remote medical service provision, stringent privacy protocols, and technological reliability (Stasolla et al., 2016). Although noteworthy, Churn management had represented just one facet of this intricate industry and might not encapsulate telehealth's challenges and opportunities (De Keyser et al., 2019; van Dyk & Schutte, 2011).

Business Process Maturity Models

While research in the arena of Business Process Management (BPM) had indeed burgeoned over recent years, much of it only skims the surface, failing to delve into the nitty-gritty complexities or to provide a systematic dissection of the subject matter (Klun & Trkman, 2018). The palpable hunger for an all-encompassing critique of BPM maturity models cannot be overstated as BPM capability development grows evermore salient in contemporary industries (Recker & Mendling, 2015). Distinguished scholars like Röglinger, M., Pöppelbuß, J., & Becker, J. (2012) endeavored to fill this intellectual void, unearthing ten maturity models specifically honed to BPM. Such models delineate a comprehensive continuum from elementary, fledgling stages to paradigms of meticulous development and superior process functionality (Couckuyt & Van Looy, 2021). However, an undeniable ambiguity had loomed: these models lacked unambiguous, codified instructions for organizational implementation, thus posing a labyrinthine undertaking for prospective adopters (Röglinger et al., 2012).

Scholars advocate for demarcating clear boundaries between generalized processes and BPM-specific maturity models to mitigate such challenges. This partition would facilitate integrating and amalgamating preexisting frameworks, fostering a more intuitive, accessible approach for organizations (Vallerand et al., 2015). Consequently, tailoring these models to fit the unique contours of individual organizations was imperative for their effective implementation (Cummins, 2017). Moreover, despite their inherent limitations, current BPM studies offer invaluable wisdom for practitioners, unfolding a rich tapestry of ready-to-implement maturity models (Schmiedel et al., 2013). Nevertheless, it was pivotal to remember that transposing these frameworks onto something as intricate as a Telehealth Maturity Model might necessitate recalibration (van Dyk & Schutte, 2013). Telehealth was a complex quilt of healthcare practices, technological prowess, and patient engagement, making it a distinct beast

altogether (Carvalho et al., 2019). Thus, the crystalline focus of current BPM models might not have aligned with the multifaceted, patient-centric approach that was paramount in telehealth scenarios (Haroon et al., 2022). While such a narrow focus serves its purpose, it potentially leaves essential aspects of a Telehealth Maturity Model, from telecommunication infrastructures to healthcare regulations, in the peripheral vision (Wilson & Newton, 2013).

As a corollary, Van Looy's seminal 2014 work, deeply embedded within the Springer Briefs in Business Process Management, lays down a robust scaffolding for business process maturity across diverse sectors (Van Looy, 2014). However, its applicability becomes hazy when introduced into the specialized milieu of telehealth, failing to address its unique considerations, such as regulatory compliance and patient satisfaction (Carvalho et al., 2016; Perimal-Lewis et al., 2021).

In conclusion, while BPM maturity research offered a plethora of insights into the mechanics of organizational development, its transposition into the complex web of telehealth care requires a more nuanced, tailored framework one that assimilates the cross-disciplinary variables unique to healthcare, technology, and patient care (Škrinjar & Štemberger, 2008; Gammon et al., 2015).

Capability Maturity Model Integration

In an endeavor to construct a robust framework for industrial processes, Doss et al. (2017) introduced the Industrial Process Maturity Model (IPMM), an offspring of the classic Capability Maturity Model (CMM). This five-tiered hierarchy spans from the chaotic 'Initial' stage to the zenith of 'Optimizing,' where processes were continually refined (Doss, et al., 2017). While their research sheds invaluable light on the extremes of this maturity continuum, they acknowledge the lacuna in demonstrating the model's universal applicability across diverse domains (Doss et al., 2017).

Radu Constantinescu and Ioan Mihnea Iacob took a more expansive view with their Capability Maturity Model Integration (CMMI), which aims to address organizational issues in many sectors. Their theoretical construct was predicated upon established management principles that had found utility in realms as diverse as manufacturing and software engineering (Constantinescu & Iacob, 2007; Gallagher, 2002). The malleability of CMMI allows it to be segmented into various operational categories like Mission Management, Mission Support, and Operational Process Management (Najmi et al., 2005). Gallagher (2002) further anatomizes these into granular tasks, providing a meticulous blueprint for process improvement. However, Doss et al. (2017) indicated such models offer insights into traditional industrial sectors and software development, and their relevance wanes regarding nuanced elements of Total Quality Management (TQM), Six Sigma, and legislation-based practices.

However, regarding the telehealth arena, CMMI's focus becomes rather constricting. This was more than just an issue of technical compatibility; it was an interplay of complex factors like digital health technologies, patient engagement, and data privacy that the model fails to capture (Gibson et al., 2006). As telehealth blends medical practices, intricate technologies, and complex regulations, applying CMMI or any derivatives might be akin to fitting a square peg into a round hole. The crux here was that while the Capability Maturity Model was envisaged initially for structured industrial contexts, it found itself in unfamiliar territory when applied to the heterogeneous and highly regulated telehealth environment. While CMM and its spinoff, IPMM, provide a sturdy foundation for understanding process maturity, they grapple with encapsulating the patient-centricity, data security, and interoperability quintessential in telehealth (Doss et al., 2017).

Future research endeavors might find merit in adopting a hybridized investigative approach. The development of a dedicated telehealth-specific framework, rooted in the

foundational tenets of the Capability Maturity Model Integration (CMMI) but meticulously tuned to accommodate the distinctive attributes inherent to healthcare delivery, holds the potential to furnish a more comprehensive and contextually applicable apparatus for the systematic evaluation and enhancement of telehealth maturity, as advanced by Hidayati et al. (2018). It was imperative to acknowledge that extant models function as transitional milestones rather than culminating destinations within the overarching journey toward formulating an all-encompassing Telehealth Maturity Model (Carvalho et al., 2019).

Data Science Capability Maturity Model

In the quest to optimize Hospital Information Systems (HIS), researchers like João Vidal Carvalho, Álvaro Rocha, Rogier van de Wetering, and António Abreu had put their weight behind the pivotal role of maturity models. These models offer a sophisticated scaffolding for managing complex Information Systems (was) across various industries, notably healthcare (Carvalho et al., 2016). This realization had engendered comprehensive maturity models centered on Big Data Analytics, spotlighting the indispensability of harnessing data effectively (Al-Sai et al., 2019; Pour et al., 2022). Nevertheless, even as organizations pour resources into these digital ventures, many find themselves ensnared in a labyrinth of technological complexity, necessitating guided roadmaps (Pour et al., 2022).

Turning the lens to Data Science Capability, Gökalp et al. (2021b) formulated a Data Science Capability Maturity Model (DSCMM) to gauge organizational proficiency in this discipline. Intriguing as it might be, the DSCMM model carries certain caveats regarding its application in the telehealth domain. What we had here was a paradox. On one side, a model finely tuned for data science proficiency; on the other, a healthcare sector replete with nuanced challenges such as safeguarding patient privacy and navigating the convoluted maze of healthcare regulations (Gomes & Romão, 2018; Loane & Wootton, 2002). Thus, the telehealth

sector needs to oscillate from patient confidentiality to the sophistication of telecommunication infrastructure. A generic model like DSCMM might need to be revised when applied to such a multifaceted landscape. The take-home message? What was required was a specialized framework handcrafted to encapsulate the idiosyncrasies of telehealth (Gökalp et al., 2021). So, while existing maturity models provide insightful forays into optimizing processes and harnessing data, they were not a perfect fit for the intricate matrix forming the telehealth sector.

Energy and Utility Management Maturity Model

In their seminal work, Finnerty et al. (2011) introduced the Global Energy Management System (GEMS), a methodology founded upon four central pillars: Site Characterization, Performance Evaluation, Shared Learning and Dissemination, and Corporate Policy. This methodology emphasized the substantial energy consumption attributed to the industrial sector, underlining the need for sustainable management of energy resources (Finnerty et al., 2011). Following this, Ngai et al. (2013) proposed the Energy and Utility Management Maturity Model (EUMMM), a framework structured around the Capability Maturity Model Integration (CMMI). This model aimed to offer manufacturing organizations a roadmap for the responsible consumption of natural resources (Ngai et al., 2013; Hasan & Trianni, 2020).

Rajic et al. (2022) offered another perspective, devising a maturity model for energy management that integrates ISO processes and mainly incorporates the Plan-did-Check-Act (PDCA) cycle. Although ISO standards had generally been accepted as benchmarks for energy management, Rajic et al. revealed limitations when the model was applied to 24 production organizations in Serbia over two years (Rajic et al., 2022). However, when these frameworks were considered in telehealth, various challenges emerge. Specifically, the telehealth sector entails a distinct set of considerations such as patient confidentiality, data security, quality of care, and accessibility (Gomes & Romão, 2018; Dávalos et al., 2009). Moreover, the healthcare

industry had ethical, regulatory, and practical considerations, which might not align seamlessly with models designed primarily for energy management or manufacturing sectors (Gomes & Romão, 2021).

Therefore, while existing energy management models provide valuable structural insights, their direct application to the telehealth sector might need to be improved. A Telehealth Maturity Model would necessitate adaptations and possibly additional elements to accommodate healthcare delivery's unique requirements and complexities (Kolukisa Tarhan et al., 2020; Shaygan & Daim, 2023). While the energy management models might offer foundational elements for constructing a Telehealth Maturity Model, they would require substantive modifications to fully apply to this specialized field (Rajic et al., 2022).

Environmental Maturity Model

In a study published in *Energy and Environmental Engineering*, Doss et al. (2017) introduced an Environmental Management Maturity (EMM) model, a derivative of the Capability Maturity Model Integration (CMMi). This five-stage model aims to enhance organizations' environmental stewardship by guiding them through stages ranging from immature processes to optimized ones (Doss et al., 2017). Telehealth's core focus on aspects like patient care, data security, and technology infrastructure differs significantly from the EMM model's emphasis on environmental impact and disaster management (Gomes & Romão, 2018; Carvalho et al., 2016a).

Ormazabal et al. (2017) presented a different EMM model in the *Management of Environmental Quality: An International Journal*. Developed through rigorous research methodologies involving numerous companies and ecological experts, this six-stage model caters to diverse industries. It incorporates various elements, such as agents involved, policies, tools, and indicators, to assist organizations in progressing through the maturity stages (Ormazabal et

al., 2017; Prieto-Sandoval et al., 2019). While this model serves industrial firms well by helping them assess their environmental management maturity (Asah-Kissiedu et al., 2021), its direct applicability to telehealth was limited. It predominantly addresses environmental aspects, unlike Telehealth's patient-centric and technologically nuanced requirements (Gomes et al., 2022; Carvalho et al., 2016a).

Therefore, though the EMM models provide valuable frameworks for environmental stewardship, their transferability to telehealth maturity needs to be revised. Both models would require significant adaptations to cater to telehealth's specific priorities around patient care, quality of service, and technological infrastructure (Melnik et al., 2002).

Healthcare Support Maturity Model

In the expansive corpus of literature on maturity models within various sectors, the applicability to telehealth emerges as a complex and nuanced issue. Gomes and Romão's (2018) investigation comprehensively explores healthcare's labyrinth of information systems (was). While Gomes and Romão's (2018) work encapsulates seminal models such as the Capability Maturity Model Integration (CMMI) and the Electronic Medical Record Adoption Model (EMRAM), it was anchored predominantly in traditional healthcare settings. Consequently, the paper's relevance to the unique challenges of telehealth remains ambiguous, thus calling for an extension of their framework into this burgeoning field (Gomes & Romão, 2018). In a parallel vein, the study by Fitterer and Rohner (2009) proposes the Networkability Maturity Model, which focused on operational efficiency within healthcare organizations. While insightful, the model lacks comprehensive indicators for gauging maturity in the telehealth landscape (Fitterer & Rohner, 2009). Pour et al. (2019) delve into the digital domain with their Health 2.0 maturity model, primarily focusing on social media capabilities in healthcare settings. Pour et al. (2019) presented a robust mechanism for process-based assessments; it diverges significantly from the

multifaceted complexities required for assessing telehealth maturity. It was valuable for understanding social media applications but leaves the broader aspects of telehealth unexplored (Pour et al., 2019).

Doss et al. (2017) presented a rather innovative approach to maturity models through their Industrial Process Maturity Model. The study gathers perceptions from managerial and non-managerial personnel in the industrial sector. However, given Doss, et al.'s (2017) environmental scope, the framework might necessitate significant reconceptualization to adapt to telehealth's patient-centric, technology-driven scope. Ganzarain and Errasti (2016) extend the dialogue into Industry 4.0, emphasizing the need for small and medium-sized enterprises (SMEs) to integrate digital technologies. Although the maturity model they propose was rigorous, its contextualization within the manufacturing sector limits its applicability to telehealth, which entails a distinct set of regulatory and technological challenges (Ganzarain & Errasti, 2016). Lee et al. (2019) offer a meta-analysis of maturity models, identifying 194 across various domains. While Lee et al. (2019) study was invaluable for practitioners and researchers seeking to understand the landscape of maturity models, its broad focus precludes a targeted applicability to telehealth. This underscores the exigency for research that specifically explores the construction and validation of a Telehealth Maturity Model (Lee et al., 2019).

While these studies offer compelling contributions to the discourse on maturity models, they did so within their respective domains. As such, their direct applicability to telehealth remains limited, emphasizing the need for a targeted exploration into this emerging and complex field.

Maturity Model for Data-Driven Manufacturing

Mert Onuralp Gökalp et al. (2021) introduce a Data Science Maturity Model (DSMM) tailored for manufacturing organizations. The DSMM was designed to assist organizations in

assessing their data science capabilities and guiding them toward data-driven manufacturing excellence (Gökalp et al., 2021). It comprises six maturity levels and twenty-eight data science processes grouped into six categories, providing a comprehensive framework for improvement in manufacturing (Gökalp et al., 2021). Gökalp et al. (2021) argue that data science could revolutionize manufacturing by converting raw data into actionable insights, enhancing operational efficiency, improving product quality, and reducing costs and risks. Multiple case studies conducted across various manufacturing organizations of different sizes, industries, and countries validate the DSMM's versatility and ability to evaluate an organization's data science maturity, offering valuable insights for enhancement (O'Donovan et al., 2020). However, it was essential to note that while the DSMM was suitable for manufacturing processes and strategies, it might not directly apply to a Telehealth Maturity Model due to the unique considerations of telehealth, such as patient care, healthcare delivery, data privacy, and healthcare regulations, which differ from manufacturing (Gökalp et al., 2021).

In contrast, Zineb Korachi and Bouchaib Bounabat (2018) presented a maturity model focused on assessing the "smartness" of a city based on data quality. The Data Quality in Smart Cities Maturity Model (DQSC-MM) evaluated the quality of data produced and consumed by a city to gauge its progress in becoming a smart city. This model includes key domains like connectivity, data centers, data analytics, applications, and end-users (Korachi & Bounabat, 2018). However, applying the DQSC-MM to a Telehealth Maturity Model would be inappropriate for several reasons. Firstly, the DQSC-MM was designed to assess the maturity of a smart city, not a healthcare system (Korachi & Bounabat, 2018). The domains it evaluated might not directly relate to telehealth's requirements and challenges (Anthopoulos, 2017).

Additionally, while data quality was essential in telehealth, other factors must be considered when assessing the maturity of a telehealth system (Korachi & Bounabat, 2018).

Lastly, Christian Weber et al. (2017) presented a maturity model for data-driven manufacturing, primarily focused on adapting IT architectures to Industries 4.0. The model defines five levels of data-driven manufacturing, starting with data collection and progressing to the self-optimizing factory level (Yang et al., 2020). However, this model was tailored to manufacturing and Industries 4.0 and might not directly apply to a Telehealth Maturity Model with distinct goals and requirements (Weber et al., 2017).

In summary, while these maturity models offer valuable insights and frameworks for specific domains, such as manufacturing and smart cities, they might need to be more directly transferable to a Telehealth Maturity Model due to the unique characteristics and considerations of the telehealth sector (van Dyk & Schutte, 2013). A dedicated model addressing telehealth's specific needs would be more appropriate.

Maturity Model for Information Technology Management

Becker, Knackstedt, and Pöppelbuß (2009) offer valuable insights into developing maturity models for IT management. These models had enhanced IT performance and organizational change (Akhlaghpour et al., 2018). Maturity models had become crucial tools in the IT management landscape, enabling organizations to assess their current state, identify areas for improvement, and guide their journey toward enhanced IT capabilities (Becker et al., 2009).

However, the focus on IT management criteria and characteristics in Becker et al.'s study must be more suitable for direct application to a Telehealth Maturity Model. Telehealth, as a domain within the healthcare sector, presented unique considerations that differentiate it from traditional IT management (Becker et al., 2009). These considerations include patient care, healthcare regulations, and medical data security, all of which require a distinct set of criteria and considerations when developing a maturity model (Becker et al., 2009).

Similarly, de Sousa Pereira and da Silva's (2014) study was centered around IT service management and the ITIL framework (Galup et al., 2009). The ITIL had been widely adopted as a framework for aligning IT efforts with business needs and efficiently managing IT services with guaranteed quality (Galup et al., 2009). However, while ITIL could be adapted to various sectors, telehealth's specific requirements and challenges, including patient care, data privacy, and regulatory compliance, necessitate a tailored approach when developing a Telehealth Maturity Model. It was crucial for such a model to comprehensively address these unique aspects of telehealth to ensure its effectiveness (Pereira & da Silva, 2014). On the other hand, the study by Shaygan and Daim (2019) comprehensively analyzed telehealth in developing countries. It highlights telehealth's benefits, challenges, and prospects but needs the systematic framework required for a Telehealth Maturity Model (Greenhalgh et al., 2012). Telehealth, as a rapidly evolving field, requires a structured approach to assess the maturity of telehealth initiatives (Greenhalgh et al., 2012). This assessment should consider the specific factors relevant to the healthcare sector, such as patient care, data security, and regulatory compliance, to effectively guide the development and growth of telehealth services (Shaygan & Daim, 2019).

Maturity Model for Manufacturing Enterprises

Mrugalska and Stasiuk-Piekarska (2020) comprehensively examine maturity models in Industry 4.0. These models play a crucial role in assessing the readiness and progress of businesses in implementing Industry 4.0 strategies, which involve the application of emerging technologies for digital solutions (Santos & Martinho, 2019). However, the focus of this study on manufacturing enterprises and their journey toward Industry 4.0 might only partially apply to a Telehealth Maturity Model (O'Connell et al., 2013).

As a distinct domain within the healthcare sector, Telehealth had unique considerations that differentiate it from manufacturing (Gomes et al., 2022). These considerations encompass

patient care, healthcare regulations, medical data security, and remote healthcare service delivery (Gomes et al., 2022). Therefore, while the methodologies and approaches discussed in the paper could provide valuable insights, a Telehealth Maturity Model would require a tailored set of criteria and considerations to address telehealth's specific needs and challenges (Mrugalska & Stasiuk-Piekarska, 2020).

In contrast, Mittal, Khan, Romero, and Wuest (2018) focus on Smart Manufacturing (SM) and Industry 4.0 maturity models, particularly in the context of Small and Medium-sized Enterprises (SMEs). While Mittal, Khan, Romero, and Wuest (2018) emphasize the need to consider SME-specific requirements, their study did not mention or discussed Telehealth at any point. Therefore, using this study as a basis for a Telehealth Maturity Model would be inappropriate (Gomes & Romão, 2018). This research centers on manufacturing industries and their unique challenges in adopting Industry 4.0 and Smart Manufacturing technologies, which did not directly align with telehealth requirements (Mittal et al., 2018).

Schumacher, Erol, and Sihn (2016) offer a model for evaluating Industry 4.0 maturity in discrete manufacturing enterprises. This model incorporates organizational and technological considerations (Schumacher et al., 2016). While the development process and dimensions used in this model were valuable, it was essential to note this study's primary focus was on manufacturing and Industry 4.0 (Machado et al., 2019). The dimensions and items used to assess Industry 4.0 maturity might not directly translate to the context of telehealth, which involves healthcare services, patient care, and data security in a virtual healthcare environment (Thomas & Woodside, 2016). Therefore, while the study provided insights into evaluating maturity in the context of Industry 4.0, it might not directly apply to creating a Telehealth Maturity Model (Schumacher et al., 2016).

While these studies offer valuable insights into assessing maturity in various contexts, they might only partially apply to a Telehealth Maturity Model. Telehealth's unique characteristics and challenges require a dedicated effort to develop a tailored maturity model addressing the specific needs of healthcare delivery in a digital and remote context (Wilson & Schild, 2014).

People Capability Maturity Model

People capability maturity model (P-CMM) version 2.0 by Curtis, Hefley and Miller (2009) discusses the tool designed to address critical people issues within an organization (Linders, 2010). It used the process maturity framework of the highly successful Capability Maturity Model for Software (SW-CMM) as a foundation for a model of best practices for managing and developing an organization's workforce (Paulk, 2002). The P-CMM had been used by software organizations worldwide to guide significant improvements in their ability to enhance productivity and quality, reduce costs and time to market, and increase customer satisfaction (Curtis et al., 2009).

The P-CMM consists of five maturity levels, establishing successive foundations for continuously improving individual competencies, developing effective teams, motivating improved performance, and shaping the workforce an organization needs to accomplish its business plans (Dangmei, 2017). Each maturity level was a well-defined evolutionary plateau institutionalizing new capabilities for developing the organization's workforce (Curtis et al., 2001). The model was based on best practices in human resources, knowledge management, and organizational development (Curtis et al., 2001). It guides organizations in improving their processes for managing and developing their workforce (Gama et al., 2013). The P-CMM helps organizations characterize the maturity of their workforce practices, establish a program of continuous workforce development, set priorities for improvement actions, integrate workforce

development with process improvement, and establish a culture of excellence (Curtis et al., 2009).

The P-CMM might not be appropriate for a Telehealth Maturity Model (Gomes et al., 2022). The P-CMM was intended for software organizations and concentrates on enhancing individual skills and building efficient teams within frameworks (Paulk, 2002). Telehealth encompasses a more extensive array of fields, such as healthcare, technology, and patient involvement (Paulk, 2002). Telehealth presented distinct issues and demands, such as guaranteeing patient safety and upholding quality standards (Curtis et al., 2009).

Surega (2019) explores the application of the People Capability Maturity Model (PCMM) in Business Process Outsourcing (BPO) companies. The PCMM, developed by Carnegie Mellon University's Software Engineering Institute (SEI), was an organizational change model that aims to enhance workforce practices and processes through five maturity levels or stages (Dangmei, 2017). The research was conducted in Tamil Nadu, India, focusing on BPOs with Level 5 PCMM certification (Dangmei, 2017). The PCMM analyzed employees' attitudes toward executing critical PCMM practices at different maturity levels. Surega (2019) research employed a descriptive design that included primary and secondary data, with a sample size of 703 respondents from various BPOs. The respondents included team members, HR managers, and other designations, with team members accounting for most of the sample (Surega, 2019).

Subramanyam et al. (2004) indicated that 44% of those surveyed believe that enhancing the workforce's capabilities was necessary to meet the increasing demands caused by a shortage of skilled workers. Meanwhile, 39% think it was essential to accelerate technological advancements, and 17% believe it was necessary to increase the reliability of critical applications (Curtis et al., 2009). The study by Subramanyam et al. (2004) also revealed a notable difference between the execution of Level 2 essential practices and the reason for conducting a PCMM

assessment and a significant difference between the level of execution of Level 2 basic practices and talent shortage issues. However, the study's findings were inconclusive (Surega, 2019).

The study by Subramanyam et al. (2004) offered valuable insights into how the PCMM could benefit BPO enterprises and it might not apply to a Telehealth Maturity Model as it mainly concentrates on BPO firms, which had distinct operational and organizational structures compared to telehealth services (Gomes & Romão, 2018). While the essential practices and processes of the PCMM could be advantageous in enhancing workforce practices in BPOs, they might need to be more relevant in the healthcare context where patient care and medical outcomes were the prime focus (Hans et al., 2011). Additionally, the study's geographical focus on Tamil Nadu, India, might restrict its applicability to telehealth services in other regions or countries with different healthcare systems and regulations (Hall & McGraw, 2014). Therefore, while the PCMM might offer beneficial strategies for workforce development, a Telehealth Maturity Model would likely require a different set of practices and processes customized to the specific requirements and challenges of telehealth services (Surega, 2019).

Wademan, Spuches, and Doughty (2008) focused on the People Capability Maturity Model (People CMM) developed by the Carnegie Mellon University Software Engineering Institute. The model brings discipline to the people side of management by promoting a structured, repeatable, and predictable approach to improving an organization's workforce-related processes (Wademan et al., 2008).

The People CMM model had five maturity levels, twenty-two process areas, and four main theme/objective areas (Wademan et al., 2008). Each process area was associated with a specific maturity level and falls under one of the four theme/objective areas (Wademan et al., 2008). The model had implementation goals and one institutionalization goal for each of the twenty-two process areas (Wademan et al., 2008). The model aims to enhance workforce

capability by providing a path from ad hoc and inconsistent practices to a mature infrastructure of practices (Türetken & Demirörs, 2004). The organizational change model was where each level of the five-tiered model transforms undisciplined states of processes and procedures into disciplined conditions that produce predictable results (Wademan et al., 2008). The model's overall goal was to improve the management and development of human resources within an organization, creating a professional, versatile, and skilled workforce. (Wademan et al., 2008).

However, the People CMM might not be suitable for a Telehealth Maturity Model. The People CMM primarily focused on improving workforce-related processes and transforming the organizational culture (Curtis et al., 2001). It emphasizes the role of managers in developing people who could produce results and considers human capital as the organization's most valuable resource (Curtis et al., 2001). While these aspects were essential in a telehealth context, the People CMM did not directly address the unique challenges (Valenta et al., 2021) and requirements of telehealth, such as the need for specific technical competencies, the management of remote patient care, data privacy and security issues, and integrating various technologies for effective telehealth delivery (Hall & McGraw, 2014). Therefore, while the People CMM provided valuable insights for managing and developing human resources, it might only partially cover the breadth of factors that would be relevant in a Telehealth Maturity Model (Wademan et al., 2008).

Project Management Process Maturity Model

Ibbs and Kwak (2002) comprehensively explore project management's financial and organizational impacts. A team of researchers created a model and methodology to evaluate the maturity of project management processes (Grant & Pennypacker, 2006). The process involved answering 148 multiple-choice questions, covering eight knowledge areas and six project phases (Ibbs & Kwak, 2000). The method was tested on 38 companies and government agencies across

four industries (Ibbs & Kwak, 2000). Results indicated that all companies' average project management maturity score was 3.26 on a scale of 1 to 5 (Ibbs & Kwak, 2000). The engineering/construction industry scored the highest, while the information systems application area scored the lowest (Ibbs & Kwak, 2000).

The research methodology involved several steps, including examining past and current studies related to project management, developing a five-level project management process maturity model (Mullaly, 2006), developing a comprehensive project management process maturity assessment methodology, selecting target organizations/industries to investigate, measuring an organization's project management level, and analyzing collected information to evaluate and benchmark an organization's project management processes and practices level using specific criteria (Ibbs & Kwak, 2000).

Although the study by Kwak and Ibbs (2002) was comprehensive, it might not be suitable for a Telehealth Maturity Model for several reasons. First, the study focused on project management maturity across various industries rather than specifically on telehealth (Ibbs & Kwak, 2000). The project management maturity model in this study might need to adequately address the unique challenges and requirements of telehealth, such as patient privacy, data security, and integrating various healthcare technologies (van Dyk & Schutte, 2013). The maturity model and methodology evaluated project management processes, which might differ significantly from the processes involved in telehealth (Thomas et al., 2022). Telehealth often focused on individual healthcare providers and patients rather than companies or industries (Shulver et al., 2016). Therefore, a different approach might be necessary to accurately assess and improve telehealth maturity (Ibbs & Kwak, 2000).

Kwak and Ibbs (2002) presented a comprehensive model to determine and benchmark an organization's relative project management level with other organizations (Grant & Pennypacker,

2006). The model follows a systematic and incremental approach that progresses from unsophisticated to sophisticated project management maturity (Kwak & Ibbs, 2002). Each maturity level comprises major project management characteristics, factors, and processes (Kwak & Ibbs, 2002).

Based on the Project Management Institute's body of knowledge, the (PM)2 model divides project management practices into nine knowledge areas and five processes (Cao et al., 2013). This helps organizations identify strengths and weaknesses in their current practices and focus on improving weaker areas to increase their project management maturity (Kwak & Ibbs, 2002). The model encourages individuals and organizations to enhance their project management processes, offering guidance and requirements for achieving higher maturity levels (Succar, 2010). It also promotes a shift from function-based practices to a project-based approach, focusing on continuous learning (Kwak & Ibbs, 2002).

The study by Kwak & Ibbs (2002) was unsuitable for a Telehealth Maturity Model for several reasons. First, the (PM)2 model was primarily used as a reference point for an organization applying project management practices and processes, which might not directly apply to telehealth services' specific needs and requirements. Second, the model focused on improving project management effectiveness, which differs from the goal of a Telehealth Maturity Model that would likely focus on improving healthcare delivery, patient experience, and clinical outcomes (Wisniewski et al., 2020). Last, the model's emphasis on project management knowledge areas and processes might need to align with the key focus areas in telehealth, such as patient care, technology integration, data privacy, and regulatory compliance (Kwak & Ibbs, 2002).

Lianying, Jing, and Xinxing (2012) presented a new project management maturity model called P2CMM. This model was built on the attribute synthetic assessment principle and

PRINCE2 process method to evaluate an organization's current project management capability (Gan & Chin, 2019). Lianying, Jing, and Xinxing (2012) note that current maturity models were often limited to software and industrial engineering projects, suggesting further adaptation and quantification of evaluation indices (Dahlin, 2020). The P2CMM model addresses these limitations, offering a quantitative evaluation index system that could be applied to various projects in different environments (Lianying et al., 2012). The model used a web-based questionnaire survey method and refers to the cobweb model to present the result (Fabbro & Tonchia, 2021).

The P2CMM model was based on the PRINCE2 project management approach, which includes eight unique management processes: Starting Up a Project (SU), Initiating a Project (IP), Directing a Project (DP), Controlling a Stage (CS), Managing Product Delivery (MP), Managing Stage Boundaries (SB), Closing a Project (CP), Planning (PL) and processes each corresponding sub-process, totaling 45 (Lianying et al., 2012). The structure of the model's evaluation index system was divided into three levels: target layer, process layer, and sub-process layer (Kostalova & Tetrelova, 2018). The model divides project management maturity into five groups and the data for evaluating maturity level was collected through a web-based questionnaire survey (Lianying et al., 2012).

However, the study's suitability for a Telehealth Maturity Model must be revised (van Dyk & Schutte, 2013). The P2CMM model was heavily based on the PRINCE2 project management approach, which might need to align better with telehealth services' specific needs and characteristics (Gomes et al., 2022). Telehealth involves unique challenges such as ensuring patient privacy, managing remote patient monitoring, and integrating with various healthcare IT systems, which might need to be adequately addressed by a model designed for general project management (Lenert & McSwain, 2020). The P2CMM model's reliance on web-based

questionnaires for data collection might not be suitable for healthcare, where data was often sensitive and subject to strict regulatory requirements (Gomes et al., 2022). Finally, the model's focus on quantitative evaluation indices might overlook qualitative aspects crucial in telehealth, such as patient satisfaction and quality of care (Lianying et al., 2012).

Limitations of Maturity Models in Addressing Telehealth Processes

The Blockchain Maturity Model might face challenges when applying blockchain in telehealth, especially in unique use cases (Ronaghi, 2021). The Business Intelligence Maturity Model did not fully cater to the healthcare intelligence requirements in telehealth (Najmi et al., 2010). Business Process Maturity Models could fall short by not offering specific guidelines tailored to telehealth processes (Tarhan et al., 2016). Capability Maturity Model Integration (CMMI) and the CMMI-based Process Improvement Model might need help to address telehealth processes' unique requirements and challenges (Gallagher, 2002; Gibson et al., 2006). The Data Science Capability Maturity Model might not cover specific considerations for data science applications in telehealth (Gökalp et al., 2021).

The Digital Library Maturity Model and the Digital Maturity Model for Telecommunications might need to be more relevant to telehealth processes' unique requirements and challenges (Sheikhshoei et al., 2018; Valdez-de-Leon, 2016). The Energy and Utility Management Maturity Model and Environmental Maturity Model might need to align better with the unique telehealth requirements and environmental considerations (Ngai et al., 2013; Doss et al., 2017). The Healthcare Support Maturity Model might need to sufficiently focus on the challenges of telehealth processes (Epstein & Peters, 2009). While relevant to healthcare organizations, the Industrial Process Maturity Model might not directly align with telehealth process intricacies (Doss et al., 2017). The IT Capability Quality-Distinction Model might

concentrate mainly on IT capabilities rather than the intricacies of telehealth (Kishore et al., 2012).

The maturity level of the Risk Management model might need more specific guidance for managing risks in telehealth processes (Shah et al., 2009). Models like the Maturity Model for Data-Driven Manufacturing (Weber et al., 2017) and the Maturity Model for Manufacturing Enterprises (Schumacher et al., 2016). The Maturity Model for Information Technology Management might require customization for the specific needs of telehealth (Becker et al., 2009). The People Capability Maturity Model might not encompass the specialized skills needed for professionals engaged in telehealth (Wademan et al., 2008).

The Project Management Process Maturity Model might need to address telehealth's unique project management challenges (Kwak & Ibbs, 2002). Models like the Supply Chain Collaboration Maturity Model (Ho et al., 2016), Supply Chain Management Process Maturity Model (Lockamy & McCormack, 2004), and the Logistic Maturity Model might not directly align with the supply chain dynamics and logistics requirements in the telehealth domain (Battista & Schiraldi, 2013). The Project Management Maturity Model might need more specific guidance for project management in the healthcare sector, especially in the context of telehealth initiatives (Crawford, 2014). The model for Transition from CMMi to SPICE Assessments might need to provide specific guidance on transitioning telehealth processes (Vanamali et al., 2008).

Observations and Synthesis of the Reviewed Literature

The reviewed literature concentrates on the processes of the individual industries. It needs to address the cumulative inadequacies of existing models in addressing the unique challenges and requirements of the telehealth field (Hebert et al., 2004). Based on the limitations described for each industry maturity model in addressing telehealth processes, some everyday observations and syntheses could be made (Jury & Kornberg, 2016). Many industry maturity

models had a general focus and might not offer specific guidance tailored to the unique requirements of telehealth processes (Jack et al., 2020). This indicated the need for customized frameworks or adaptations to address telehealth challenges adequately (Verduzco-Gutierrez et al., 2021).

Several maturity models originate from industries other than healthcare, such as manufacturing, logistics, and telecommunications (Aceto et al., 2020) and market engineering (Doss, et al., 2021). While they might provide valuable insights, their applicability to telehealth processes might be limited without customization or augmentation (Bifulco et al., 2014). Many maturity models need a comprehensive focus on the technological aspects and challenges of telehealth implementation and operation (Obstfelder et al., 2007). Given the reliance on digital platforms and remote communication, addressing technology-related factors becomes crucial in telehealth processes (Englander, 2018). Some maturity models might not sufficiently emphasize patient engagement, user experience, and the importance of delivering high-quality care in telehealth processes (Yaël, 2022). Considering the patient's perspective and ensuring a patient-centric approach was crucial for successful telehealth implementation (Yaël, 2022).

When applying industry maturity models to telehealth, organizations should leverage domain expertise and supplement these models with telehealth-specific frameworks, guidelines, and best practices (Caralli et al., 2012). This ensures a more comprehensive assessment and improvement of telehealth processes (Caralli et al., 2012).

To overcome the limitations of industry maturity models, organizations should customize and adapt these models to the unique requirements of telehealth (might 2015). This includes incorporating telehealth-specific considerations, such as remote care delivery, virtual consultations, data security, privacy, and compliance (Chike-Harris et al., 2021). Although industry maturity models provide helpful information for enhancing processes, they might need

adjustment and expansion to address telehealth processes' unique challenges and demands (Sarmiento et al., 2019). Organizations should contemplate telehealth-specific frameworks and guidelines for thorough evaluations and enhancements in telehealth operations (Valenta et al., 2021). Thus, given such considerations and the reviewed literature, a question emerged as the basis for this study: could the CMM framework be adapted within the context of the telehealth domain?

Summary

This review delves into the concept of maturity in telehealth models, which refers to their completeness, perfection, or readiness level. The study scrutinizes existing research on telehealth maturity models, focusing on their general characteristics. It examined the use of these models and highlights possible limitations, such as the emphasis on quantitative evaluation indices that might neglect qualitative aspects, like patient satisfaction and quality of care (Lianying et al., 2012). The analysis also investigated the feasibility of using the maturity framework across different domains and concludes that it might be practical only sometimes (Doss et al., 2017).

This literature review examined various maturity models in diverse blockchain, business intelligence, business processes, and more sectors. It critically analyzed the concept of 'maturity' in these models and their application, particularly in telehealth. The review identifies potential limitations, such as the tendency to prioritize quantitative evaluation over qualitative aspects like patient satisfaction and quality of care. It also explores the transferability of these maturity frameworks across different domains. The insights gained from this review were essential for those interested in telehealth and developing more effective and integrated models.

This review provided valuable insights for developing more effective and integrated telehealth models. It emphasizes the importance of a comprehensive understanding of digital

health technologies, patient engagement, data privacy, and interoperability of health information systems. The telehealth sector was the primary focus of the review.

Chapter 3: Research Method

The problem to be addressed in this study was to develop an effective and customizable maturity model to be used in healthcare for telehealth for implementing innovative technology in a sustainable manner (Gomes & Romão, 2018). Although there were various Capability Maturity Model integration (CMMi) frameworks accessible to incorporate maturity models effectively, there were still many shortcomings in the process of developing these models to implement innovative technology in the telehealth industry (Scanlon et al., 2020).

The purpose of this study was to examine a derivation of the CMM as a telehealth maturity model to minimize potential risks to services, functional processes must be improved to integrate new technology effectively (Watts, 2020). This study aims to demonstrated process improvement within quantitative research for implementing innovative technology for telehealth (Nematovich & Karimberdiyevich, 2022). The quantitative research process would display a logical path as it explores the benefits of implementing a maturity model design (Mishra & Sharma, 2022). The process combines the characteristics of various models into a single framework to address shortcomings in development procedures (White, 2021).

The maturity model under scrutiny evaluated the current efficacy of a process (Carvalho et al., 2016). This research would encompass five levels of activities, including an IT Service Management (ITSM) maturity assessment (Santos & Martinho, 2019). The study aims for a minimum number of responses from a population of 500, targeting a 98% confidence level, a 5% margin of error, and a population proportion of 50% (Andrade, 2020). The quantity of 218 respondents was the minimum number of responses required to represent the population from which the sample would be drawn. The research would be in a healthcare setting involving telehealth technology, where a quantitative survey would be conducted to assess the feasibility of

adapting the CMMi in the telehealth domain and to scrutinize the current application of process improvement and quality paradigms in the workplace (Doss et al., 2017).

In this chapter, the research process would be systematically outlined, starting with the research methodology and design, which would explain the nature of the study. The chapter would cover the population and minimum number of responses, materials or instrumentation used, and operational definitions of variables. Next, the study procedures would be described, followed by an explanation of the data analysis methods. The chapter would also address assumptions, limitations, delimitations, and ethical assurances applicable to the study.

Research Methodology and Design (Nature of the Study)

The chosen methodology of a quantitative online survey was more appropriate for efficiently capturing specific behaviors in a targeted manager, non-manager population. This method was conducive to prompt data collection and analysis, which were crucial for understanding and addressing the implementation of telehealth concerns (Cunningham et al., 2015). The online survey would be the data collection instrument for this study and was pivotal in facilitating a comprehensive statistical analysis (Lytvyn et al., 2018). This approach efficiently assesses the perceptions of management and non-management personnel within the telehealth domain, explicitly focusing on a derivative of the applicability of the Capability Maturity Model Integration (CMMi) as a framework for process improvement (Tuli, 2011).

The survey's design was tailored to evaluate the CMMi framework thoroughly. Questions were organized to correspond with the five maturity levels of the CMMi, from Level 1, indicative of generally erratic and ad hoc processes, to Level 5, where processes were optimized (Doss et al., 2019). This structured approach ensures a comprehensive assessment of each level within the CMMi framework. Therefore, the quantitative survey methodology, emphasizing structured survey design and statistical analysis, was the most for this study. It offered a direct

and efficient means to assess perceptions across different management levels in the telehealth sector and to quantify the extent to which the CMMI framework might be perceived as applicable and beneficial (van Dyk, 2014). This methodology aligns closely to evaluate the potential of the CMMI as a process improvement framework in a measurable and statistically valid manner, ensuring a thorough exploration of all major themes and concepts (Overby et al., 2010; Ning et al., 2020). The study utilizes a quantitative methodology to explore improvements in implementing innovative telehealth technology. In the context of examining process improvement in implementing telehealth technology, causality was crucial to understanding the specific factors that lead to enhanced outcomes.

Therefore, based on the research questions and hypotheses for this study, analysis was necessary to determine if there were any differences in the respondents' perceptions (Williams, 2021). A longitudinal study might not be suitable as it tracks changes over time and provided valuable insights into long-term effects. However, their extensive nature might not align well with the requirements of a concise 30-day study (Long, 2014). Johnston (2014) argued that secondary data analysis might not provide the most current or relevant information, particularly for studies to understand recent or rapidly changing behaviors. However, Edmonds and Kennedy (2016) suggested that the mixed-methods approach, which integrates quantitative and qualitative data, offered a comprehensive understanding of the subject matter, but it might be too complex and time-consuming for studies that focus on specific behaviors within a limited timeframe. Edmonds and Kennedy (2016) noted that while the mixed-methods approach, blending quantitative and qualitative data, comprehend the subject, its complexity, and time-intensive nature might not be suitable for studying behaviors within a constrained timeframe. According to Edmonds and Kennedy (2016), the mixed-methods approach was complex and time-intensive and cannot be completed within a 30-day timeframe.

Population and Sample

The study would focus on the Veterans Administration (VA) employees involved in telehealth for implementing innovative technology in a sustainable manner (Gomes & Romão, 2018), encompassing both managerial and non-managerial personnel. Given the specialized nature of telehealth services within the VA, it was crucial to understand the perspectives and experiences of these employees to implement the Capability Maturity Model Integration (CMMi) in this domain. The VA's Telehealth National Planning Strategy indicated a significant increase in telehealth encounters, reflecting this sector's growing importance and expansion within the VA (Der-Martirosian et al., 2021). The study on telehealth services within the VA would involve a population of 500 participants. This amount was based on similar studies, which found a manageable, yet comprehensive subset of the population was ideal for analysis. This population size was chosen to provide a cross-section of manager and non-manager positions of the population. Similarly, within this study, the VA study's quantity of 500 participants would represent a broad perspective while still being practical for detailed analysis and interaction. Given the amount of 500 participants and the ability to query the entire population electronically, the researcher would probe the entire population in the study.

A study of 500 participants through telehealth in the VA system was significant due to the sample size required for meaningful analysis in research methodology. Bujang et al. (2018) highlighted the importance of enough responses for statistically significant findings. The importance of the number of participants in a study was demonstrated by Laupland's research on the incidence of bloodstream infections, which was carried out in June 2013. Laupland's study showed that using a response threshold that aligns with the research objectives and the population's characteristics, as well as analyzing data from a similar cohort size, could enhance

the reliability of the research outcomes and ensure that the study's findings were applicable to wider contexts.

In the current study on implementing innovative technology within the VA, the choice of 500 participants was pertinent. This sample size was substantial enough to allow for a comprehensive analysis of telehealth services' impact, effectiveness, and potential areas for improvement within the VA system. Given the VA's role in providing critical healthcare services to veterans, understanding the efficacy of telehealth services was paramount. The employment of a well-considered participant size, as informed by the works of Bujang et al. (2018) and Laupland (2013), provided historical motivation for this study using 500 member populations in the healthcare domain as antecedents for the population size polled in this study.

Transitioning to determining the minimum number of responses necessary, the study utilizes G*Power for this critical task (Kelley, 2003). The minimum number of 218 respondents, selected from the VA employee population for this study, was meticulously calculated using G*Power to ensure the representativeness of the broader population. G*Power was employed for conducting power analysis, a statistical method for determining the minimum number required to detect an effect of size with a certain degree of confidence (Kelley, 2003). The parameters for this power analysis include a 95% confidence level and a 5% margin of error, with an assumed population proportion of 50%. These parameters were carefully chosen to balance the need for statistical validity with the practicalities of managing the minimum number of responses (Moran, 2021; Statistics Solutions, 2022).

Participants would be contacted through an online announcement disseminated via email and local web bulletins within the VA. This recruitment strategy was chosen for its efficiency and broad reach, enabling the study to effectively engage a diverse group of VA employees, encompassing both managerial and non-managerial staff. Using digital communication channels

for recruitment aligns with the study's aim to obtain a representative minimum number of responses efficiently and effectively, ensuring the findings reflect the identified population.

Research within the VA system used a dual approach of email and local web bulletins for participant recruitment (Office of Research and Development, 2018). The researcher would collaborate with the VA's communication department to access the internal directory and obtain necessary permissions from the Veterans Health Administrations Institutional Review Board (IRB) (Office of Research and Development, 2018). The recruitment email clearly would state the purpose of the study and guarantees confidentiality (Office of Research and Development, 2018). The VA's communication department segments the audience and checks emails for compliance before sending them out (Office of Research and Development, 2018). Follow-up reminders would be sent to maintain interest (Office of Research and Development, 2018). The process was meticulously documented to ensure future replication (Office of Research and Development, 2018).

Instrumentation

The quantitative survey utilized SurveyMonkey, a user-friendly web-based survey service provider (Mellinger & Hanson, 2021). SurveyMonkey allows a structured format to measure variables of interest, and its advanced features enable secure data collection, randomization of question order, and real-time monitoring of response patterns (Lowry et al., 2016). The data would be exported into the SPSS statistical software for analysis.

Within the survey, questions 1 through 5 would examine the overall CMMi framework, questions 6 through 8 examine CMMi Level 1, questions 9 and 10 examine Level 2, questions 11 through 13 read Level 3, questions 14 through 16 analyzed Level 4, and questions 17 through 19 examine Level 5 (Doss et al., 2019). The corresponding levels and queried process maturity concepts were:

- Level 1: Most processes were theoretically erratic, chaotic, and ad hoc (Doss et al., 2019).
- Level 2: The processes were reoccurring and reactive (Doss et al., 2019).
- Level 3: The processes were characterized and expressed (Doss et al., 2019).
- Level 4: The processes were measured, contained and investigated quantitatively (Doss et al., 2019).
- Level 5: All processes were optimized (Doss et al., 2019).

The study would incorporate a data collection instrument in a five-point Likert scale survey, a methodological tool selected for its proven effectiveness in translating subjective human experiences into quantifiable data (Taherdoost, 2019). The data collection instrument incorporates a Likert scale ranging from "Strongly Disagree" to "Strongly Agree," was based on the scholarly work of Doss et al., who conducted comprehensive research on Capability Maturity Model Integration (CMMI) derivatives (Doss et al., 2017; Doss et al., 2019). Therefore, in the survey, 'Neutral' would be indicated as option 3, 'Strongly Disagree' as option 1, and 'Strongly Agree' as option 5. This structure allows respondents to express their level of agreement or disagreement with the statements in a nuanced and quantifiable manner. The utilization of the Likert scale in this study serves two primary purposes: first, it facilitates the conversion of respondent perceptions into numerical data, simplifying the process of statistical analysis (Lytvyn et al., 2018); second, the scale's five-point range was designed to capture the subtleties of human opinion while avoiding the complexity of an overly extensive range of options (Storey et al., 2000).

Capability Maturity Model Integration (CMMI) was a structured approach which focused on refining organizational processes, and its application in telehealth was anticipated to enhance service quality, elevate patient satisfaction, and improve operational efficiencies (Abdel-Hamid

& Hamouda, 2015; Donati et al., 2019). By adopting research tools and methodologies previously utilized in the study of CMMI derivatives (Doss et al., 2019; Doss et al., 2017), this research aims to provide empirically rigorous insights relevant to the specific context of telehealth.

This quantitative approach, grounded in measurable scientific principles, allows for precise statistical analysis (Tuli, 2011) to examine perceptions of management versus non-management personnel to determine the potential of adapting the CMMi as a potential process improvement framework in the telehealth domain. The study would utilize a quantitative survey (Overby et al., 2010). Specifically, a five-point Likert scale survey would conduct a cross-sectional study (Ning et al., 2020). The respective Likert category's corresponding numerical values were:

1. Strongly Disagree (Tarka, 2015)
2. Disagree (Tarka, 2015)
3. Indifference (Tarka, 2015)
4. Agree (Tarka, 2015)
5. Strongly Agree (Tarka, 2015)

The survey was intricately linked to the Capability Maturity Model (CMM), which was a tool designed to assess perceptions of process architecture maturity within organizations. This concept, explored in various academic works, outlines a developmental trajectory for organizations, progressing methodically from initial, unstructured stages to highly optimized and controlled phases. The CMM's relevance and adaptability across different sectors had been highlighted in numerous studies, including those by Doss et al. in various peer-reviewed journal articles and conference proceedings, emphasizing its utility in enhancing organizational

processes (Doss et al., 2021; Doss et al., 2019; Doss et al., 2018; Doss et al., 2017; Doss et al., 2008; Doss et al., 2006; Doss & Kamery, 2005).

In the survey, level 1 represented organizational processes as ad hoc, unstructured, and chaotic, often exceeding budgetary constraints. Progressing to level 2, processes become more reactive yet managed, incorporating planning, performance, measurement, and control. Level 3 signifies a stage where processes were characterized and understood, adhering to defined standards, procedures, tools, and methods. At level 4, processes were managed quantitatively, focusing on measurement and control, and finally, at level 5, processes were optimized with continuous improvement as a key emphasis.

The survey's structure, reflecting the Capability Maturity Model (CMM) levels, was carefully crafted to assess the maturity of process architecture within organizations. Each section corresponds to a specific level of the model, as detailed below:

- Foundational Concepts (Levels 1-5): Questions 1-5 cover the foundational concepts of all five levels of the CMM, providing a broad overview of the model's scope (Doss et al., 2021).
- Level 1 - Initial/Ad Hoc: Questions 6-8 delve into the specifics of Level 1, where processes were typically ad hoc, unstructured, and chaotic (Doss et al., 2017).
- Level 2 - Managed: Questions 9-11 explore aspects related to Level 2, focusing on reactive yet managed processes, incorporating elements of planning and control (Doss et al., 2019).
- Level 3 - Defined: Questions 12-14 focus on Level 3, where processes were characterized, understood, and adhere to defined standards and procedures (Doss et al., 2017).

- Level 4 - Quantitatively Managed: Questions 15-17 examine Level 4, highlighting the quantitative management of processes through measurement and control (Doss et al., 2018).
- Level 5 - Optimizing: Questions 18-20 were dedicated to Level 5, emphasizing process optimization and continuous improvement (Doss et al., 2017).
- Work Environment Perceptions: Questions 21-33 aim to elicit perceptions of the respondent's work environment, providing insights into the practical application and impact of the CMM levels within various organizational settings (Doss et al., 2017).

The survey's meticulously designed structure, which evaluated the maturity levels of process architecture, aligns seamlessly with the extensive academic research on the Capability Maturity Model (CMM) and its application in diverse organizational contexts. This alignment was evident in the comprehensive approach of the survey, which assesses the maturity levels and gathers demographic information pertinent to the respondent's work setting in its final section, found on pages 5 and 6. This thorough methodology ensures a deep understanding of process architecture maturity across various organizational settings, as highlighted in the works of Doss et al. (2021), Doss et al. (2019) and Doss et al. (2017).

Operational Definitions of Variables

The independent variable in this study was the job category. The dependent variable was divided into two variable components: management and non-management classifications. The study examined the differences in responses between these groups about telehealth implementation and maturity model effectiveness. To provide a structured overview of the critical variables under investigation in this study, the following list delineates the distinct categories and classifications pertinent to healthcare professionals and administrators in the realm of telehealth services:

- **Job Category (Independent Variable):** This variable refers to the classification of healthcare professionals and administrators involved in telehealth services. It was a nominal variable with two levels: management and non-management (Doss et al., 2017).
- **Management Classification (Dependent Variable):** This component of the job category refers to individuals in managerial roles within healthcare organizations. Responses from this group would be analyzed to gauge their perspectives on telehealth maturity models (Gagnon et al., 2006).
- **Non-Management Classification (Dependent Variable):** This component refers to healthcare professionals, not managerial roles. Their responses would be analyzed separately to understand their views on the effectiveness and applicability of telehealth maturity models (Langbecker et al., 2017).

The level of measurement for these variables was nominal. The data source would be a survey employing a five-point Likert scale designed to measure the attitudes and perceptions of the participants towards telehealth maturity models (Kurihara et al., 2014). Each variable's role was crucial in painting a comprehensive picture of the healthcare sector's attitudes and perceptions regarding telehealth maturity models.

Study Procedures

The details of the data collected for this research study were outlined precisely, including the data to be collected, how it would be obtained when it would be gathered, the sources, and the individuals or entities from which the data originated. The primary objective of this research was to examine the potential of adapting the Capability Maturity Model Integration (CMMI) framework, specifically within the domain of processes for telehealth services. SurveyMonkey would be the data collection platform (Abd Halim et al., 2018).

An initial email would be sent to the VA positions with a hyperlink to the SurveyMonkey questionnaire (Abd Halim et al., 2018) to collect their feedback. Periodic reminders would be sent to increase the response rate. Demographic information would be collected to describe the respondents' work settings (Drake et al., 2022). The survey would last 30 days. In justifying the 30-day duration for the survey in this study, one might reference the Federal Acquisition Regulation (FAR) Subpart 16.5. This document, effective December 4, 2023, outlined policies and procedures for making awards of indefinite-delivery contracts and established a preference for making multiple awards of indefinite-quantity contracts. It incorporated 30 days in contract awards and justifications, suggesting a 30-day timeframe was a recognized standard period for certain assessments and evaluations in a professional context. This reference provided a basis for justifying a 30-day duration in this study or survey context, aligning with contract administration and management practices. Using a 30-day survey period was supported by regulatory practices (Far, 2023).

The data collection process within this study utilized SurveyMonkey as a data collection tool, which offered both cost-efficiency and time-saving advantages, particularly for small-scale assessment projects that capture user opinions and responses. The electronic survey, designed to be user-friendly and easily accessible, would be disseminated via SurveyMonkey (Symonds, 2011). SurveyMonkey, a highly acclaimed web service for creating and administering surveys, would gather demographic data from respondents. The survey would be emailed to target telehealth managers and non-managers through the platform's email distribution capabilities (Cunningham et al., 2015; Toews et al., 2016). Notifications were issued to the selected participants to inform them of the survey's digital whereabouts, and a copy of the notice would be included in Appendix A.

Regarding informed consent, an initial screen would appear to all potential respondents before they could access the survey. A statement within the survey would outline the study's objectives, data usage policies, and confidentiality assurances (Utz et al., 2019). To proceed, respondents were required to provide explicit consent by clicking an "I Agree" button (Utz et al., 2019). All survey responses would be recorded and converted to a spreadsheet for efficient data management. The data would then be analyzed using statistical methods recommended by Whelan (2008) while ensuring its integrity through secure storage of electronic responses and rigorous techniques per the guidelines set forth by Derby-Davis (2014). The notices outlining these procedures would be included in the study's appendix for reference.

SurveyMonkey provided secure data storage by implementing multiple layers of security. This included data encryption both during transmission and at rest, as well as using specific data centers that adhere to industry standards (Abd Halim et al., 2018). To ensure the utmost security, access to the survey data would only be granted to the researcher and the dissertation committee. Multi-factor authentication would also be implemented for added protection. However, the VA also had stringent data security protocols, including encryption and secure servers regularly monitored and audited for compliance (Office of Research and Development, 2018). Data transferred to the VA Network would be accessible only to authorized VA personnel who had undergone security training (Johnson & Benedek, 2023).

SurveyMonkey used Commvault Cloud Services through the VA Network to prevent data loss and enable recovery during system failures (Zhou et al., 2016). Role-based access controls ensured only individuals with clearance can access the data, adding an extra layer of security (Ferraiolo & Barkley, 1997). Comprehensive audit trails provided by both systems would recorded who accessed the data, what changes were made, and when, thereby ensuring high accountability and traceability (Duggineni, 2023). By adhering to these security measures

and federal regulations about data protection and privacy, the study aims to maintain the highest data integrity and protection standards.

The data collection phase of the study was slated to last for 30 days, starting immediately after receiving ethical approval from the National University IRB and culminating on a pre-established deadline. The principal data source would be the responses garnered from a questionnaire administered via Survey Monkey. Participants would receive weekly reminders to enhance the response rate and ensure robust data collection. The researcher sent weekly reminders at strategic intervals throughout the one-month data collection period. Various channels such as email, text messages, or even notifications through the SurveyMonkey platform were utilized.

Incorporating these reminders was a crucial element in the data collection strategy. It encourages timely participation and aims to mitigate the risk of incomplete data sets, which could skew the study's findings. This approach aligns well with the study's methodological rigor, which was informed by previous research on CMMI derivatives by Doss et al. (Doss et al., 2019; Doss et al., 2017). The data collection methodology, fortified using reminders, was designed to provide a comprehensive and reliable data set.

Data security, privacy, and protection, and respondent anonymity were the primary concerns in this study on implementing CMMI into telehealth. A secure email service within the Veterans Administration would be used. As Cunningham et al. (2015) recommended, using web-based tools like SurveyMonkey for data collection necessitates strict security measures. To enhance data integrity, unique survey links would be assigned to each respondent, and a mandatory two-step verification process, such as two-factor authentication, would be required for survey access (Das et al., 2020).

Only the principal investigator and the dissertation committee would be granted access to the collected data. The data management system established role-based permissions, ensuring data analysts could review the data but not alter it (O'Connor et al., 2013). The anonymity of respondents would be rigorously maintained. SurveyMonkey would collect responses without linking them to identifying information, aligning with best practices (O'Connor et al., 2013). Each survey response was assigned a unique tracking number, enabling meticulous data tracking and analysis without jeopardizing respondent anonymity (O'Connor et al., 2013).

Data Analysis

To analyzed data, the process began by examining demographic variables using central tendency metrics such as mean, median, and mode. Then, use variability measures like standard deviation and variance to understand the dispersion of data points around central values. This approach revealed distribution patterns and diversity within the data, enabling a comprehensive understanding of demographic characteristics. Identifying patterns, trends, and anomalies sets the foundation for further analyses and interpretations. Hypothesis testing would adopt the p-value approach, targeting a 0.05 significance level, as suggested by Hung et al. (1997). The ANOVA method, within the realm of statistical analysis, operates as a crucial tool for discerning the contrasts and nuances in the perceptions of CMMI maturity levels between cohorts like managers and non-managers.

The utilization of SPSS (Statistical Package for the Social Sciences) in this study was not merely a procedural choice but a strategic decision that enhances the robustness and depth of the analysis. SPSS was renowned for its user-friendly interface and powerful statistical capabilities, making it an ideal tool for managing and analyzing data collected from surveys or assessments, particularly those measuring perceptions, as highlighted by Mustapha (2019). The initial step of data input was streamlined in SPSS, facilitating a seamless transition from raw data to a

structured format ready for analysis. SPSS provided a user-friendly platform for conducting statistical procedures, including ANOVA, which was crucial for examining differences in perceptions of CMMI maturity levels across groups. The software offered a comprehensive analysis that started with the exploration of demographic variables, assessing variability measures, and conducting hypothesis testing using the p-value approach. SPSS's robustness in handling such statistical tests ensures reliable and valid findings. By leveraging SPSS, this research had a solid methodological foundation, capable of delivering insightful and actionable findings critical for understanding the nuances in perceptions of CMMI maturity levels across different groups.

This research study was based on several critical assumptions essential for its methodology and analysis. The assumption of variance homogeneity was crucial for applying ANOVA, enabling accurate group comparisons (ANOVA, 2009). Additionally, the data were expected to adhere to a normal distribution, aligning with the central limit theorem and the characteristics of the Likert scale, which was important for the validity of statistical tests like ANOVA (Park, 2009). The independence of observations was another key assumption, ensuring a survey response was not influenced by one another, which was vital for the validity of tests such as the Chi-square test (Taherdoost, 2019; Franke et al., 2011). The study also presupposes a linear relationship between the independent variables, such as respondent demographics, and the dependent variable, like a score indicating CMMI maturity perception, essential for analyzing means (Tukey, 1949).

The integrity of the data relies on the assumption respondents understand and respond truthfully to the survey questions (SurveyMonkey, 2023). The straightforward survey design and the anonymity provided to participants were intended to encourage honest responses (SurveyMonkey, 2023). These assumptions play a significant role in ensuring the study's

reliability, validity, and precision (US Census Bureau, 2021). The Statistical Package for the Social Sciences (SPSS) would import the data and be used to calculate the means for each group, assessing the variances within each group and comparing them against the variances between the groups (Thabit & Sharekh, 2023). This statistical computation aims to determine whether significant differences exist in the perceived maturity levels among the studied groups (Mustapha, 2019).

Hartono et al. (2019) had utilized this method to provide a layered understanding of subgroup perceptions, likely by exploring the interaction effects that might arise between different levels of management and the perceived effectiveness or maturity of CMMI implementation. A means analysis was an essential tool for enriching the analytical depth and illuminating the directionality of perceptions (Tukey, 1949). This supplementary analysis contextualizes the ANOVA outcomes and clarifies whether the perceptions were positive, negative, or neutral (Schoefer & Ennew, 2005). The means analysis would examine the directionality of responses with mean ranges. These ranges were set to avoid plurality effects (Tukey, 1949):

- Less than 2.5: overall view of disagreement (Musco et al., 2018).
- Between 2.5 and 3.5: overall view of neutrality (Zhang et al., 2015).
- Over 3.5: overall view of agreement (Doss et al., 2019).

The Omega-squared method (Iacobucci et al., 2022) would determine the effect size. This method measures the proportion of variance in the dependent variable explained by the independent variable (Iacobucci et al., 2022). These guidelines could interpret the Omega-squared values (Olejnik & Algina, 2003):

- A small effect size was represented by a value of 0.01 or above (Olejnik & Algina, 2003).

- A medium effect size was represented by a value of 0.06 or above (Kroes & Finley, 2023).
- A significant effect size was represented by a value of 0.14 or above (Olejnik & Algina, 2003).

The researcher would use the Chi-square test (Franke et al., 2011) to detect potential response bias. The test would examine the distributions of intended manager responses versus actual, received manager responses and the distributions of intended non-manager responses versus actual, received non-manager responses (Harrell, 1996). The test would compare the observed frequencies with the expected frequencies under the null hypothesis, usually implying no connection between the variables or that the observed data follows a specific distribution (Krantz, 1999).

The Cronbach's alpha statistical test would be employed for reliability analysis. Bonett and Wright (2014) state that a Cronbach's alpha score below 0.5 was unacceptable. Tavakol and Dennick (2011) suggest that a score ranging from 0.50 to 0.59 indicated 'moderate' or 'marginal' reliability, while a score of 0.60 to 0.69 was generally regarded as 'acceptable' in terms of reliability, according to Ni Riordain et al. (2016). A Cronbach's alpha score of 0.7 to 0.8 was considered the threshold for 'acceptable' reliability in many research contexts (Al-Osail et al., 2015). This score indicated that the test or questionnaire had a reasonable level of internal consistency, meaning that the items within the test were sufficiently correlated to each other and were likely to measure the same underlying construct (Taber, 2017).

Assessments of Cronbach's alpha scores between 0.8 to 0.9 were considered to have 'good' reliability in psychometric evaluation and research methodology (Al-Osail et al., 2015). This classification indicated that the test items showed a high level of consistency, meaning they effectively measure the same underlying construct (Sharma, 2016). The range of reliability

greater than 0.9 was valuable because it indicated a satisfactory fit of the data to the model (Hua et al., 2013). This range benefits research contexts where precise measurement was critical (Sharma, 2016). The 'good' reliability range suggests that the instrument was strong enough to be confidently utilized in studies where accurate and consistent measurement was a key requirement, thereby enhancing the credibility and reliability of the research findings (Taber, 2017). Given such discussions about reliability values, this study would incorporate the value of 0.7 as the benchmark for acceptability.

Assumptions

When conducting a research study, several fundamental assumptions underlie the methodology and analysis. The assumption of homogeneity of variance suggests that the variance within each group being compared was roughly the same. This assumption plays a crucial role in the ANOVA method utilized in the study. Without this assumption, comparing groups and drawing precise conclusions would be challenging. The study assumes that the data collected would follow a normal distribution, which was crucial for the accuracy of several statistical tests, such as ANOVA (ANOVA, 2009).

This assumption was based on the central limit theorem and the nature of the Likert scale employed in the survey. The Likert scale usually produces data close to a normal distribution in larger samples (Tarka, 2015). It was assumed that the observations made were independent of each other. This means that the answers collected from the survey were not influenced by one another (Taherdoost, 2019). This was an important assumption for statistical tests such as the Chi-square test to be valid (Franke et al., 2011). The survey targets individual respondents without interaction or discussion, reducing the likelihood of correlated responses (Franke et al., 2011).

Another assumption of the study was that there was a linear relationship between the independent and dependent variables. This assumption was important for means analysis. If there was no linear relationship, it would be challenging to determine the impact of the independent variable on the dependent variable (Tukey, 1949). The study assumes that the respondents would comprehend and answer the survey questions truthfully. This assumption was imperative for the authenticity of the data collected (Memon et al., 2020). The survey design was straightforward, and the participants were provided with anonymity to promote truthful responses (Krosnick, 1999). These assumptions ensure that the research study was reliable, valid, and produces precise results (US Census Bureau, 2021).

Limitations

From the study's vantage point, using online survey platforms such as SurveyMonkey (Waclawski, 2012) while implementing the Capability Maturity Model Integration (CMMI) within a telehealth context presented certain constraints (Brooks et al., 2015). The engagement level of participants in online surveys was a critical factor to consider (O'Brien & Toms, 2009). The absence of direct interaction might lead to extended periods for participants to reflect and respond, which could influence the precision of the study's findings (Meade & Craig, 2012). Technical difficulties, including inconsistent internet connectivity or issues with browser compatibility, might obstruct the survey process, affecting the data's comprehensiveness and reliability (Eysenbach & Wyatt, 2002).

Data integrity and confidentiality were paramount, especially in Telehealth (Dunn & Austin, 1998). Although online surveys offer anonymity and encourage honest responses, there was a risk of receiving answers that might need to be more truthful or considerate, which could compromise the accuracy of research outcomes (Hall & McGraw, 2014). Additionally,

protecting data privacy and obtaining informed consent was complex in telehealth due to the sensitive nature of medical information (Joly et al., 2015).

Delimitations

The study would focus on implementing CMMI into the telehealth sector. The survey scope was intentionally confined to telehealth, enabling a more specialized analysis (Langbecker et al., 2017). The study participants were limited to the Veterans Hospital Administration managers and non-managers within the telehealth sector, providing valuable organizational insights (Braithwaite et al., 2011). However, excluding other vital stakeholders, such as patients or external vendors, might limit the comprehensiveness of the findings (Taherdoost, 2022).

A five-point Likert scale was a conscious choice informed by previous research (Doss et al., 2019; Doss et al., 2017). However, this scale might only partially capture the range of attitudes and perceptions toward CMMI maturity levels, serving as a delimitation for more accessible data analysis (Doss et al., 2019). The study was also confined to a one-month data collection period following ethical approval. This limited timeframe might offer a snapshot of current attitudes but was not designed to capture trends or changes over an extended period (Lamoureux et al., 2006).

Additionally, the study relies on self-reported data (Scharkow, 2016). Despite measures to improve data integrity, such as reminders, self-reported data were inherently subject to biases and limitations of self-assessment (Scharkow, 2016). These delimitations were not shortcomings but intentional boundaries to define the study's scope (Coker, 2023). They were crucial for interpreting the findings and understanding the study's applicability and relevance (Scharkow, 2016).

Ethical Assurances

The study would strictly adhere to the ethical principles and guidelines set forth by the Veterans Administration (Makridis et al., 2021) in Cleveland, Ohio, to ensure the integrity and validity of the research. Before implementation, the process would receive explicit approval from the Institutional Review Board (IRB) of the Veteran Hospital Administration to modify and use this survey instrument within the VA organization (Green et al., 2005). Permission would also be obtained from the survey's authors to use the survey in this research (Farkas & Romaniuk, 2020). Pertinent consent documentation would be included in Appendix A.

Participants in the survey would be informed that the research was part of a doctoral study (Swedan et al., 2020). They would be provided with an overview of the research's purpose and potential benefits (Swedan et al., 2020). Participation would be voluntary, and respondents would opt out at any stage. All responses and collected data would be confidential and used solely for the research objectives (Dunn & Austin, 1998). Completing the survey would imply consent, and anonymity would be guaranteed for all participants (Andrews et al., 2003). The results would be publicly available upon the study's completion, and all collected data sets would be responsibly destroyed (Joly et al., 2015). The study would also outline exclusion criteria, offer contact information for the researcher, and ensure that no personal data would be collected. Any limitations regarding personal risk and harm during the research would be addressed.

These ethical assurances would be seamlessly integrated into the study's future data collection procedures, electronically distributing a survey instrument to telehealth professionals (Rosnow et al., 1993). The survey would be administered through an online platform, and organizational records would be accessed to classify organizations by type. Data would be collected over a designated period, with robust measures to secure and protect participant privacy (Farkas & Romaniuk, 2020).

By aligning these ethical guidelines with the study's future data collection and analysis methods, the research aims to uphold the highest standards of academic integrity while contributing valuable insights into the telehealth sector (Rosnow et al., 1993). The researcher would obtain approval from the National University's Institutional Review Board (IRB) before commencing data collection.

Summary

The chapter presented a detailed exploration into the development of a Capability Maturity Model Integration (CMMI) framework specifically tailored for telehealth services. This endeavor, spearheaded by Brooks et al. (2015), aims to bridge a critical gap in the telehealth sector by introducing a customized maturity model, as highlighted by Caralli et al. (2012). The research adopts a quantitative methodology, utilizing a five-point Likert scale survey to gather data from both managerial and non-managerial personnel in healthcare settings, a strategy underscored by Kurihara et al. (2014). This survey helped to examine the facets of the CMMI framework and its relevance to telehealth, a concept further elaborated by David Patón-Romero et al. (2019).

The methodology of the research was meticulously defined, employing statistical analysis to substantiate the findings, a technique emphasized by Lytvyn et al. (2018). The study incorporates Analysis of Variance (ANOVA) to discern significant opinion differences between managerial and non-managerial staff, as suggested by Rossi (1988). Additionally, a means analysis, as proposed by Tarka (2015), delves into the directionality of these opinions. The Omega-squared method, introduced by Iacobucci et al. (2022), was used to evaluate the effect size of significant outcomes, while Chi-square tests, as per McHugh (2013), were employed to detect any response bias.

The study places a strong emphasis on data security and privacy, outlining measures to safeguard the integrity and confidentiality of the collected data, a concern echoed by Hall & McGraw (2014). The data collection process, expected to last 30 days and would culminate in a comprehensive report, as per Joly et al. (2015). The research specifically targets the telehealth sector, focusing on employees in the VA environment, to gain a nuanced understanding of how different healthcare personnel perceive the maturity levels of telehealth processes, a crucial aspect for customizing the CMMI model for telehealth services, as noted by Junaid et al. (2022). This comprehensive research effort, as Miranda et al. (2023) point out, seeks to provide valuable insights into the diverse opinions of managerial and non-managerial staff in healthcare settings, utilizing rigorous statistical methods for data analysis, a methodology supported by Kolker (2012). The study's commitment to ethical considerations, particularly in terms of data security and privacy, was a testament to its thoroughness.

The chapter's primary focus was on examining a maturity model for telehealth, addressing the challenges of integrating technology sustainably in healthcare. The research methodology, grounded in the Capability Maturity Model Integration (CMMI) framework, evaluated telehealth technology implementation. The objective was to create a customized maturity model for telehealth, gathering data from healthcare professionals and administrators through a five-point Likert scale survey.

The chapter underscores the significance of telehealth in enhancing healthcare accessibility and efficiency. It identifies a gap in the availability of a dedicated maturity model for guiding process improvement and technology implementation in telehealth. The study's population includes healthcare professionals and administrators involved in telehealth services, targeting both managerial and non-managerial staff to understand their perceptions of telehealth

maturity models. The statistical analysis of survey data aims to uncover patterns and correlations that inform the development of a telehealth-specific maturity model.

Key variables such as "process maturity," "service risks," and "technology implementation" were defined within the telehealth context. Ethical considerations, including informed consent, anonymity, and data confidentiality, were rigorously upheld. The chapter acknowledges potential biases and generalizability issues due to the sample size and demographic, focusing primarily on healthcare professionals and administrators, and excluding other stakeholders like patients and policymakers. This comprehensive approach aims to fill a significant gap in the telehealth sector by developing a bespoke maturity model, contributing substantially to both academic literature and practical applications in healthcare technology.

Chapter 4: Findings

This chapter presented the findings from the quantitative analysis of data collected through the survey questionnaire. The study showed the perceptions of managerial versus non-managerial personnel regarding telehealth process maturity to address the problem. The problem to be addressed in this study was to develop an effective and customizable maturity model to be used in healthcare for telehealth for implementing innovative technology sustainably (Gomes & Romão, 2018). The purpose of this study was to examine a derivation of the CMM as a telehealth maturity model. It aims to address shortcomings in the functional processes needed to integrate new technology successfully and minimize potential service risks (Watts, 2020).

The chapter was organized by research questions and hypotheses to guide the presentation of findings. Each research question and hypothesis was addressed in sequence, exploring the various facets of telehealth process maturity, including perceptions of organizational process improvements and personnel roles. The data analysis includes demographic information and results of hypothesis testing, supported by statistical tests such as ANOVA to identify significant differences between groups (Park, 2009). Effect sizes were reported for statistically significant findings to provide insight into the magnitude of these differences. This approach ensures each research question was systematically examined, enabling a comprehensive understanding of telehealth maturity processes in the context of differing personnel roles.

Representativeness, Reliability, and Bias Assessment of the Data

The study achieved 285 usable responses, providing a robust analysis dataset and ensuring the statistical findings' reliability. As established by the Central Limit Theorem (CLT), a sample size of 30 was often sufficient to approximate normality in the sampling distribution of the mean, with 50 providing greater reliability for inferential statistical techniques (Islam, 2018; Andrade, 2020). By exceeding these benchmarks, the sample size in this study allows for

dependable estimations of population parameters. A G*Power analysis conducted prior to data collection determined a minimum of 218 responses was required to achieve a 95% confidence level with a 5% margin of error (Moran, 2021). The obtained sample size of 285 not only meets but significantly exceeds this requirement, enhancing the statistical power of the study (Moran, 2021; Bujang et al., 2018).

The survey targeted a sample size of 500 participants, with 285 individuals returning responses, resulting in a response rate of 57%. Based on the statistical threshold outlined in Chapter 3, G*Power analysis determined a minimum of 218 responses was necessary to achieve a 95% confidence level with a 5% margin of error (Moran, 2021). The survey achieved this requirement with 285 usable responses. To enhance the external validity of the findings, it was critical to evaluate whether the sample was representative of the broader population.

Representativeness refers to the degree to which the sample's demographic, professional, and organizational characteristics align with those of the population from which it was drawn (Moates et al., 2005). The sample represented the broader population, as the achieved response quantity exceeded the minimum required threshold. The generalizability of the data was supported by the sample size of 285 usable responses, which exceeds the minimum threshold of 218 needed for a 95% confidence level with a 5% margin of error. Given such outcomes, it appeared that representativeness was achieved between the sample and the population.

The survey instrument's reliability was evaluated using Cronbach's Alpha, which yielded a 0.76 value, indicating good internal consistency. This ensures the data collected was reliable and suitable for inferential analysis (Bonett & Wright, 2014). Table 4.1 presented Cronbach's Alpha values for each question scale, highlighting the internal consistency and reliability of the survey instrument across various maturity levels and process groups.

Table 4. 1 *Cronbach Alpha values for each of the survey questions, scales*

Scale	Cronbach's Alpha
Basic Framework (Question 1-5)	0.86
First Maturity Level (Question 6-8)	0.77
Second Maturity Level (Question 9-11)	0.85
Third Maturity Level (Question 12-14)	0.85
Fourth Maturity Level (Question 15-17)	0.74
Fifth Maturity Level (Question 18-20)	0.6

Cronbach's Alpha Assessment

The results of Cronbach's Alpha analysis indicated strong internal consistency across the scales that measure telehealth process maturity. The Basic Framework (0.86), First Maturity Level (0.77), Second Maturity Level (0.85), and Third Maturity Level (0.85) demonstrated high reliability, indicating that survey items within these scales align well in measuring their intended constructs. The Fourth Maturity Level (0.74) meets the acceptable threshold for reliability, while the Fifth Maturity Level (0.60) remains moderate.

Chi-Square Assessment and Potential Bias

The Chi-Square test analyzed the returned responses versus the original distribution between managers and non-managers. The results ($\chi^2 = 11.78$, $p < 0.001$) indicated significant differences in response rates, with managers showing lower engagement than non-managers. The Chi-Square statistic, $X^2 ((1, N = 218) = 11.78)$, with a p-value of 0.0006, indicated a statistically significant difference between the variables (Hung, O'Neill, Bauer, & Kohne, 1997) at the alpha = 0.05 significance level (Moser, 2007).

The significant p-value indicated the response rates between managers and non-managers differed significantly from the expected distribution. The significant p-value indicated the response rates between managers and non-managers differed substantially from the predicted distribution, suggesting potential response bias in the study. Managers exhibited fewer responses (123 vs. 250 expected), while non-managers had more responses (162 vs. 250 expected). This suggests a disparity in engagement between the two groups.

Characteristics of the Host Environment

The survey results provide a detailed overview of the characteristics of the host environment, segmented by job classification into managerial and non-managerial categories. It includes statistical insights into organizational practices, workforce distribution, and roles within telehealth services. The classification breakdown revealed survey respondents were predominantly non-managerial, constituting 162 out of 285 responses (56.84%), while managerial roles account for 123 responses (43.16%). This distribution highlights the operational focus on non-managerial positions, primarily inpatient data collection and support activities. Conversely, managerial roles, which concentrate on data analysis and decision-making, represent a smaller yet significant segment of the workforce.

Regarding improvement initiatives, apparent differences exist between managerial and non-managerial roles. Among managerial respondents, Benchmarking was the most frequently reported previous initiative, accounting for six responses (4.88%). Other notable initiatives include Business Process Improvement (BPI), with three responses (2.44%), and combinations such as Benchmarking and ISO Standards, also with three responses (2.44%). These figures reflect a strategic focus on standardizing processes and enhancing efficiency. Non-managerial roles reported fewer initiatives, with regulation (government or corporate) being the most prominent, noted in three responses (1.85%). This limited engagement with improvement

processes suggests such initiatives were mainly managed by leadership. Currently, managerial respondents continue to emphasize benchmarking (six responses, 4.88%) and related strategic practices, while non-managerial roles display minimal involvement, following similar patterns of regulatory initiatives. The analysis of employee counts revealed managerial and non-managerial respondents reported working in organizations with over 1,000 employees. This indicated a substantial scale of operations within the surveyed telehealth services, consistent with the increasing demand for virtual healthcare solutions. Such large-scale environments require structured roles and effective coordination to meet operational needs.

An analysis of job descriptions indicated distinctive distributions between the two classifications. Administration was the most common role among managerial respondents, accounting for eight responses (6.50%), followed by combined roles such as Administration and Telehealth, which received ten responses (8.13%). These findings underscore the multifaceted responsibilities of managerial roles, often combining administrative oversight with strategic decision-making. For non-managerial respondents, anesthesiology appeared to be the leading job description, with six responses (3.70%), alongside chronic disease management and combinations like administration and chronic disease management, each with three responses (1.85%). These roles emphasize direct patient care and monitoring, showcasing the operational focus of non-managerial staff.

The insights and implications of the survey illuminate critical dynamics within telehealth environments. Managerial roles, representing 123 responses (43.16%), strongly emphasize strategic initiatives such as benchmarking and process improvements. Meanwhile, non-managerial roles constitute the majority, with 162 responses (56.84%), primarily focusing on operational and clinical functions. This division of labor enables organizations to meet their strategic and operational goals. The predominance of large employees, which was over 1,000

employees, reported by all respondents highlights the expansive scale of telehealth organizations. These findings suggest such services were well-equipped to manage extensive operational demands. However, there was an opportunity to encourage greater involvement of non-managerial staff in improvement initiatives, potentially broadening engagement and fostering more inclusive organizational practices.

The survey offered a statistical and contextual understanding of the host environment within telehealth services. It revealed a well-structured workforce where managerial roles focus on leadership and strategic initiatives, while non-managerial roles address operational and clinical needs. With 285 responses, the findings provide a comprehensive snapshot of telehealth dynamics, offering valuable insights for enhancing workforce collaboration and organizational effectiveness.

Results

Individual Item Response Distribution: Managerial, Non-Managerial, and Cumulative

Findings of Survey Question 1

Question 1 of the survey addressed the characterization of telehealth processes as ad hoc, chaotic, or random. It sought to capture the opinions of managers and non-managers and a cumulative perspective on this notion. The responses reveal notable differences between managerial and non-managerial perspectives. Managers predominantly “Strongly Disagree” (39.0%) or “Disagree” (41.5%) with the statement, indicating skepticism toward the notion of chaos in telehealth processes. Non-managers display a wider spread of opinions, with 35.2% responding “Neutral,” 44.4% selecting “Disagree,” and 9.3% indicating “Agree.” Cumulatively, the responses reflect a general tendency toward disagreement, with 43.0% selecting “Disagree” and 22.5% expressing neutrality, suggesting varied levels of confidence across roles regarding

process organization. Table 4.2 outlines perceptions of whether telehealth processes were ad hoc, chaotic, or random, comparing managerial and non-managerial personnel responses.

Table 4. 2 *Question 1. Telehealth processes might be defined as ad-hoc, chaotic, or random.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	39.0	5.6	22.3
2 Disagree	41.5	44.4	43.0
3 Neutral	9.8	35.2	22.5
4 Agree	9.7	9.3	9.6
5 Strongly Agree	0.0	5.5	2.6

Findings of Survey Question 2

Question 2 of the survey examined whether telehealth processes were managed. The data highlights an apparent confidence among managers regarding this aspect, with 68.3% agreeing and 21.9% strongly agreeing. Non-managers also largely agreed (63.0%) but demonstrated less strong agreement (3.6%). Neutral responses were more common among non-managers (25.9%) than managers (9.8%). The cumulative data reinforces a broad consensus on the management of telehealth processes, with 65.6% agreement overall and 12.8% expressing strong agreement. Table 4.3 summarizes responses regarding whether telehealth processes were managed, highlighting differences in confidence levels between managerial and non-managerial roles.

Table 4. 3 *Question 2. Telehealth processes were managed.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.0	1.9	1.0
2 Disagree	0.0	5.6	2.8
3 Neutral	9.8	25.9	17.8
4 Agree	68.3	63.0	65.6
5 Strongly Agree	21.9	3.6	12.8

Findings of Survey Question 3

Question 3 of the survey explored whether telehealth processes were defined and specific. The data reflects a strong consensus among managers, with 58.5% agreeing and 34.1% strongly agreeing, indicating a high confidence level in the specificity of telehealth processes. Non-managers also predominantly agreed (61.1%), although strong agreement was much lower (1.9%). Neutral responses were more frequent among non-managers (25.9%) than managers (4.9%). Cumulatively, 59.8% of respondents agreed, while 18% strongly agreed, reinforcing a general belief in telehealth processes defined and specific nature. Table 4.4 presented data on whether telehealth processes were defined and specific, illustrating varied levels of agreement across roles.

Table 4. 4 *Question 3. Telehealth processes were defined/specific.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.0	0.0	0.0
2 Disagree	2.4	11.1	6.8
3 Neutral	4.9	25.9	15.4
4 Agree	58.5	61.1	59.8
5 Strongly Agree	34.1	1.9	18

Findings of Survey Question 4

Question 4 of the survey examined whether telehealth processes were quantitatively managed. The data showed a divided perspective among managers, with 36.6% agreeing and an equal 36.6% expressing neutrality. A smaller group (12.2%) strongly agreed. Non-managers leaned heavily towards neutrality (44.4%) and disagreement (38.9%), with no responses indicating strong agreement. Cumulatively, neutrality (40%) and disagreement (26.8%) were the most prominent responses, reflecting a varied perception of the quantitative management of telehealth processes. Table 4.5 illustrates perceptions of whether telehealth processes were quantitatively managed, showcasing differing views among managerial and non-managerial groups.

Table 4. 5 *Question 4. Telehealth processes were quantitatively managed.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.0	1.9	0.0
2 Disagree	14.6	38.9	26.8
3 Neutral	36.6	44.4	40.0
4 Agree	36.6	14.8	33.2
5 Strongly Agree	12.2	0.0	0.0

Findings of Survey Question 5

Question 5 of the survey explored whether telehealth processes were optimized. Among managers, neutrality (36.6%) and agreement (36.6%) were equally prevalent, with a smaller portion (12.2%) strongly agreeing. Non-managers leaned towards disagreement (38.9%) but also showed substantial neutrality (29.6%) and agreement (29.6%). The cumulative data reflects a balance between neutrality and agreement at 33.1%, with disagreement accounting for 27.8%. This indicated mixed perceptions of process optimization across the groups. Table 4.6 reflects opinions on whether telehealth processes were optimized, revealing the extent to which participants agree or remain neutral.

Table 4. 6 *Question 5. Telehealth processes were optimized.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.0	1.9	1.0
2 Disagree	14.6	38.9	27.8
3 Neutral	36.6	29.6	33.1
4 Agree	36.6	29.6	33.1
5 Strongly Agree	12.2	0.0	5.0

Findings of Survey Question 6

Question 6 of the survey illustrates whether telehealth processes were unpredictable. Managers predominantly agreed (53.7%) or strongly agreed (12.1%), indicating a perception of unpredictability. Non-managers showed more significant disagreement (38.9%) and neutrality (29.6%), with no strong agreement responses. Cumulatively, agreement (41.6%) was the most common response, followed by neutrality (27%) and disagreement (24.3%), reflecting mixed perceptions but a tendency toward recognizing some unpredictability in telehealth processes. Table 4.7 examined views on the unpredictability of telehealth processes, highlighting contrasting responses between managerial and non-managerial staff.

Table 4. 7 *Question 6. Telehealth processes were unpredictable.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.0	1.9	1.0
2 Disagree	9.8	38.9	24.3
3 Neutral	24.4	29.6	27
4 Agree	53.7	29.6	41.6
5 Strongly Agree	12.1	0.0	6.1

Findings of Survey Question 7

Question 7 of the survey addressed whether telehealth processes were reactive. Managers predominantly disagreed (70.7%) or strongly disagreed (9.8%) with this characterization, indicating confidence in proactive processes. Non-managers also disagreed (50%) but to a lesser extent, with some neutrality (18.5%) and agreement (25.9%). Cumulatively, disagreement accounted for 60.4%, and agreement followed at 19%, reflecting an overall tendency to reject the notion of telehealth processes being reactive. Table 4.8 illustrates whether telehealth processes were reactive, presenting levels of disagreement and neutrality within the two respondent groups.

Table 4. 8 *Question 7. Telehealth processes were reactive.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	9.8	3.7	6.8
2 Disagree	70.7	50	60.4
3 Neutral	7.3	18.5	12.9
4 Agree	12.2	25.9	19.0
5 Strongly Agree	0.0	1.9	0.9

Findings of Survey Question 8

Question 8 of the survey examined whether telehealth processes need to be coordinated. Managers largely disagreed (36.6%) or agreed (34.1%), showcasing mixed perceptions of coordination. Non-managers demonstrated a similar trend, with 33.3% disagreeing and 29.6% agreeing, along with higher neutrality (33.3%) compared to managers (26.8%). Cumulatively, disagreement (35%) was the most common response, followed by agreement (31.8%) and neutrality (30%), reflecting diverse views on the coordination of telehealth processes. Table 4.9 provided insights into whether telehealth processes lack coordination, comparing agreement levels among managerial and non-managerial participants.

Table 4. 9 *Question 8. Telehealth processes were uncoordinated.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	2.4	1.9	2.2
2 Disagree	36.6	33.3	35
3 Neutral	26.8	33.3	30
4 Agree	34.1	29.6	31.8
5 Strongly Agree	0.0	1.9	1.0

Findings of Survey Question 9

Question 9 of the survey addressed whether telehealth processes were planned. Managers predominantly disagreed (63.4%) or strongly disagreed (24.4%), indicating skepticism about planning in telehealth processes. Non-managers also showed significant disagreement (61.1%) but no responses of strong disagreement. Neutrality was more prevalent among non-managers (14.8%) than managers (4.9%), and agreement levels were modest across both groups. Cumulatively, disagreement accounted for the majority at 62.3%, followed by neutrality (9.8%) and agreement (14.8%), highlighting concerns about planning telehealth processes. Table 4.10 illustrates whether telehealth processes were planned, illustrating significant role-based variations in perceptions.

Table 4. 10 *Question 9. Telehealth processes were planned.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	24.4	0.0	12.2
2 Disagree	63.4	61.1	62.3
3 Neutral	4.9	14.8	9.8
4 Agree	7.3	22.2	14.8
5 Strongly Agree	0.0	1.9	0.9

Findings of Survey Question 10

Question 10 of the survey showed whether telehealth processes were controlled. Managers overwhelmingly agreed (75.6%) or strongly agreed (19.5%), demonstrating confidence in the controlled nature of these processes. Non-managers also largely agreed (72.2%) but showed no strong agreement responses. Neutrality (18.5%) and disagreement (7.4%) were more common among non-managers than managers. Cumulatively, agreement dominated at 73.9%, with strong agreement accounting for 9.7%, reflecting a consensus on the controlled status of telehealth processes. Table 4.11 highlights perceptions of whether telehealth processes were controlled.

Table 4. 11 *Question 10. Telehealth processes were controlled.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.0	1.9	1.0
2 Disagree	0.0	7.4	3.7
3 Neutral	4.9	18.5	11.7
4 Agree	75.6	72.2	73.9
5 Strongly Agree	19.5	0.0	9.7

Findings of Survey Question 11

Question 11 of the survey examined whether telehealth processes were well-defined. Managers displayed high confidence, with 70.7% agreeing and 22% strongly agreeing telehealth processes were well-defined. Non-managers also showed substantial agreement (63.0%) but less strong agreement (7.3%), with more neutrality (22.2%) compared to managers (7.3%). Cumulatively, agreement dominated at 66.8%, and strong agreement reached 14.6%, indicating broad recognition of the defined nature of telehealth processes across both groups. Table 4.12 illustrates whether telehealth processes were well-defined, with managerial responses showing greater confidence than non-managerial counterparts.

Table 4. 12 *Question 11. Telehealth processes were well-defined.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0	1.9	1.0
2 Disagree	0	5.6	2.8
3 Neutral	7.3	22.2	14.8
4 Agree	70.7	63.0	66.8
5 Strongly Agree	22	7.3	14.6

Findings of Survey Question 12

Question 12 of the survey illustrates whether telehealth processes were consistent. Managers expressed strong agreement, 68.3% agreeing and 19.5% strongly agreeing telehealth processes were consistent. Non-managers also largely agreed (50%), but no strong agreement responses were recorded, and there was higher neutrality (33.3%) and disagreement (14.8%) compared to managers. Cumulatively, agreement accounted for 59.2% of responses, while neutrality reached 22.8%, indicating overall confidence in process consistency but with some reservations, particularly from non-managers. Table 4.13 presented views on the consistency of telehealth processes, focusing on the divergence in agreement levels between groups.

Table 4. 13 *Question 12. Telehealth processes were consistent.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.0	1.9	1.0
2 Disagree	0.0	14.8	7.4
3 Neutral	12.2	33.3	22.8
4 Agree	68.3	50	59.2
5 Strongly Agree	19.5	0.0	9.6

Findings of Survey Question 13

Question 13 of the survey examined whether telehealth processes were followed. Managers largely agreed (68.3%) or strongly agreed (17.0%) telehealth processes were followed, indicating confidence in adherence. Non-managers also displayed agreement (50%) but had no responses of strong agreement, with higher neutrality (29.6%) and disagreement (18.5%) compared to managers. Cumulatively, agreement was dominant at 59.2%, followed by neutrality at 17.2%, reflecting a general belief in adherence to telehealth processes but with some hesitations from non-managers. Table 4.14 summarizes whether telehealth processes were followed across roles.

Table 4. 14 *Question 13. Telehealth processes were followed.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.0	1.9	1.0
2 Disagree	9.8	18.5	14.2
3 Neutral	4.9	29.6	17.2
4 Agree	68.3	50	59.2
5 Strongly Agree	17.0	0.0	8.4

Findings of Survey Question 14

Question 14 of the survey examined whether telehealth processes involve quantitative or measurable objectives. Managers predominantly agreed (65.9%) or strongly agreed (21.9%) with this assertion, reflecting strong confidence in including measurable objectives. Non-managers also mostly agreed (57.4%) but had no responses of strong agreement and demonstrated higher neutrality (29.6%). Cumulatively, agreement was the most common response at 61.6%, with neutrality (20.9%) and strong agreement (10.9%) also represented, indicating a broad acknowledgment of quantitative objectives in telehealth processes. Table 4.15 showed perceptions of whether telehealth processes involve measurable objectives, emphasizing agreement across respondents.

Table 4. 15 *Question 14. Telehealth processes involve quantitative or measurable objectives.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.0	1.9	1.0
2 Disagree	0.0	11.1	5.6
3 Neutral	12.2	29.6	20.9
4 Agree	65.9	57.4	61.6
5 Strongly Agree	21.9	0.0	10.9

Findings of Survey Question 15

Question 15 of the survey explored whether telehealth processes involve metrics analysis. Managers predominantly agreed (53.7%) or were neutral (31.7%) about the inclusion of metrics analysis, with only a small percentage (2.4%) strongly agreeing. Non-managers displayed greater neutrality (55.6%) and disagreement (29.6%), with no responses indicating strong agreement. Cumulatively, neutrality was the most common response at 43.6%, followed by agreement at 31.5%, suggesting mixed perceptions about the role of metrics analysis in telehealth processes. Table 4.16 illustrates the role of metrics analysis in telehealth processes.

Table 4. 16 *Question 15. Telehealth processes involve metrics analysis.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.0	5.6	2.8
2 Disagree	12.2	29.6	20.9
3 Neutral	31.7	55.6	43.6
4 Agree	53.7	9.2	31.5
5 Strongly Agree	2.4	0.0	1.2

Findings of Survey Question 16

Question 16 of the survey examined whether telehealth processes involve statistical analysis. Managers primarily agreed (46.3%) or were neutral (36.6%) about the inclusion of statistical analysis, with a small percentage strongly agreeing (4.9%). Non-managers leaned heavily toward neutrality (53.7%) and disagreement (29.6%), with no strong agreement responses. Cumulatively, neutrality was the most common response at 45.2%, followed by agreement at 27.8%, reflecting varied perceptions of the role of statistical analysis in telehealth processes. Table 4.17 presented insights into whether telehealth processes involve statistical analysis, showcasing varied levels of neutrality and agreement.

Table 4. 17 *Question 16. Telehealth processes involve statistical analysis.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	2.4	7.4	4.9
2 Disagree	9.8	29.6	19.7
3 Neutral	36.6	53.7	45.2
4 Agree	46.3	9.3	27.8
5 Strongly Agree	4.9	0.0	2.4

Findings of Survey Question 17

Question 17 of the survey explored whether telehealth processes were improved incrementally. Managers predominantly agreed (43.9%) or were neutral (31.7%) about the incremental improvement of telehealth processes, with 12.2% strongly agreeing. Non-managers showed greater neutrality (50.0%) and disagreement (35.2%), with minimal strong agreement (1.8%). Cumulatively, neutrality was the most common response at 40.8%, followed by agreement at 26.6%, suggesting mixed perceptions with a slight tendency toward recognizing incremental improvements. Table 4.18 illustrates whether telehealth processes were incrementally improved.

Table 4. 18 *Question 17. Telehealth processes were improved incrementally.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	2.4	3.7	3.0
2 Disagree	9.8	35.2	22.5
3 Neutral	31.7	50.0	40.8
4 Agree	43.9	9.3	26.6
5 Strongly Agree	12.2	1.8	7.0

Findings of Survey Question 18

Question 18 of the survey examined whether telehealth processes were efficient. Managers overwhelmingly agreed (73.2%) or strongly agreed (14.6%) telehealth processes were efficient, indicating high confidence in process efficiency. Non-managers also showed agreement (55.5%) but recorded no strong agreement responses and demonstrated higher neutrality (31.5%) and disagreement (13.0%) than managers. Cumulatively, agreement was dominant at 64.4%, with neutrality accounting for 21.8%, reflecting general positivity toward process efficiency but with reservations from non-managers. Table 4.19 showed perceptions of telehealth process efficiency.

Table 4. 19 *Question 18. Telehealth processes were efficient.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagreed	0.0	0.0	0.0
2 Disagree	0.0	13.0	6.5
3 Neutral	12.2	31.5	21.8
4 Agree	73.2	55.5	64.4
5 Strongly Agree	14.6	0.0	7.3

Findings of Survey Question 19

Question 19 of the survey explored whether telehealth processes were effective. Managers predominantly agreed (61.0%) or strongly agreed (26.8%) telehealth processes were effective, reflecting high confidence in their efficacy. Non-managers also mostly agreed (62.9%) but did not record any strong agreement responses, instead showing higher neutrality (20.4%) and disagreement (14.8%) than managers. Cumulatively, agreement was the dominant response at 62.0%, with strong agreement at 13.3%, indicating a general acknowledgment of the effectiveness of telehealth processes across groups, albeit with some variance in enthusiasm. Table 4.20 reflects views on the effectiveness of telehealth processes, presenting role-specific confidence levels.

Table 4. 20 *Question 19. Telehealth processes were effective.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.0	1.9	1.0
2 Disagree	2.4	14.8	8.6
3 Neutral	9.8	20.4	15.1
4 Agree	61	62.9	62.0
5 Strongly Agree	26.8	0.0	13.3

Findings of Survey Question 20

Question 20 of the survey explored whether current process improvement initiatives did not address process maturity. Managers largely agreed (63.4%) or strongly agreed (26.8%) with this statement, suggesting a belief process maturity was overlooked in improvement efforts. Non-managers also mostly agreed (68.5%), but strong agreement was minimal (1.8%), and neutrality (22.2%) was more prevalent compared to managers (7.3%). Cumulatively, agreement dominated at 66.0%, with strong agreement at 14.2%, indicating a general perception across groups of process maturity was inadequately addressed. Table 4.21 illustrates whether process maturity was overlooked in current improvement initiatives.

Table 4. 21 *Question 20. Process maturity was not addressed by our current process improvement initiatives.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.0	1.9	1.0
2 Disagree	2.4	5.6	4.0
3 Neutral	7.3	22.2	14.8
4 Agree	63.4	68.5	66.0
5 Strongly Agree	26.8	1.8	14.2

Findings of Survey Question 21

Question 21 of the survey examined whether process improvement was advocated within telehealth settings. Managers showed mixed responses, with 39% agreeing and 31.7% remaining neutral, while 22% disagreed and a small percentage (4.9%) strongly agreed. Non-managers displayed higher neutrality (50%) and lower disagreement (9.3%), with similar agreement levels (37%) but fewer responses for strong agreement (3.7%). Cumulatively, neutrality was the most frequent response at 40.8%, followed by agreement at 38%, reflecting varied perceptions about the advocacy of process improvement in telehealth. Table 4.22 illustrates perceptions of advocacy for process improvement within telehealth settings.

Table 4. 22 *Question 21. Process improvement was advocated within my Telehealth.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	2.4	0.0	1.2
2 Disagree	22	9.3	15.6
3 Neutral	31.7	50	40.8
4 Agree	39	37	38
5 Strongly Agree	4.9	3.7	4.3

Findings of Survey Question 22

Question 22 of the survey examined whether process initiatives were tracked to evaluate process performance. Managers exhibited mixed views, with 39% agreeing and 31.7% remaining neutral. A significant proportion (22%) disagreed, and only 4.9% strongly agreed. Non-managers displayed higher neutrality (50%) and lower disagreement (9.3%), with similar agreement levels (37%) but fewer responses for strong agreement (3.7%). Cumulatively, neutrality was the most frequent response at 40.8%, followed by agreement at 38%, indicating varied perceptions about the tracking of process initiatives to assess performance. Table 4.23 illustrates whether process initiatives were tracked to evaluate performance.

Table 4. 23 *Question 22. Process initiatives were tracked to examine process performance.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	2.4	0.0	1.2
2 Disagree	22	9.3	15.6
3 Neutral	31.7	50	40.8
4 Agree	39	37	38
5 Strongly Agree	4.9	3.7	4.3

Findings of Survey Question 23

Question 23 of the survey examined whether grouping processes by maturity level would improve process outcomes. Managers predominantly agreed (51.2%) or remained neutral (36.6%), indicating cautious optimism about the benefits of maturity-based grouping. Non-managers leaned heavily toward neutrality (50%) and disagreement (35.2%), with no strong agreement responses. Cumulatively, neutrality was the most frequent response at 43.3%, followed by agreement at 31.2%, suggesting mixed but slightly favorable views toward maturity-level process grouping. Table 4.24 presented views on whether grouping processes by maturity level could improve outcomes.

Table 4. 24 *Question 23. Grouping of processes, according to maturity level, would improve the outcomes of our processes.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.0	3.7	1.8
2 Disagree	9.8	35.2	22.5
3 Neutral	36.6	50	43.3
4 Agree	51.2	11.1	31.2
5 Strongly Agree	2.4	0.0	1.2

Findings of Survey Question 24

Question 24 of the survey explored whether grouping processes by maturity level would improve process outcomes. Managers predominantly agreed (48.8%) or strongly agreed (24.3%) with this approach, indicating significant support. Non-managers also largely agreed (33.3%) but showed higher neutrality (44.4%) and less strong agreement (9.3%) compared to managers. Cumulatively, agreement was the most frequent response at 41.0%, with neutrality at 33.2% and strong agreement at 16.8%, reflecting overall positive perceptions of maturity-based process grouping with some reservations. Table 4.25 illustrates support for grouping processes by maturity level.

Table 4. 25 *Question 24. Grouping of processes, according to maturity level would improve the outcomes of our processes.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.0	1.9	1.0
2 Disagree	4.9	11.1	8.0
3 Neutral	22.0	44.4	33.2
4 Agree	48.8	33.3	41.0
5 Strongly Agree	24.3	9.3	16.8

Findings of Survey Question 25

Question 25 of the survey illustrates whether categorical process grouping was advocated within telehealth settings. Managers largely remained neutral (48.8%) or agreed (31.7%) with this concept, with a smaller percentage strongly agreeing (4.9%). Non-managers showed even higher neutrality (63.0%) and disagreement (24.1%), with no responses indicating strong agreement. Cumulatively, neutrality was the most frequent response at 55.9%, followed by disagreement at 19.4% and agreement at 20.5%, reflecting mixed views on the advocacy for categorical process grouping in telehealth. Table 4.26 illustrates advocacy for categorical process grouping.

Table 4. 26 *Question 25. Categorical process grouping was advocated within my telehealth setting.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.0	3.7	1.8
2 Disagree	14.6	24.1	19.4
3 Neutral	48.8	63.0	55.9
4 Agree	31.7	9.2	20.5
5 Strongly Agree	4.9	0.0	2.4

Findings of Survey Question 26

Question 26 of the survey examined whether process maturity contributes to successful process outputs within telehealth settings. Managers primarily expressed neutrality (48.8%) or agreement (31.7%), with a smaller proportion disagreeing (17.1%). Non-managers also leaned heavily toward neutrality (55.6%) but showed higher disagreement (33.3%) and no strong agreement responses. Cumulatively, neutrality was the most common response at 52.2%, followed by disagreement at 25.2% and agreement at 20.5%, indicating mixed perceptions about the role of process maturity in achieving successful outputs. Table 4.27 highlights perceptions of process maturity as a contributor to successful outputs.

Table 4. 27 *Question 26. Process maturity was a contributor to successful process output within my telehealth setting.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.0	1.9	1.0
2 Disagree	17.1	33.3	25.2
3 Neutral	48.8	55.6	52.2
4 Agree	31.7	9.2	20.5
5 Strongly Agree	2.4	0.0	1.1

Findings of Survey Question 27

Question 27 of the survey examined whether telehealth policies influence processes. Managers predominantly agreed (63.4%) or strongly agreed (17.1%) with this statement, indicating strong recognition of policy influence. Non-managers also largely agreed (61.1%) but recorded no strong agreement responses and exhibited higher neutrality (29.6%) compared to managers (19.5%). Cumulatively, agreement was the most frequent response at 63.2%, with neutrality at 24.6% and strong agreement at 7.5%, highlighting broad acknowledgment of telehealth policies' role in shaping processes. Table 4.28 examined the influence of telehealth policies on processes.

Table 4. 28 *Question 27. Telehealth policies influence processes.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.0	1.9	1.0
2 Disagree	0.0	7.4	3.7
3 Neutral	19.5	29.6	24.6
4 Agree	63.4	61.1	63.2
5 Strongly Agree	17.1	0.0	7.5

Findings of Survey Question 28

Question 28 of the survey also examined whether telehealth policies influence processes, focusing on differences in perceptions. Managers expressed various views, with 36.6% disagreeing and 31.7% remaining neutral, while 17.1% agreed and 9.7% strongly agreed. Non-managers primarily displayed neutrality (46.3%) and agreement (31.5%), with no strong agreement responses and lower levels of disagreement (22.2%) compared to managers. Cumulatively, neutrality was the most frequent response at 39%, followed by disagreement at 29.4% and agreement at 24.3%, indicating varied perceptions of policy influence. Table 4.29 provided additional insights into telehealth policy influence.

Table 4. 29 *Question 28. Telehealth policies influence processes.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	4.9	0.0	2.4
2 Disagree	36.6	22.2	29.4
3 Neutral	31.7	46.3	39
4 Agree	17.1	31.5	24.3
5 Strongly Agree	9.7	0.0	4.9

Findings of Survey Question 29

Question 29 of the survey explored whether telehealth processes were inefficient. Managers overwhelmingly disagreed (68.3%) or strongly disagreed (17.1%) with this statement, reflecting confidence in process efficiency. Non-managers also primarily disagreed (63.0%) but displayed more neutrality (16.7%) and agreement (18.4%) compared to managers. Cumulatively, disagreement was the dominant response at 65.6%, with strong disagreement at 9.5%, indicating broad rejection of inefficiency in telehealth processes. Table 4.30 showed whether telehealth processes were inefficient.

Table 4. 30 *Question 29. Telehealth processes were inefficient.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	17.1	1.9	9.5
2 Disagree	68.3	63.0	65.6
3 Neutral	12.2	16.7	14.4
4 Agree	2.4	18.4	10.4
5 Strongly Agree	0.0	0.0	0.0

Findings of Survey Question 30

Question 30 of the survey explored whether telehealth processes were ineffective. Managers predominantly disagreed (63.4%) or strongly disagreed (22.0%) with this statement, reflecting strong confidence in process effectiveness. Non-managers also largely disagreed (64.8%) but exhibited lower levels of strong disagreement (1.9%) and slightly higher neutrality (14.8%) and agreement (18.5%). Cumulatively, disagreement was the most frequent response at 64.1%, with strong disagreement at 12.0%, reinforcing a general rejection of the notion telehealth processes were ineffective. Table 4.31 illustrates the perceived stability of telehealth processes, showcasing the levels of agreement or disagreement across managerial and non-managerial roles.

Table 4. 31 *Question 30. Telehealth processes were ineffective.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	22.0	1.9	12.0
2 Disagree	63.4	64.8	64.1
3 Neutral	12.2	14.8	13.5
4 Agree	2.4	18.5	10.4
5 Strongly Agree	0.0	0.0	0.0

Findings of Survey Question 31

Question 31 of the survey examined whether telehealth processes change frequently. Managers overwhelmingly disagreed (63.4%) or strongly disagreed (22.0%) with this statement, indicating a perceived stability in telehealth processes. Non-managers also largely disagreed (64.8%) but showed less strong disagreement (1.9%) and slightly higher neutrality (14.8%) and agreement (18.5%). Cumulatively, disagreement was the dominant response at 64.1%, with strong disagreement at 12.0%, suggesting a general belief in the consistency of telehealth processes across groups. Table 4.32 illustrates perceptions of frequent changes in telehealth processes, highlighting trends among managers and non-managers.

Table 4. 32 *Question 31. Telehealth processes change frequently.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	22.0	1.9	12.0
2 Disagree	63.4	64.8	64.1
3 Neutral	12.2	14.8	13.5
4 Agree	2.4	18.5	10.4
5 Strongly Agree	0.0	0.0	0.0

Findings of Survey Question 32

Question 32 of the survey explored whether telehealth settings advocate process training. Managers predominantly disagreed (43.9%) or strongly disagreed (12.2%) with this statement, reflecting skepticism about the advocacy for process training. Non-managers also largely disagreed (38.9%) but did not record any strong disagreement responses, instead displaying higher neutrality (40.7%) compared to managers (29.3%). Cumulatively, disagreement was the most frequent response at 41.4%, with neutrality at 35% and agreement at 15.1%, indicating a perception advocacy for process training in telehealth settings was limited. Table 4.32 presented insights into advocacy for process training within telehealth settings, comparing managerial and non-managerial perspectives.

Table 4. 33 *Question 32. My telehealth environment advocates process training.*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	12.2	0.0	6.1
2 Disagree	43.9	38.9	41.4
3 Neutral	29.3	40.7	35
4 Agree	9.8	20.4	15.1
5 Strongly Agree	4.8	0.0	2.4

Scaled Item Response Distribution: Managerial, Non-Managerial, and Cumulative

Findings of the First Scale (Questions Q1-Q5)

The first scale represented perceptions of the foundational framework of telehealth processes, capturing how these processes were experienced across managerial and non-managerial roles. Scale 1 (Questions Q1-Q5) The basic framework illustrates the response distributions across managerial, non-managerial, and cumulative groups. In the managerial responses, 1.95% strongly disagree, 19.51% disagree, 16.59% remain neutral, 45.85% agree, and 16.10% strongly agree. Similarly, the non-managerial responses showed 1.85% strongly disagree, 28.89% disagree, 28.89% were neutral, 38.52% agree, and 1.85% strongly agree. When these groups were combined, the cumulative percentages reveal 1.89% strongly disagree, 24.84% disagree, 23.58% were neutral, 41.68% agree, and 8.00% strongly agree. Table 4.34 presented the managerial, non-managerial, and cumulative groups.

Table 4. 34 *Scale 1: Questions 1-5; Basic Framework*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	1.95	1.85	1.89
2 Disagree	19.51	28.89	24.84
3 Neutral	16.59	28.89	23.58
4 Agree	45.85	38.52	41.68
5 Strongly Agree	16.1	1.85	8.0

Findings of the Second Scale (Questions Q6-Q8)

The second scale displayed perceptions of telehealth processes at their initial maturity level, highlighting how processes transition from chaotic and unpredictable to early stages of control. Scale 2 (Questions Q6-Q8) displayed the response distributions across the managerial, non-managerial, and cumulative groups. The managerial responses indicated 8.94% strongly disagree, 33.33% disagree, 12.20% were neutral, 39.02% agree, and 6.50% strongly agree. Similarly, the non-managerial responses showed 1.23% strongly disagree, 33.95% disagree, 22.22% were neutral, 41.36% agree, and 1.23% strongly agree. When these groups were combined, the cumulative percentages showed 4.56% strongly disagree, 33.68% disagree, 17.89% were neutral, 40.35% agree, and 3.51% strongly agree. Table 4.35 presented the managerial, non-managerial, and cumulative groups.

Table 4. 35 *Scale 2: Questions 6 through 8; First Maturity Level*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	8.94	1.23	4.56
2 Disagree	33.33	33.95	33.68
3 Neutral	12.2	22.22	17.89
4 Agree	39.02	41.36	40.35
5 Strongly Agree	6.5	1.23	3.51

Findings of the Third Scale (Questions Q9-Q11)

The third scale represented perceptions of telehealth processes at the second maturity level, focusing on the transition toward defined and planned processes to establish greater consistency and control. Scale 3 (Questions Q9 through Q11) illustrates the response distributions among managerial, non-managerial, and cumulative groups. The managerial responses indicated 0.00% strongly disagree, 3.25% disagree, 8.13% were neutral, 69.11% agree, and 19.51% strongly agree. Similarly, the non-managerial responses showed 1.85% strongly disagree, 12.96% disagree, 28.40% were neutral, 54.32% agree, and 2.47% strongly agree. When these groups were combined, the cumulative percentages reveal 1.05% strongly disagree, 8.77% disagree, 19.65% were neutral, 60.70% agree, and 9.82% strongly agree. Table 4.36 features managerial, non-managerial, and cumulative groups.

Table 4. 36 *Scale 3: Questions 9 through 11; Second Maturity Level*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0	1.85	1.05
2 Disagree	3.25	12.96	8.77
3 Neutral	8.13	28.4	19.65
4 Agree	69.11	54.32	60.7
5 Strongly Agree	19.51	2.47	9.82

Findings of the Fourth Scale (Questions Q12-Q14)

The fourth scale illustrates perceptions of telehealth processes at the third maturity level, emphasizing the establishment of defined, consistent, and systematically followed processes. Scale 4 (Questions Q12-Q14) presented the response distributions across managerial, non-managerial, and cumulative groups. The managerial responses indicated 0.81% strongly disagree, 7.32% disagree, 26.83% were neutral, 55.28% agree, and 9.76% strongly agree. Similarly, the non-managerial responses reveal 4.94% strongly disagree, 23.46% disagree, 46.30% were neutral, 25.31% agree, and 0.00% strongly agree. When combining these groups, the cumulative percentages showed 3.16% strongly disagree, 16.49% disagree, 37.89% were neutral, 38.25% agree, and 4.21% strongly agree. Table 4.37 includes managerial, non-managerial, and cumulative groups.

Table 4. 37 *Scale 4: Questions 12 through 14; Third Maturity Level*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.81	4.94	3.16
2 Disagree	7.32	23.46	16.49
3 Neutral	26.83	46.3	37.89
4 Agree	55.28	25.31	38.25
5 Strongly Agree	9.76	0	4.21

Findings of the Fifth Scale (Questions Q15-Q17)

The fifth scale assesses perceptions of telehealth processes at the fourth maturity level, highlighting the incorporation of metrics and quantitative analysis to monitor and improve performance. Scale 5 (Questions Q15-Q17) illustrates the response distributions across managerial, non-managerial, and cumulative groups. The managerial responses indicated 0.81% strongly disagree, 4.07% disagree, 17.89% were neutral, 59.35% agree, and 17.89% strongly agree. In contrast, the non-managerial responses reveal 1.23% strongly disagree, 20.99% disagree, 33.95% were neutral, 42.59% agree, and 1.23% strongly agree. When combining these groups, the cumulative percentages demonstrated 1.05% strongly disagree, 13.68% disagree, 27.02% were neutral, 49.82% agree, and 8.42% strongly agree. Table 4.39 encompasses managerial, non-managerial, and cumulative groups.

Table 4. 38 *Scale 5: Questions 15 through 17; Fourth Maturity Level*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.81	1.23	1.05
2 Disagree	4.07	20.99	13.68
3 Neutral	17.89	33.95	27.02
4 Agree	59.35	42.59	49.82
5 Strongly Agree	17.89	1.23	8.42

Findings of the Sixth Scale (Questions Q18-Q20)

The sixth scale evaluated perceptions of telehealth processes at the fifth maturity level, emphasizing optimization through continuous improvements, efficiency, and adaptability. Scale 6 (Questions Q18-Q20) displayed the response distributions for managerial, non-managerial, and cumulative groups. The managerial responses indicated 0.81% strongly disagree, 9.76% disagree, 25.20% were neutral, 51.22% agree, and 13.01% strongly agree. Similarly, the non-managerial responses reveal 1.23% strongly disagree, 16.05% disagree, 40.74% were neutral, 39.51% agree, and 2.47% strongly agree. When combining these groups, the cumulative percentages showed 1.05% strongly disagree, 13.33% disagree, 34.04% were neutral, 44.56% agree, and 7.02% strongly agree. Table 4.39 presented the managerial, non-managerial, and cumulative groups.

Table 4. 39 *Scale 6: Questions 18 through 20; Fifth Maturity Level*

Response Category	Managerial (%)	Non-Managerial (%)	Cumulative Percentage (%)
1 Strongly Disagree	0.81	1.23	1.05
2 Disagree	9.76	16.05	13.33
3 Neutral	25.2	40.74	34.04
4 Agree	51.22	39.51	44.56
5 Strongly Agree	13.01	2.47	7.02

Analysis of the Means: Considerations of the Individual Questions

Means Analysis: Question 1

From the survey, Question 1 asks whether telehealth processes might be defined as ad-hoc, chaotic, or random. The cumulative mean was 3.83, with a standard deviation (SD) of 0.71 and a direction of agreement. The manager group mean was 4.12, with an SD of 0.55 and a direction of agreement. The non-manager group mean was 3.61, with an SD of 0.73 and a direction of agreement. Table 4.40 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 40 *Question 1. Telehealth processes might be defined as ad-hoc, chaotic, or random.*

Descriptor	Mean	SD	Directionality
Cumulative	3.83	0.71	Agreement
Manager	4.12	0.55	Agreement
Non-Manager	3.61	0.73	Agreement

Means Analysis: Question 2

From the survey, Question 2 asks if telehealth processes were managed. The cumulative mean was 3.84, with a standard deviation (SD) of 0.77 and a direction of agreement. The manager group mean was 4.24, with an SD of 0.66 and a direction of agreement. The non-manager group mean was 3.54, with an SD of 0.71 and a direction of agreement. Table 4.41 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 41 *Question 2. Telehealth processes were managed.*

Descriptor	Mean	SD	Directionality
Cumulative	3.84	0.77	Agreement
Manager	4.24	0.66	Agreement
Non-Manager	3.54	0.71	Agreement

Means Analysis: Question 3

From the survey, Question 3 asks whether telehealth processes were defined/specific. The cumulative mean was 3.05, with a standard deviation (SD) of 0.9 and a direction of neutrality. The manager group mean was 3.46, with an SD of 0.89 and a direction of neutrality. The non-manager group mean was 2.74, with an SD of 0.78 and a direction of neutrality. Table 4.42 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 42 *Question 3. Telehealth processes were defined/specific.*

Descriptor	Mean	SD	Directionality
Cumulative	3.05	0.9	Neutrality
Manager	3.46	0.89	Neutrality
Non-Manager	2.74	0.78	Neutrality

Means Analysis: Question 4

From the survey, Question 4 asks whether telehealth processes were quantitatively managed. The cumulative mean was 3.22, with a standard deviation (SD) of 0.93 and a direction of neutrality. The manager group mean was 3.68, with an SD of 0.81 and a direction of agreement. The non-manager group mean was 2.87, with an SD of 0.86 and a neutral direction. Table 4.43 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 43 *Question 4. Telehealth processes were quantitatively managed.*

Descriptor	Mean	SD	Directionality
Cumulative	3.22	0.93	Neutrality
Manager	3.68	0.81	Agreement
Non-Manager	2.87	0.86	Neutrality

Means Analysis: Question 5

From the survey, Question 5 asks about cumulative. The cumulative mean was 2.51, with a standard deviation (SD) of 0.92 and a direction of neutrality. The manager group mean was 2.22, with an SD of 0.78 and a direction of disagreement. The non-manager group mean was

2.72, with an SD of 0.95 and a direction of neutrality. Table 4.44 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 44 *Question 5. Telehealth processes were optimized.*

Descriptor	Mean	SD	Directionality
Cumulative	2.51	0.92	Neutrality
Manager	2.22	0.78	Disagreement
Non-Manager	2.72	0.95	Neutrality

Means Analysis: Question 6

From the survey, Question 6 asks if telehealth processes were unpredictable. The cumulative mean was 2.95, with a standard deviation (SD) of 0.89 and a direction of neutrality. The manager group mean was 2.93, with an SD of 0.90 and a direction of neutrality. The non-manager group mean was 2.96, with an SD of 0.88 and a direction of neutrality. Table 4.45 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 45 *Question 6. Telehealth processes were unpredictable.*

Descriptor	Mean	SD	Directionality
Cumulative	2.95	0.89	Neutrality
Manager	2.93	0.90	Neutrality
Non-Manager	2.96	0.88	Neutrality

Means Analysis: Question 7

From the survey, Question 7 asks whether telehealth processes were reactive. The cumulative mean was 2.35, with a standard deviation (SD) of 0.91 and a direction of

disagreement. The manager group mean was 1.95, with an SD of 0.77 and a direction of disagreement. The non-manager group mean was 2.65, with an SD of 0.89 and a direction of neutrality. Table 4.46 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 46 *Question 7. Telehealth processes were reactive.*

Descriptor	Mean	SD	Directionality
Cumulative	2.35	0.91	Disagreement
Manager	1.95	0.77	Disagreement
Non-Manager	2.65	0.89	Neutrality

Means Analysis: Question 8

From the survey, Question 8 asks whether telehealth processes were uncoordinated. The cumulative mean was 3.84, with a standard deviation (SD) of 0.67 and a direction of agreement. The manager group mean was 4.15, with an SD of 0.47 and a direction of agreement. The non-manager group mean was 3.61, with an SD of 0.71 and a direction of agreement. Table 4.47 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 47 *Question 8. Telehealth processes were uncoordinated.*

Descriptor	Mean	SD	Directionality
Cumulative	3.84	0.67	Agreement
Manager	4.15	0.47	Agreement
Non-Manager	3.61	0.71	Agreement

Means Analysis: Question 9

From the survey, Question 9 asks if telehealth processes were planned. The cumulative mean was 3.88, with a standard deviation (SD) of 0.71 and a direction of agreement. The manager group mean was 4.15, with an SD of 0.52 and a direction of agreement. The non-manager group mean was 3.69, with an SD of 0.77 and a direction of agreement. Table 4.48 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 48 *Question 9. Telehealth processes were planned.*

Descriptor	Mean	SD	Directionality
Cumulative	3.88	0.71	Agreement
Manager	4.15	0.52	Agreement
Non-Manager	3.69	0.77	Agreement

Means Analysis: Question 10

From the survey, Question 10 asks if telehealth processes were controlled. The cumulative mean was 3.64, with a standard deviation (SD) of 0.79 and a direction of agreement. The manager group mean was 4.07, with an SD of 0.56 and a direction of agreement. The non-manager group mean was 3.31, with an SD of 0.79 and a direction of neutrality. Table 4.49 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 49 *Question 10. Telehealth processes were controlled.*

Descriptor	Mean	SD	Directionality
Cumulative	3.64	0.79	Agreement
Manager	4.07	0.56	Agreement
Non-Manager	3.31	0.79	Neutrality

Means Analysis: Question 11

From the survey, Question 11 asks about cumulative. The cumulative mean was 3.56, with a standard deviation (SD) of 0.87 and a direction of agreement. The manager group mean was 3.93, with an SD of 0.78 and a direction of agreement. The non-manager group mean was 3.28, with an SD of 0.83 and a direction of neutrality. Table 4.50 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 50 *Question 11. Telehealth processes were well-defined.*

Descriptor	Mean	SD	Directionality
Cumulative	3.56	0.87	Agreement
Manager	3.93	0.78	Agreement
Non-Manager	3.28	0.83	Neutrality

Means Analysis: Question 12

From the survey, Question 12 asks whether telehealth processes were consistent. The cumulative mean was 3.72, with a standard deviation (SD) of 0.76 and a direction of agreement. The manager group mean was 4.10, with an SD of 0.58 and a direction of agreement. The non-manager group mean was 3.43, with an SD of 0.76 and a direction of neutrality. Table 4.51

illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 51 *Question 12. Telehealth processes were consistent.*

Descriptor	Mean	SD	Directionality
Cumulative	3.72	0.76	Agreement
Manager	4.10	0.58	Agreement
Non-Manager	3.43	0.76	Neutrality

Means Analysis: Question 13

From the survey, Question 13 asks whether telehealth processes were followed. The cumulative mean was 3.02, with a standard deviation (SD) of 0.82 and a direction of neutrality. The manager group mean was 3.46, with an SD of 0.74 and a direction of neutrality. The non-manager group mean was 2.69, with an SD of 0.72 and a direction of neutrality. Table 4.52 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 52 *Question 13. Telehealth processes were followed.*

Descriptor	Mean	SD	Directionality
Cumulative	3.02	0.82	Neutrality
Manager	3.46	0.74	Neutrality
Non-Manager	2.69	0.72	Neutrality

Means Analysis: Question 14

From the survey, Question 14 asks if telehealth processes involve quantitative or measurable objectives. The cumulative mean was 2.98, with a standard deviation (SD) of 0.87

and a direction of neutrality. The manager group mean was 3.41, with an SD of 0.83 and a direction of neutrality. The non-manager group mean was 2.65, with an SD of 0.75 and a direction of neutrality. Table 4.53 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 53 *Question 14. Telehealth processes involve quantitative or measurable objectives.*

Descriptor	Mean	SD	Directionality
Cumulative	2.98	0.87	Neutrality
Manager	3.41	0.83	Neutrality
Non-Manager	2.65	0.75	Neutrality

Means Analysis: Question 15

From the survey, Question 15 asks whether telehealth processes involve metrics analysis. The cumulative mean was 3.06, with a standard deviation (SD) of 0.93 and a direction of neutrality. The manager group mean was 3.54, with an SD of 0.92 and a direction of agreement. The non-manager group mean was 2.70, with an SD of 0.76 and a direction of neutrality. Table 4.54 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 54 *Question 15. Telehealth processes involve metrics analysis.*

Descriptor	Mean	SD	Directionality
Cumulative	3.06	0.93	Neutrality
Manager	3.54	0.92	Agreement
Non-Manager	2.70	0.76	Neutrality

Means Analysis: Question 16

From the survey, Question 16 asks whether telehealth processes involve statistical analysis. The cumulative mean was 3.68, with a standard deviation (SD) of 0.70 and a direction of agreement. The manager group mean was 4.02, with an SD of 0.52 and a direction of agreement. The non-manager group mean was 3.43, with an SD of 0.71 and a direction of neutrality. Table 4.55 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 55 *Question 16. Telehealth processes involve statistical analysis.*

Descriptor	Mean	SD	Directionality
Cumulative	3.68	0.70	Agreement
Manager	4.02	0.52	Agreement
Non-Manager	3.43	0.71	Neutrality

Means Analysis: Question 17

From the survey, Question 17 asks whether telehealth processes were improved incrementally. The cumulative mean was 3.78, with a standard deviation (SD) of 0.79 and a direction of agreement. The manager group mean was 4.12, with an SD of 0.67 and a direction of agreement. The non-manager group mean was 3.52, with an SD of 0.77 and a direction of agreement. Table 4.56 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 56 *Question 17. Telehealth processes were improved incrementally.*

Descriptor	Mean	SD	Directionality
Cumulative	3.78	0.79	Agreement
Manager	4.12	0.67	Agreement
Non-Manager	3.52	0.77	Agreement

Means Analysis: Question 18

From the survey, Question 18 asks if telehealth processes were efficient. The cumulative mean was 3.89, with a standard deviation (SD) of 0.67 and a direction of agreement. The manager group mean was 4.15, with an SD of 0.65 and a direction of agreement. The non-manager group mean was 3.70, with an SD of 0.63 and a direction of agreement. Table 4.57 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 57 *Question 18. Telehealth processes were efficient.*

Descriptor	Mean	SD	Directionality
Cumulative	3.89	0.67	Agreement
Manager	4.15	0.65	Agreement
Non-Manager	3.70	0.63	Agreement

Means Analysis: Question 19

From the survey, Question 19 asks if telehealth processes were effective. The cumulative mean was 3.29, with a standard deviation (SD) of 0.81 and a direction of neutrality. The manager group mean was 3.22, with an SD of 0.93 and a direction of neutrality. The non-manager group

mean was 3.35, with an SD of 0.70 and a direction of neutrality. Table 4.58 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 58 *Question 19. Telehealth processes were effective.*

Descriptor	Mean	SD	Directionality
Cumulative	3.29	0.81	Neutrality
Manager	3.22	0.93	Neutrality
Non-Manager	3.35	0.70	Neutrality

Means Analysis: Question 20

From the survey, Question 20 asks whether process maturity was not addressed by our current process improvement initiatives. The cumulative mean was 3.11, with a standard deviation (SD) of 0.84 and a direction of neutrality. The manager group mean was 3.61, with an SD of 0.70 and a direction of agreement. The non-manager group mean was 2.72, with an SD of 0.73 and a direction of neutrality. Table 4.59 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 59 *Question 20. Process maturity was not addressed by our current process improvement initiatives.*

Descriptor	Mean	SD	Directionality
Cumulative	3.11	0.84	Neutrality
Manager	3.61	0.70	Agreement
Non-Manager	2.72	0.73	Neutrality

Means Analysis: Question 21

From the survey, Question 21 asks whether process improvement was advocated within my telehealth. The cumulative mean was 3.02, with a standard deviation (SD) of 0.81 and a direction of neutrality. The manager group mean was 3.46, with an SD of 0.70 and a direction of neutrality. The non-manager group mean was 2.69, with an SD of 0.72 and a direction of neutrality. Table 4.60 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 60 *Question 21. Process improvement was advocated within my Telehealth.*

Descriptor	Mean	SD	Directionality
Cumulative	3.02	0.81	Neutrality
Manager	3.46	0.70	Neutrality
Non-Manager	2.69	0.72	Neutrality

Means Analysis: Question 22

From the survey, Question 22 asks whether process initiatives were tracked to examine process performance. The cumulative mean was 3.65, with a standard deviation (SD) of 0.86 and a direction of agreement. The manager group mean was 3.93, with an SD of 0.81 and a direction of agreement. The non-manager group mean was 3.44, with an SD of 0.83 and a direction of neutrality. Table 4.61 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 61 *Question 22. Process initiatives were tracked to examine process performance.*

Descriptor	Mean	SD	Directionality
Cumulative	3.65	0.86	Agreement
Manager	3.93	0.81	Agreement
Non-Manager	3.44	0.83	Neutrality

Means Analysis: Question 23

From the survey, Question 23 asks about grouping of processes according to maturity level, would improve the outcomes of our processes. The cumulative mean was 2.99, with a standard deviation (SD) of 0.75 and a direction of neutrality. The manager group mean was 3.27, with an SD of 0.77 and a direction of neutrality. The non-manager group mean was 2.78, with an SD of 0.66 and a direction of neutrality. Table 4.62 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 62 *Question 23. Grouping of processes according to maturity level, would improve the outcomes of our processes.*

Descriptor	Mean	SD	Directionality
Cumulative	2.99	0.75	Neutrality
Manager	3.27	0.77	Neutrality
Non-Manager	2.78	0.66	Neutrality

Means Analysis: Question 24

From the survey, Question 24 asks whether categorical process grouping was advocated within my telehealth. The cumulative mean was 2.93, with a standard deviation (SD) of 0.73 and a direction of neutrality. The manager group mean was 3.2, with an SD of 0.74 and a direction of

neutrality. The non-manager group mean was 2.72, with an SD of 0.65 and a direction of neutrality. Table 4.63 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 63 *Question 24. Categorical process grouping was advocated within my Telehealth.*

Descriptor	Mean	SD	Directionality
Cumulative	2.93	0.73	Neutrality
Manager	3.2	0.74	Neutrality
Non-Manager	2.72	0.65	Neutrality

Means Analysis: Question 25

From the survey, Question 25 asks whether process maturity was a contributor to successful process output within my telehealth. The cumulative mean was 2.59, with a standard deviation (SD) of 0.83 and a direction of neutrality. The manager group mean was 2.37, with an SD of 0.82 and a direction of disagreement. The non-manager group mean was 2.76, with an SD of 0.79 and a direction of neutrality. Table 4.64 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 64 *Question 25. Process maturity was a contributor to successful process output within my Telehealth.*

Descriptor	Mean	SD	Directionality
Cumulative	2.59	0.83	Neutrality
Manager	2.37	0.82	Disagreement
Non-Manager	2.76	0.79	Neutrality

Means Analysis: Question 26

From the survey, Question 26 asks whether processes were informal within my telehealth. The cumulative mean was 3.74, with a standard deviation (SD) of 0.65 and a direction of agreement. The manager group mean was 3.98, with an SD of 0.61 and a direction of agreement. The non-manager group mean was 3.56, with an SD of 0.63 and a direction of agreement. Table 4.65 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 65 *Question 26. Processes were informal within my Telehealth.*

Descriptor	Mean	SD	Directionality
Cumulative	3.74	0.65	Agreement
Manager	3.98	0.61	Agreement
Non-Manager	3.56	0.63	Agreement

Means Analysis: Question 27

From the survey, Question 27 asks whether telehealth policies influence processes. The cumulative mean was 3.01, with a standard deviation (SD) of 0.89 and a direction of neutrality. The manager group mean was 2.90, with an SD of 1.06 and a direction of neutrality. The non-manager group mean was 3.09, with an SD of 0.73 and a direction of neutrality. Table 4.66 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 66 *Question 27. Telehealth policies influence processes.*

Descriptor	Mean	SD	Directionality
Cumulative	3.01	0.89	Neutrality
Manager	2.90	1.06	Neutrality
Non-Manager	3.09	0.73	Neutrality

Means Analysis: Question 28

From the survey, Question 28 asks if methods of managing processes vary within my telehealth. The cumulative mean was 2.29, with a standard deviation (SD) of 0.78 and a direction of disagreement. The manager group mean was 2.00, with an SD of 0.63 and a direction of disagreement. The non-manager group mean was 2.52, with an SD of 0.81 and a direction of neutrality. Table 4.67 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 67 *Question 28. Methods of managing processes vary within my Telehealth.*

Descriptor	Mean	SD	Directionality
Cumulative	2.29	0.78	Disagreement
Manager	2.00	0.63	Disagreement
Non-Manager	2.52	0.81	Neutrality

Means Analysis: Question 29

From the survey, Question 29 asks if telehealth processes were inefficient. The cumulative mean was 2.26, with a standard deviation (SD) of 0.80 and a direction of disagreement. The manager group mean was 1.95, with an SD of 0.66 and a direction of disagreement. The non-manager group mean was 2.50, with an SD of 0.81 and a direction of

neutrality. Table 4.68 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 68 *Question 29. Telehealth processes were inefficient.*

Descriptor	Mean	SD	Directionality
Cumulative	2.26	0.80	Disagreement
Manager	1.95	0.66	Disagreement
Non-Manager	2.50	0.81	Neutrality

Means Analysis: Question 30

From the survey, Question 30 asks if telehealth processes were ineffective. The cumulative mean was 2.68, with a standard deviation (SD) of 0.88 and a direction of neutrality. The manager group mean was 2.51, with an SD of 0.99 and a direction of neutrality. The non-manager group mean was 2.81, with an SD of 0.75 and a direction of neutrality. Table 4.69 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 69 *Question 30. Telehealth processes were ineffective.*

Descriptor	Mean	SD	Directionality
Cumulative	2.68	0.88	Neutrality
Manager	2.51	0.99	Neutrality
Non-Manager	2.81	0.75	Neutrality

Means Analysis: Question 31

From the survey, Question 31 asks whether telehealth processes change frequently. The cumulative mean was 3.11, with a standard deviation (SD) of 0.85 and a direction of neutrality.

The manager group mean was 3.51, with an SD of 0.83 and a direction of agreement. The non-manager group mean was 2.80, with an SD of 0.73 and a direction of neutrality. Table 4.70 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 70 *Question 31. Telehealth processes change frequently.*

Descriptor	Mean	SD	Directionality
Cumulative	3.11	0.85	Neutrality
Manager	3.51	0.83	Agreement
Non-Manager	2.80	0.73	Neutrality

Means Analysis: Question 32

From the survey, Question 32 asks whether my telehealth advocates process training. The cumulative mean was 3.00, with a standard deviation (SD) of 0.00 and a direction of neutrality. The manager group mean was 3.00, with an SD of 0.00 and a direction of neutrality. The non-manager group mean was 3.00, with an SD of 0.00 and a direction of neutrality. Table 4.71 illustrates the means, SDs, and the respective directions for cumulative, manager, and non-manager groups.

Table 4. 71 *Question 32. My Telehealth advocates process training.*

Descriptor	Mean	SD	Directionality
Cumulative	3.00	0.00	Neutrality
Manager	3.00	0.00	Neutrality
Non-Manager	3.00	0.00	Neutrality

Analysis of the Means: Findings of the Scaled Items

Findings of the First Scale (Questions 1-5)

The "Analysis of the Means: Findings of the Basic Framework Scale" delves into the perceptions of process maturity across the CMMI framework and its five maturity levels. By analyzing responses to survey questions grouped according to these levels, the study provided insights into the alignment and divergence of perspectives between managers and non-managers. Each maturity level, from the initial chaotic and ad hoc processes of Level 1 to the optimized processes of Level 5, was evaluated through mean scores and standard deviations. This comparative analysis highlights the directionality of responses, ranging from agreement to disagreement, offering a nuanced view of organizational process maturity and its perceived effectiveness.

Telehealth processes demonstrated varying levels of maturity, from ad-hoc and chaotic beginnings to controlled and optimized outcomes, as perceived differently by managers and non-managers. Scale 1, the Basic Framework, questions 1 through 5 in this table, provided a comparative analysis of perceptions regarding the overall CMMI framework, emphasizing general organizational process maturity. The manager mean was 4.00, with a standard deviation (SD) of 0.50 and a direction of agreement. The non-manager mean was 3.60, with an SD of 0.70 and a direction of agreement. The cumulative mean was 3.80, with an SD of 0.60 and a direction of agreement. Table 4.72 illustrates the means, SDs, and directionality for managers, non-managers, and the cumulative group.

Table 4. 72 *Scale 1: Questions 1-5; Basic Framework*

Descriptor	Mean	SD	Directionality
Managers	4.00	0.50	Agreement
Non-Managers	3.60	0.70	Agreement
Cumulative	3.80	0.60	Agreement

Findings of the Second Scale (Questions 6-8)

At Level 1, processes were largely unpredictable and chaotic, with minimal standardization or coordination, reflecting an ad hoc operational state. In Scale 2, the First Maturity Level, questions 6 through 8 examine CMMi Level 1. The manager mean was 3.90, with a standard deviation (SD) of 0.60 and a direction of agreement. The non-manager mean was 3.50, with an SD of 0.80 and a direction of neutrality. The cumulative mean was 3.70, with an SD of 0.70 and a direction of agreement. Table 4.73 illustrates the means, SDs, and directionality for managers, non-managers, and the cumulative group.

Table 4. 73 *Scale 2: Questions 6 through 8; First Maturity Level*

Descriptor	Mean	SD	Directionality
Managers	3.90	0.60	Agreement
Non-Managers	3.50	0.80	Neutrality
Cumulative	3.70	0.70	Agreement

Findings of the Third Scale (Questions 9-10)

Processes at Level 2 transition into being reactive and repetitive, showing early signs of control but still lacking comprehensive planning or measurable objectives. In Scale 3, Questions 9 and 10, the Second Maturity Level, the manager mean was 3.80, with a standard deviation

(SD) of 0.70 and a direction of agreement. The non-manager mean was 3.40, with an SD of 0.90 and a direction of neutrality. The cumulative mean was 3.60, with an SD of 0.80 and a direction of agreement. Table 4.74 illustrates the means, SDs, and directionality for managers, non-managers, and the cumulative group.

Table 4. 74 *Scale 3: Questions 9 and 10; Second Maturity Level*

Descriptor	Mean	SD	Directionality
Managers	3.80	0.70	Agreement
Non-Managers	3.40	0.90	Neutrality
Cumulative	3.60	0.80	Agreement

Findings of the Fourth Scale (Questions 11-13)

By Level 3, processes were clearly defined and consistently followed, providing a structured approach emphasizing process adherence across roles. In Scale 4, Questions 11 through 13, the Third Maturity Level, the manager mean was 3.70, with a standard deviation (SD) of 0.80 and a direction of agreement. The non-manager mean was 3.30, with an SD of 1.00 and a direction of neutrality. The cumulative mean was 3.50, with an SD of 0.90 and a direction of neutrality. Table 4.75 illustrates the means, SDs, and directionality for managers, non-managers, and the cumulative group.

Table 4. 75 *Scale 4: Questions 11 through 13; Third Maturity Level*

Descriptor	Mean	SD	Directionality
Managers	3.70	0.80	Agreement
Non-Managers	3.30	1.00	Neutrality
Cumulative	3.50	0.90	Neutrality

Findings of the Fifth Scale (Questions 14-16)

Level 4 processes focus on metrics and quantitative analysis, where statistical tools were applied to monitor and improve performance outcomes. In Scale 5, the Fourth Maturity Level, the manager mean was 3.60, with a standard deviation (SD) of 0.90 and a direction of agreement. The non-manager mean was 3.20, with an SD of 1.10 and a direction of neutrality. The cumulative mean was 3.40, with an SD of 1.00 and a direction of neutrality. Table 4.76 illustrates the means, SDs, and directionality for managers, non-managers, and the cumulative group.

Table 4. 76 *Scale 5: Questions 14 through 16; Fourth Maturity Level*

Descriptor	Mean	SD	Directionality
Managers	3.60	0.90	Agreement
Non-Managers	3.20	1.10	Neutrality
Cumulative	3.40	1.00	Neutrality

Findings of the Sixth Scale (Questions 17-19)

At Level 5, telehealth processes achieve optimization, characterized by continuous incremental improvements, efficiency, and adaptability to changing requirements. In Scale 6, the Fifth Maturity Level, the manager mean was 3.50, with a standard deviation (SD) of 1.00 and a direction of neutrality. The non-manager mean was 3.10, with an SD of 1.20 and a direction of neutrality. The cumulative mean was 3.30, with an SD of 1.10 and a direction of neutrality. Table 4.77 illustrates the means, SDs, and directionality for Managers, Non-Managers, and the cumulative group.

Table 4.77 Scale 6: Questions 17 through 19; Fifth Maturity Level

Descriptor	Mean	SD	Directionality
Managers	3.50	1.00	Neutrality
Non-Managers	3.10	1.20	Neutrality
Cumulative	3.30	1.10	Neutrality

Hypothesis Testing: Individual Question Items

Question 1: Individual ANOVA Assessment

Survey question 1, "Telehealth processes might be defined as ad hoc, chaotic, or random," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis revealed significant differences between groups ($F = 41.75$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.126$, indicating a medium effect. Managerial respondents ($M = 4.12$) agreed with the statement, while non-managerial respondents ($M = 3.61$) agreed.

Question 2: Individual ANOVA Assessment

Survey question 2, "Telehealth processes were managed," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis revealed significant differences between groups ($F = 73.25$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.203$, indicating a large effect. Managerial respondents ($M = 4.24$) agreed with the statement, while non-managerial respondents ($M = 3.54$) agreed.

Question 3: Individual ANOVA Assessment

Survey question 3, "Telehealth processes were defined and specific," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis revealed significant differences between groups ($F = 53.35$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The

effect size for the significant outcome was $\omega^2 = 0.156$, indicating a large effect. Managerial respondents ($M = 3.46$) were neutral to the statement, while non-managerial respondents ($M = 2.74$) were neutral.

Question 4: Individual ANOVA Assessment

Survey question 4, "Telehealth processes were quantitatively managed," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis revealed significant differences between groups ($F = 65.02$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.185$, indicating a large effect. Managerial respondents ($M = 3.68$) agreed with the statement, while non-managerial respondents ($M = 2.87$) were neutral.

Question 5: Individual ANOVA Assessment

Survey question 5, "Telehealth processes were optimized," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis revealed significant differences between groups ($F = 22.57$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.071$, indicating a medium effect. Managerial respondents ($M = 2.22$) disagreed with the statement, while non-managerial respondents ($M = 2.72$) were neutral.

Question 6: Individual ANOVA Assessment

Survey question 6, "Telehealth processes were unpredictable," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis revealed no significant differences between groups ($F = 0.12$, $F\text{-Critical} = 3.87$, $p = 0.73$, $\alpha = 0.05$). The effect size for the insignificant outcome was $\omega^2 = -0.003$, indicating no meaningful effect. Managerial respondents ($M = 2.93$) were neutral with the statement, while non-managerial respondents ($M = 2.96$) were neutral.

Question 7: Individual ANOVA Assessment

Survey question 7, "Telehealth processes were reactive," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis revealed significant differences between groups ($F = 48.38$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.143$, indicating a large effect. Managerial respondents ($M = 1.95$) disagreed with the statement, while non-managerial respondents ($M = 2.65$) were neutral.

Question 8: Individual ANOVA Assessment

Survey question 8, "Telehealth processes were uncoordinated," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis revealed significant differences between groups ($F = 52.55$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.154$, indicating a large effect. Managerial respondents ($M = 4.15$) agreed with the statement, while non-managerial respondents ($M = 3.61$) agreed.

Question 9: Individual ANOVA Assessment

Survey question 9, "Telehealth processes were planned," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis revealed significant differences between groups ($F = 32.79$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.101$, indicating a medium effect. Managerial respondents ($M = 4.15$) agreed with the statement, while non-managerial respondents ($M = 3.69$) agreed.

Question 10: Individual ANOVA Assessment

Survey question 10, "Telehealth processes were controlled," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis revealed significant differences between groups ($F = 81.70$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.222$, indicating a large effect. Managerial

respondents ($M = 4.07$) agreed with the statement, while non-managerial respondents ($M = 3.31$) were neutral with the statement.

Question 11: Individual ANOVA Assessment

Survey question 11, "Telehealth processes were well-defined," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis revealed significant differences between groups ($F = 45.09$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.135$, indicating a medium effect. Managerial respondents ($M = 3.93$) agreed with the statement, while non-managerial respondents ($M = 3.28$) were neutral.

Question 12: Individual ANOVA Assessment

Survey question 12, "Telehealth processes were consistent," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis revealed significant differences between groups ($F = 66.39$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.188$, indicating a large effect. Managerial respondents ($M = 4.10$) agreed with the statement, while non-managerial respondents ($M = 3.43$) were neutral.

Question 13: Individual ANOVA Assessment

Survey question 13, "Telehealth processes were followed," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis revealed significant differences between groups ($F = 80.14$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.219$, indicating a large effect. Managerial respondents ($M = 3.46$) agreed with the statement, while non-managerial respondents ($M = 2.69$) were neutral.

Question 14: Individual ANOVA Assessment

Survey question 14, "Telehealth processes involve quantitative or measurable objectives," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis revealed significant differences between groups ($F = 66.50$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.188$, indicating a large effect. Managerial respondents ($M = 3.41$) were neutral with the statement, while non-managerial respondents ($M = 2.65$) were neutral.

Question 15: Individual ANOVA Assessment

Survey question 15, "Telehealth processes involve metrics analysis," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis revealed significant differences between groups ($F = 69.90$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.196$, indicating a large effect. Managerial respondents ($M = 3.54$) were neutral with the statement, while non-managerial respondents ($M = 2.70$) were neutral.

Question 16: Individual ANOVA Assessment

Survey question 16, "Telehealth processes involve statistical analysis," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis revealed significant differences between groups ($F = 61.87$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.177$, indicating a large effect. Managerial respondents ($M = 4.02$) agreed with the statement, while non-managerial respondents ($M = 3.43$) were neutral.

Question 17: Individual ANOVA Assessment

Survey question 17, "Telehealth processes were improved incrementally," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis

revealed significant differences between groups ($F = 48.16$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.143$, indicating a large effect. Managerial respondents ($M = 4.12$) agreed with the statement, while non-managerial respondents ($M = 3.52$) agreed.

Question 18: Individual ANOVA Assessment

Survey question 18, "Telehealth processes were efficient," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis revealed significant differences between groups ($F = 33.67$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.104$, indicating a medium effect. Managerial respondents ($M = 4.15$) agreed with the statement, while non-managerial respondents ($M = 3.70$) agreed.

Question 19: Individual ANOVA Assessment

Survey question 19, "Telehealth processes were effective," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis revealed no significant differences between groups ($F = 1.88$, $F\text{-Critical} = 3.87$, $p = 0.17$, $\alpha = 0.05$). The effect size for the insignificant outcome was $\omega^2 = 0.003$, indicating a small effect. Managerial respondents ($M = 3.22$) were neutral with the statement, while non-managerial respondents ($M = 3.35$) were neutral.

Question 20: Individual ANOVA Assessment

Survey question 20, "Process maturity was not addressed by our current process improvement initiatives," was analyzed to explore group perceptions between managers and non-managers. The ANOVA analysis revealed significant differences between groups ($F = 106.92$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.272$, indicating a large effect. Managerial respondents ($M = 3.61$) agreed with the statement, while non-managerial respondents ($M = 2.72$) were neutral.

Hypothesis Testing: Scaled Response Items

Question Scale 1: Overall Telehealth Maturity Model

Scale 1, Basic Framework evaluated the foundational characteristics of telehealth processes, including their initial state, chaotic or ad hoc nature, and early standardization efforts. ANOVA analysis revealed significant differences between groups ($F = 82.76$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.21$, indicating a large effect. Managerial respondents ($M = 4.30$) agreed with the statement, while non-managerial respondents ($M = 3.70$) agreed with the statement.

Question Scale 2: Overall Telehealth Maturity Model

Scale 2, The First Maturity Level, assesses the early maturity of telehealth processes, emphasizing unpredictability, reactivity, and efforts to establish initial consistency. ANOVA analysis revealed no significant differences between groups ($F = 1.72$, $F\text{-Critical} = 3.87$, $p = 0.19$, $\alpha = 0.05$). The effect size for the insignificant outcome was $\omega^2 = 0.01$, indicating a small effect. Managerial respondents ($M = 3.50$) were neutral with the statement, while non-managerial respondents ($M = 3.40$) were neutral.

Question Scale 3: Overall Telehealth Maturity Model

Scale 3, The Second Maturity Level, assesses the transition to defined and repeatable telehealth processes, highlighting the importance of planning and coordination. ANOVA analysis revealed significant differences between groups ($F = 68.96$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.19$, indicating a large effect. Managerial respondents ($M = 4.10$) agreed with the statement, while non-managerial respondents ($M = 3.60$) agreed.

Question Scale 4: Overall Telehealth Maturity Model

Scale 4, The Third Maturity Level, evaluated the structure and consistency of telehealth processes, including their adherence to clearly defined protocols. ANOVA analysis revealed significant differences between groups ($F = 99.12$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.25$, indicating a large effect. Managerial respondents ($M = 4.40$) agreed with the statement, while non-managerial respondents ($M = 3.80$) agreed.

Question Scale 5: Overall Telehealth Maturity Model

Scale 5, The Fourth Maturity Level, assesses the quantitative management of telehealth processes, emphasizing metrics, statistical analysis, and ongoing improvements. ANOVA analysis revealed significant differences between groups ($F = 101.44$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.26$, indicating a large effect. Managerial respondents ($M = 4.50$) agreed with the statement, while non-managerial respondents ($M = 3.90$) agreed.

Question Scale 6: Overall Telehealth Maturity Model

Scale 6, The Fifth Maturity Level, reflects the optimization and continuous improvement of telehealth processes, emphasizing efficiency, adaptability, and innovation. ANOVA analysis revealed significant differences between groups ($F = 74.92$, $F\text{-Critical} = 3.87$, $p = 0.00$, $\alpha = 0.05$). The effect size for the significant outcome was $\omega^2 = 0.22$, indicating a large effect. Managerial respondents ($M = 4.20$) agreed with the statement, while non-managerial respondents ($M = 3.70$) agreed.

Synopsis of Statistically Significant Findings

Appendix A, Table A.1 presented 17 statistically significant findings and specific questions demonstrating notable differences between managers and non-managers.

Significant findings for question scales were summarized here in Table 4.78.

Table 4. 78 *Significant Findings for the Question Scales*

Scale	Topic	F Value	F-Critical	p-Value	Effect	Significance
Item					Size (ω^2)	Level
Scale 1	Basic	82.76	3.87	0.00	0.21	0.05
	Framework					
Scale 3	Second	68.96	3.87	0.00	0.19	0.05
	Maturity					
	Level					
Scale 4	Third	99.12	3.87	0.00	0.25	0.05
	Maturity					
	Level,					
Scale 5	Fourth	101.44	3.87	0.00	0.26	0.05
	Maturity					
	Level					
Scale 6	Fifth	74.92	3.87	0.00	0.22	0.05
	Maturity					
	Level					

Synopsis of Statistically Insignificant Findings

Insignificant findings for the individual questions were detailed in Table 4.79, indicating areas where statistical differences were not observed.

Table 4. 79 *Insignificant Findings for the Individual Questions 1 – 20*

Question Item	Topic	F Value	F-Critical	p-Value	Effect Size (ω^2)	Significance Level
Question 6	Telehealth processes are unpredictable.	0.12	3.87	0.73	0.003	0.05
Question 19	Telehealth processes are effective.	1.88	3.87	0.17	0.003	0.05

Insignificant findings for the question scales were outlined in Table 4.80, showing elements with no significant variance.

Table 4. 80 *Insignificant Findings for the Question Scales*

Scale Item	Topic	F Value	F-Critical	p-Value	Effect Size (ω^2)	Significance Level
Scale 2	First Maturity Level	1.72	3.87	0.19	0.01	0.05

Evaluation of the Findings

Research Question 1 (Q1-Q5): Basic Framework of CMM Portability in Telehealth

The study examined the applicability of the Capability Maturity Model (CMM) to telehealth, evaluating its portability through different maturity levels. The Basic Framework (Q1-Q5) results showed a cumulative mean of 3.83, indicating a general agreement that fundamental maturity principles apply to telehealth across managerial and non-managerial roles. Managers

reported a higher mean (4.12) than non-managers (3.61), suggesting a discrepancy in perceived process standardization.

These findings align with previous research on CMM's adaptability across industries (Watts, 2020), which supports the assertion that a structured process model improves predictability in new technological implementations (Poeppelbuss et al., 2011). However, the variance in agreement between managerial and non-managerial groups suggests that process implementation might not be uniformly understood or applied across roles, a concern also noted in studies of healthcare process standardization (David Patón-Romero et al., 2019).

Research Question 2 (Q6-Q8): Predictability and Management of Telehealth Processes

The First Maturity Level focused on process predictability and structured management. Managers reported a mean of 3.68, reflecting agreement with structured processes, while non-managers leaned toward neutrality (mean of 2.87). The cumulative mean of 3.22 suggests that telehealth processes were perceived as moderately structured but lack uniform management perception.

This result was consistent with findings from Jokela & Lalli (2003), who assert that early-stage maturity models require strong organizational buy-in to ensure effectiveness. The discrepancy highlights potential implementation gaps, reinforcing the argument that formal process training might be necessary for non-managerial roles to perceive structure similarly to managers.

Research Question 3 (Q9-Q11): Definition and Standardization of Telehealth Processes

The Second Maturity Level emphasizes process definition and standardization, with a cumulative mean of 3.56. Managers showed more substantial agreement (3.93) than non-managers (3.28), indicating that while defined processes exist, their clarity and enforcement vary across organizational levels.

This supports prior literature emphasizing that structured frameworks like CMM lead to more transparent operational procedures (Royce, 2002). However, similar to previous levels, non-managers' lower agreement suggests potential challenges in translating theoretical process maturity into practice, possibly due to gaps in training or communication (White, 2021).

Research Question 4 (Q12-Q14): Consistency and Adherence in Telehealth Maturity

The Third Maturity Level, focusing on consistency and adherence, reported a cumulative mean of 3.72. Managers exhibited strong agreement (4.10), whereas non-managers were more neutral (3.43). This indicated that managerial perspectives align closely with structured implementation, while non-managers might not perceive process consistency as strongly.

This divergence was supported by previous research in healthcare IT maturity models, which highlights that process adherence often varies depending on role-based interaction with the system (Bashshur et al., 2011). The results further validate findings from Gibson et al. (2006), which suggest that telehealth frameworks require additional standardization efforts at lower hierarchical levels to achieve uniform compliance.

Research Question 5 (Q15-Q20): Scalability, Transferability, and Optimization of CMM in Telehealth

The Fourth and Fifth Maturity Levels examine scalability and continuous improvement in telehealth implementation. The Fourth Level recorded a cumulative mean of 3.68, with managers (4.02) showing higher agreement than non-managers (3.43). The Fifth Level, which focused on process optimization, had a cumulative mean of 3.78, again with managers (4.15) expressing greater agreement than non-managers (3.52).

These findings reinforce the theoretical framework proposed by Watts (2020), which states that optimization and scalability in CMM-based frameworks depend on organizational commitment to continuous improvement. The higher agreement among managers suggests they

recognize process refinement efforts. Still, the lower non-managerial scores indicated that direct practitioners might not experience or engage with these optimizations at the same level.

Evaluation of the CMM's Portability to the Telehealth Domain

The cumulative responses across all maturity levels confirm the partial portability of the Capability Maturity Model (CMM) within telehealth settings. Specifically, the scaled items representing portability include the Basic Framework (Q1-Q5), which demonstrated partial portability with cumulative agreement (M = 3.83), managerial agreement (M = 4.12), and non-managerial agreement (M = 3.61). The First Maturity Level (Q6-Q8) indicated partial portability, with managers agreeing (M = 3.68) while non-managers were neutral (M = 2.87), leading to a cumulative mean of 3.22. Similarly, the Second Maturity Level (Q9-Q11) reflected agreement among managers (M = 3.93) and neutrality among non-managers (M = 3.28), with an overall cumulative mean of 3.56. The Third Maturity Level (Q12-Q14) further supported portability with a cumulative mean of 3.72, showing managerial agreement (M = 4.10) and non-managerial neutrality (M = 3.43). The Fourth Maturity Level (Q15-Q17) revealed partial portability, evidenced by a cumulative mean of 3.68 and managerial agreement (M = 4.02), contrasting with non-managerial neutrality (M = 3.43). Finally, the Fifth Maturity Level (Q18-Q20) demonstrated portability, with an overall mean of 3.78, reflecting agreement across all respondent groups.

Table 4.81 represented the portability of the Capability Maturity Model in telehealth managerial, non-managerial, and cumulative perspectives.

Table 4. 81 *Portability of the Capability Maturity Model in telehealth*

Scale	Manager Mean	Manager Agreement Level	Non- Manager Mean	Non- Manager Agreement Level	Cumulative Mean	Cumulative Agreement Level
Basic Framework (Q1-Q5)	4.12	Agreement	3.61	Agreement	3.83	Agreement
First Maturity Level (Q6-Q8)	3.68	Agreement	2.87	Neutrality	3.22	Neutrality
Second Maturity Level (Q9-Q11)	3.93	Agreement	3.28	Neutrality	3.56	Agreement
Third Maturity Level (Q12-Q14)	4.1	Agreement	3.43	Neutrality	3.72	Agreement
Fourth Maturity Level (Q15-Q17)	4.02	Agreement	3.43	Neutrality	3.68	Agreement

Fifth Maturity Level (Q18-Q20)	4.15	Agreement	3.52	Agreement	3.78	Agreement
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Consistency with Existing Research and Theoretical Framework

Overall, the study provided substantial evidence supporting the partial portability of the CMM framework to telehealth, which aligns with existing research. Structured maturity frameworks appear to enhance process reliability, but disparities in agreement between managerial and non-managerial roles highlight implementation challenges. These discrepancies align with prior studies on organizational adoption of structured process models (Lee et al., 2019), which suggest that effective maturity model deployment requires strong communication and training to bridge perception gaps across roles.

While the results support the theoretical assumptions of CMM's adaptability, they also emphasize that telehealth's unique operational demands such as patient-provider interactions and technology usability might require tailored modifications to the standard CMM framework to ensure full portability.

Summary

Chapter 4 illustrates an analysis of the quantitative data collected through the survey, focusing on the perceptions of managerial and non-managerial personnel regarding the maturity of telehealth processes. The study showed the limited portability of the Capability Maturity Model (CMM) within telehealth environments, offering insights into process standardization, adherence, and optimization. The findings were systematically presented based on research questions and hypotheses, highlighting significant perspective differences through statistical analyses such as means comparisons and ANOVA testing.

The study identified 17 significant and three insignificant outcomes across various maturity levels. Significant findings were observed within the Basic Framework (Q1-Q5) and the Second (Q9-Q11), Third (Q12-Q14), Fourth (Q15-Q17), and Fifth Maturity Levels (Q18-Q20), indicating substantial agreement from managerial respondents regarding process standardization and optimization. However, the First Maturity Level (Q6-Q8) yielded statistically insignificant results, suggesting the need for further refinement to improve process predictability at early stages.

Regarding the portability of the CMM within telehealth settings, the findings confirm its partial portability, with higher maturity levels showing broader acceptance and applicability. Managerial respondents consistently reported higher levels of agreement compared to their non-managerial counterparts. The study underscores the limited potential of the CMM to be adapted for telehealth process improvements while identifying key areas for achieving complete portability.

This study's findings align with Lee et al. (2019) on how a maturity model could enhance telehealth processes. Bashshur et al. (2011) support this by highlighting the value of structured assessments in improving management in telehealth. The results relate to the CMMi framework, as van Dyk and Schutte (2012) noted, confirming the model's role in optimizing vendor-supplier interactions, reducing process inefficiencies, and mitigating telehealth implementation risks. Chapter 5 discusses the implications of these findings, offering recommendations for telehealth implementation and future research. The final chapter synthesizes key insights, addressing both theoretical and practical aspects while exploring strategies to improve telehealth maturity frameworks in evolving healthcare environments.

Chapter 5: Implications, Recommendations, and Conclusions

This chapter presented the implications, recommendations, and conclusions drawn from this study on implementing a maturity model for telehealth technology at the Veterans Administration (VA). Within the study, the author aimed to evaluate the portability of the Capability Maturity Model Integration (CMMI) framework in a hospital setting, specifically assessing how organizations at different maturity levels manage process optimization, standardization, and scalability. The findings contribute to understanding telehealth maturity and its impact on healthcare delivery within the VA system. This study addressed the need for an effective and customizable maturity model for the sustainable implementation of innovative telehealth technology. Current Capability Maturity Model Integration (CMMI) frameworks pose challenges for telehealth adoption due to a lack of standardization and integration of process improvements. The study aimed to develop a telehealth-specific maturity model to enhance implementation efficiency and reduce the risks linked to adopting new technology.

This study employed a quantitative methodology within a VA hospital setting, gathering managerial and non-managerial personnel data to assess perspectives on telehealth process maturity. Statistical analyses, including Cronbach's Alpha, Chi-Square tests, and ANOVA assessments, were conducted to evaluate variations in perceptions and the applicability of the CMMI framework. The findings indicated statistically significant differences in how managers and non-managers perceive telehealth process maturity. Additionally, the study demonstrated that higher maturity levels were associated with improved telehealth service reliability and efficiency.

While the study provided valuable insights into telehealth process maturity, its findings were specific to the VA and might not be generalizable to other healthcare environments. The reliance on self-reported survey data introduces the potential for response bias, which might

impact the interpretation of results. Future research should consider expanding the scope to additional healthcare systems to validate the portability of the CMMI framework beyond the VA. This chapter discusses the study's implications for each research question, followed by recommendations for practice and future research. The chapter concludes with a synthesis of the study's contributions to telehealth maturity and its relevance to digital healthcare transformation within the VA.

Implications

The Veterans Administration (VA) was the largest integrated healthcare system in the United States, offering healthcare services to over nine million enrolled veterans each year (Sullivan & Hughes, 2021). The VA operates more than 1,200 healthcare facilities, including medical centers, outpatient clinics, and community-based centers (Administration, 2008). With a workforce of around 371,000 personnel, encompassing healthcare professionals, administrative staff, and support workers, the VA serves as a vital healthcare provider nationwide (Der-Martirosian et al., 2021). The complexity of managing such a large healthcare system necessitates the implementation of structured frameworks like the Capability Maturity Model Integration (CMMI) to enhance operational efficiency, improve the quality of telehealth services, and standardize efforts across all VA facilities.

The findings from this study hold significant implications for the VA, its patients, employees, and stakeholders. By implementing a structured telehealth maturity model grounded in CMMI principles, the VA could enhance the efficiency and scalability of its telehealth services (Gagnon et al., 2006). A clear telehealth maturity model promotes better coordination of digital healthcare initiatives, streamlines processes, and improves the standardization of telehealth services across various VA medical centers. Standardizing processes enables interoperability, reduces redundancy, and ensures the effective deployment of telehealth

technologies to underserved populations (Baker & Bufka, 2011). These enhancements could result in greater access to healthcare services, shorter wait times for veterans, and increased overall patient satisfaction with virtual medical consultations (Butzner & Cuffee, 2021).

For veterans, adopting a telehealth maturity model ensures reliable, high-quality remote healthcare services. Many veterans, especially those living in rural or underserved areas, rely on telehealth to access timely medical care. A structured telehealth framework could bridge healthcare disparities by improving appointment availability, reducing the need for extensive travel, and fostering continuity of care (Greenhalgh et al., 2012). By implementing process-driven telehealth services, veterans would benefit from enhanced virtual consultations, improved chronic disease management, and better integration with primary care providers (Der-Martirosian et al., 2021). Furthermore, decreasing reliance on in-person visits through effective telehealth solutions could ease the burden on VA hospitals and clinics, allowing healthcare professionals to focus on complex and urgent cases (Cartwright et al., 2013).

Veterans Administration (VA) employees, including healthcare providers and administrative staff, would experience operational improvements through the integration of a structured telehealth maturity model. Standardized telehealth workflows help reduce administrative burdens, increase efficiency, and enhance clarity in roles and responsibilities (Chike-Harris et al., 2021). Additionally, telehealth training programs and best-practice protocols would empower VA healthcare providers to deliver high-quality remote care while ensuring compliance with evolving telehealth regulations (Hilty et al., 2018). A data-driven approach to telehealth implementation allows providers to make informed decisions regarding patient care and resource allocation, thereby improving workforce productivity and job satisfaction (Kho et al., 2020).

From a stakeholder perspective, including policymakers, healthcare administrators, and government agencies, implementing a structured telehealth maturity model provided strategic advantages in healthcare management. A mature telehealth system allows stakeholders to evaluate the effectiveness of telehealth policies, pinpoint areas for improvement, and apply data-driven strategies for resource allocation (Kidholm et al., 2012). Additionally, integrating a process maturity framework empowers stakeholders to advocate for policy refinements that align telehealth services with best practices in digital healthcare expansion (Bashshur et al., 2011). As the VA continues to enhance its telehealth capabilities, decision-makers could utilize the findings of this study to optimize funding, infrastructure development, and regulatory support for telehealth initiatives across the nation.

The work environment within the VA was set for transformation with the adoption of a structured telehealth maturity model. Digital health technologies would increasingly be integrated into routine operations, necessitating comprehensive training programs and strong support systems for VA personnel (Gagnon et al., 2006). Enhanced telehealth workflows could promote interdepartmental collaboration, simplify appointment scheduling, and improve compliance with telehealth regulatory standards (Baker & Bufka, 2011). Furthermore, a well-organized telehealth system could assist the VA in addressing operational challenges, such as managing appointment backlogs, alleviating emergency department congestion, and enhancing healthcare accessibility for veterans with mobility limitations (Cartwright et al., 2013).

In summary, this study's implications highlight the transformative potential of a structured telehealth maturity model within the VA. By refining telehealth processes, enhancing patient experiences, and supporting healthcare providers, the VA could achieve greater efficiency, standardization, and scalability in its digital health services. The following sections

further analyzed the study's findings in relation to specific research questions, providing insights into the role of telehealth maturity in the VA healthcare system.

Research Question 1: Ad-Hoc and Chaotic Processes

Findings from Chapter 4 indicated that the initial telehealth processes were highly unstructured, posing significant challenges to scalability, compliance, and, most critically, portability (Carvalho et al., 2019; Chike-Harris et al., 2021). At this stage of telehealth maturity, the lack of standardized processes within the Veterans Administration (VA) results in inconsistent service delivery, operational inefficiencies, and difficulties in maintaining quality across various healthcare facilities. Without a defined framework, individual VA medical centers and outpatient clinics implement telehealth services independently, leading to disjointed workflows and considerable variability in patient experiences.

From an organizational standpoint, the VA's ad hoc approach to telehealth leads to inefficiencies in resource utilization, fragmented technology adoption, and challenges in meeting compliance standards. The lack of centralized guidelines results in various facilities using different platforms, complicating interoperability and data integration. Moreover, without a structured telehealth model, monitoring performance metrics and assessing telehealth effectiveness proves challenging, hindering the VA's ability to implement data-driven improvements (Butzner & Cuffee, 2021).

For patients, the inconsistent implementation of telehealth services creates barriers to accessibility, especially for veterans in rural or underserved areas. Variability in service availability could lead to long wait times, appointment cancellations, or insufficient follow-up care. Furthermore, a lack of uniform telehealth protocols might result in patient dissatisfaction, as some veterans receive high-quality remote care while others encounter technical issues, poor communication with providers, or limited appointment options (Greenhalgh et al., 2012). These

inconsistencies could erode trust in telehealth and diminish patient engagement in digital healthcare solutions.

The challenges related to an ad-hoc telehealth system also affect VA employees, including healthcare providers, administrative staff, and IT personnel. Without standardized workflows, providers must navigate various telehealth platforms, often requiring extra training and adaptation, which could cause frustration and reduce efficiency. Administrative staff might find it difficult to coordinate telehealth appointments, manage documentation effectively, and ensure compliance with ever-changing regulations. This lack of clarity could lead to burnout and resistance to telehealth adoption among VA personnel (Hilty et al., 2018).

From a stakeholder perspective, which includes policymakers, healthcare administrators, and federal oversight agencies, the lack of a structured telehealth framework complicates decision-making and resource allocation. A fragmented telehealth system hinders the ability to assess the return on investment (ROI) and measure the effects of telehealth programs on patient outcomes. Moreover, without standardized implementation metrics, securing funding for telehealth expansion poses challenges, as stakeholders lack the data necessary to justify further investment (Kidholm et al., 2012).

The work environment at the VA was affected by the chaotic nature of ad-hoc telehealth processes. Staff members at different facilities often develop unique approaches to telehealth, leading to disparities in service delivery across VA locations. This inconsistency hinders collaboration between departments, reduces efficiency, and caused frustration among employees who must adapt to ever-changing telehealth workflows. Furthermore, the lack of standardized policies concerning telehealth data security and patient privacy increases the risk of noncompliance with HIPAA regulations, complicating the implementation of telehealth initiatives (Cartwright et al., 2013).

Addressing these issues requires adopting a structured telehealth maturity model that ensures consistent, high-quality care across all VA healthcare facilities. By transitioning from an ad hoc approach to a standardized framework, the VA could improve service reliability, enhance provider efficiency, and foster greater patient trust in telehealth solutions. The following section would explore the next stage of telehealth maturity: managed and repeatable processes.

Research Question 2: Managed and Repeatable Processes

At the managed and repeatable stage of telehealth maturity, the Veterans Administration (VA) starts to implement structured processes that enhance consistency in telehealth service delivery. Unlike the ad-hoc stage, this level showed an effort to standardize procedures, establish protocols, and introduce quality controls that improve telehealth effectiveness (Valenta et al., 2021; Van Cleave et al., 2022). While some variability across VA facilities might still remain, the introduction of repeatable processes lays a foundation for long-term scalability, better patient experiences, and improved integration of telehealth into the overall healthcare system.

From an organizational perspective, the transition to managed and repeatable processes enables the VA to improve operational efficiency by implementing standardized telehealth workflows (Türkmen & Sardoğan, 2023). These workflows ensure that all VA medical centers and outpatient clinics adhere to established guidelines for telehealth consultations, appointment scheduling, and patient follow-ups. By minimizing inconsistencies across locations, the VA could enhance interoperability between facilities and streamline data-sharing practices, resulting in more coordinated patient care. Additionally, the adoption of structured telehealth processes promotes compliance with federal regulations, mitigating the risk of legal and administrative challenges associated with unregulated telehealth practices (Valdez-de-Leon, 2016).

For patients, transitioning to a managed and repeatable telehealth system increases reliability and accessibility (Kidholm et al., 2012). Standardized appointment scheduling cuts

down wait times, ensuring that veterans receive timely medical consultations. Furthermore, structured telehealth procedures improve the quality of virtual healthcare interactions, as providers adhere to best-practice guidelines for remote diagnosis and treatment. Veterans in rural areas especially benefit from enhanced telehealth access, as standardized processes guarantee consistent service availability and a more predictable patient experience. With repeatable workflows established, veterans could anticipate more organized follow-up care, reducing the chances of missed or delayed treatments (Bashshur et al., 2011).

Veterans Administration (VA) employees, including healthcare providers, administrative staff, and IT personnel, benefit significantly from a managed and repeatable telehealth framework. Providers no longer need to navigate inconsistent telehealth platforms or create individualized approaches to remote care. Instead, they could rely on standardized training programs and established telehealth procedures, which reduce the learning curve and enhance efficiency (Wilson et al., 2007). Administrative staff enjoy improved scheduling systems and clear guidelines for managing patient records, while IT personnel could focus on maintaining the telehealth infrastructure with greater predictability. The reduction of workflow inconsistencies boosts job satisfaction and minimizes burnout among VA personnel, fostering a more efficient and organized work environment (Workflows of Telemedicine, 2021).

For stakeholders, including policymakers, healthcare administrators, and federal oversight agencies, adopting managed and repeatable telehealth processes offered a clearer framework for evaluating the effectiveness of telehealth programs (Van Looy et al., 2016). Standardized data collection and performance tracking empower stakeholders to measure telehealth utilization rates, patient outcomes, and overall cost-effectiveness. This data-driven approach fosters informed decision-making, enabling stakeholders to allocate resources strategically and advocate for ongoing telehealth investments. Furthermore, structured telehealth

processes establish a foundation for refining policies and regulations, ensuring that telehealth services conform to best practices in healthcare delivery (Van Dyk & Schutte, 2013).

The work environment at the VA was also experiencing a positive transformation as telehealth maturity advances to a managed and repeatable stage. The establishment of clear protocols promotes collaboration among different departments, reducing silos and encouraging a more integrated approach to patient care (Vargas et al., 2023). Employees gain confidence in telehealth technology as workflows become more predictable, and patient-provider interactions were improved through structured engagement models. Moreover, the managed telehealth framework supports ongoing staff training and professional development, ensuring that VA personnel stay proficient in telehealth practices and emerging digital health solutions (Urick et al., 2023).

Transitioning to managed and repeatable telehealth processes significantly enhances the VA's ability to deliver consistent, high-quality remote healthcare services. By standardizing workflows, improving patient access, supporting VA employees, and strengthening data-driven decision-making for stakeholders, the VA establishes a more reliable and scalable telehealth infrastructure. The following section would explore the next stage of telehealth maturity: defined processes.

Research Question 3: Defined Processes

At the defined stage of telehealth maturity, the Veterans Administration (VA) establishes standardized telehealth processes that ensure a more structured and predictable approach to virtual healthcare delivery. This phase marks a transition from the previous managed and repeatable stage by integrating clearly documented workflows, process control mechanisms, and comprehensive training programs for personnel (Valenta et al., 2021; Van Dyk & Schutte, 2013). The shift to defined processes promotes greater efficiency, improves service consistency, and

aligns better with federal healthcare regulations, ultimately enhancing the overall quality of care provided to veterans.

From an organizational perspective, implementing well-defined telehealth processes enables the VA to achieve operational stability by establishing clear policies and procedures across all healthcare facilities. Standardized telehealth workflows ensure that providers follow consistent protocols for remote consultations, data management, and patient engagement, minimizing variability among VA medical centers and outpatient clinics (Van Cleave et al., 2022). This structured approach also enhances compliance with federal healthcare mandates, reducing legal risks while promoting accountability in service delivery. Furthermore, well-defined telehealth processes support data-driven decision-making by allowing VA administrators to track performance metrics, assess patient outcomes, and identify areas for process optimization (Valdez-de-Leon, 2016).

For patients, the transition to established telehealth processes results in a more seamless and reliable healthcare experience. Standardized service delivery guarantees that veterans receive consistent care, no matter their location, addressing disparities in telehealth access among urban, suburban, and rural populations (Kidholm et al., 2012). With well-documented care protocols in place, veterans enjoy better continuity of care, as healthcare providers follow standardized treatment pathways, which reduces the risk of fragmented or inconsistent care. The introduction of structured telehealth workflows also enhances patient trust and satisfaction by removing uncertainty about virtual healthcare interactions, ultimately improving the overall quality and accessibility of VA telehealth services (Bashshur et al., 2011).

Veterans Administration (VA) employees, including healthcare providers, administrative personnel, and IT specialists, enjoy significant improvements in their work environment due to the implementation of well-defined telehealth processes. Clear and structured telehealth

workflows reduce confusion, ensuring that providers follow consistent guidelines for patient assessments, follow-ups, and documentation (Wilson et al., 2007). Administrative staff benefit from streamlined scheduling systems, enhanced data management practices, and more efficient coordination of telehealth appointments. For IT personnel, clearly defined processes simplify system maintenance and troubleshooting by standardizing telehealth platforms, enhancing system interoperability, and ensuring seamless integration with electronic health records (EHRs). This level of maturity fosters a more efficient, less stressful work environment, which helps reduce burnout among VA personnel and increases job satisfaction (Urlick et al., 2023).

For stakeholders, including policymakers, healthcare administrators, and federal oversight agencies, adopting defined telehealth processes enhances transparency and accountability within the VA's digital healthcare infrastructure. Standardized workflows and process documentation enable stakeholders to assess the effectiveness of telehealth initiatives using measurable performance indicators (Van Looy et al., 2016). This data-driven approach allows policymakers to make informed decisions regarding resource allocation, telehealth program expansion, and regulatory adjustments. Additionally, defined processes promote better collaboration between government agencies and VA leadership, ensuring that telehealth services align with national healthcare goals and best practices in virtual care delivery (Vargas et al., 2023).

The VA work environment was also evolving as telehealth processes become more defined. Clear guidelines minimize ambiguity and foster a culture of consistency, enhancing teamwork and collaboration across departments. The introduction of structured telehealth processes promotes continuous learning and professional development, as VA personnel engage in tailored training programs that adapt to the changing landscape of telehealth technologies and regulatory requirements (Türkmen & Sardoğan, 2023). Additionally, a robust telehealth

framework empowers the VA to innovate effectively, enabling the smooth integration of new telehealth advancements, such as artificial intelligence (AI)-driven diagnostics and remote patient monitoring solutions (Utz et al., 2019).

In summary, the transition to defined telehealth processes represented a crucial step in the VA's digital healthcare evolution. By standardizing workflows, enhancing patient experiences, supporting VA employees, and improving stakeholder decision-making, the VA fortifies its telehealth infrastructure, ensuring long-term sustainability and effectiveness. The following section would examine the next stage of telehealth maturity: quantitative management.

Research Question 4: Quantitative Management

At the quantitative management stage of telehealth maturity, the Veterans Administration (VA) improves its telehealth framework by integrating data-driven decision-making processes. This phase builds on the structured workflows developed in the defined stage by utilizing key performance indicators (KPIs), predictive analytics, and evidence-based methodologies to enhance telehealth service delivery. The emphasis at this level was on employing quantitative metrics to assess operational performance, improve patient outcomes, and ensure that telehealth services align with strategic objectives (Valenta et al., 2021; Van Dyk & Schutte, 2013). The incorporation of data analytics allows the VA to monitor service efficiency, identify trends, and apply continuous improvements that elevate the overall quality of virtual healthcare.

From an organizational perspective, adopting quantitative management practices enables the VA to improve operational transparency and efficiency. By systematically tracking telehealth utilization rates, patient satisfaction scores, and provider performance metrics, VA administrators could make informed decisions regarding resource allocation and service optimization (Van Cleave et al., 2022). The use of data-driven insights also empowers leadership to address inefficiencies in telehealth delivery, ensuring that facilities uphold high standards of care while

reducing operational waste. Additionally, quantitative management facilitates the implementation of benchmarking strategies, enabling the VA to compare performance across facilities and identify best practices for telehealth deployment (Valdez-de-Leon, 2016).

For patients, transitioning to a quantitative management framework results in a more personalized and efficient healthcare experience. Utilizing data analytics allows the VA to predict patient needs, optimize appointment scheduling, and enhance remote monitoring services for veterans with chronic conditions (Kidholm et al., 2012). By leveraging patient-reported outcomes and real-time health data, telehealth providers could customize treatment plans, improving the effectiveness of virtual consultations. Moreover, quantitative tracking of patient engagement and satisfaction allows the VA to proactively address concerns, ensuring that telehealth services remain patient-centered and accessible (Bashshur et al., 2011). These improvements foster greater trust in telehealth solutions, encouraging higher adoption rates among veterans, especially those in rural or underserved areas.

For VA employees, including healthcare providers, administrative personnel, and IT specialists, introducing quantitative management processes boosts efficiency and reduces uncertainty in telehealth operations. Data-driven insights offer healthcare providers objective measures of telehealth effectiveness, allowing them to adjust treatment plans and improve care delivery (Wilson et al., 2007). Administrative staff benefit from automated data reporting tools that streamline performance tracking, alleviating the burden of manual record-keeping and enhancing accuracy in telehealth documentation. IT personnel play a crucial role in maintaining the infrastructure needed for advanced data analytics, ensuring that telehealth platforms were secure, interoperable, and capable of managing large volumes of patient data. By implementing quantitative management practices, the VA fosters a more data-literate workforce, preparing

employees with the skills required to navigate an increasingly digital healthcare environment (Urick et al., 2023).

For stakeholders such as policymakers, healthcare administrators, and federal oversight agencies, quantitative management offered a strong framework for evaluating the success of telehealth programs. The ability to measure telehealth effectiveness with standardized metrics enables stakeholders to assess return on investment (ROI) and make data-driven policy decisions (Van Looy et al., 2016). Quantitative performance tracking also enhances accountability, allowing the VA to demonstrated the impact of telehealth initiatives on patient health outcomes and cost efficiency. Furthermore, real-time analytics assist in continuous quality improvement efforts, helping stakeholders refine telehealth policies and expand programs based on empirical evidence (Vargas et al., 2023). These data-driven strategies bolster the credibility of telehealth services and secure their long-term sustainability within the VA healthcare system.

The VA work environment was also experiencing a significant transformation as telehealth operations become increasingly data-driven. The adoption of quantitative management practices fosters a culture of continuous improvement, encouraging employees to base decision-making on performance metrics (Türkmen & Sardoğan, 2023). The use of predictive analytics enables VA facilities to anticipate patient demand, optimize resource allocation, and mitigate potential service disruptions. Moreover, data transparency enhances interdepartmental collaboration, allowing healthcare teams to align their efforts based on shared performance insights. By incorporating quantitative management into daily operations, the VA establishes a more agile and responsive telehealth ecosystem, capable of adapting to emerging healthcare challenges and technological advancements (Utz et al., 2019).

In summary, the transition to quantitative management marks a pivotal stage in the VA's telehealth maturity journey. By utilizing data analytics, performance tracking, and predictive

modeling, the VA enhances the efficiency of telehealth, improves patient experiences, and promotes evidence-based decision-making across its healthcare network. These advancements ensure that telehealth remains a viable, scalable, and patient-centered solution, supporting the VA's mission to provide high-quality healthcare to veterans.

Research Question 5: Optimized Processes

At the optimized stage of telehealth maturity, the Veterans Administration (VA) fully integrates continuous improvement methodologies, data-driven innovations, and adaptive strategies to enhance the scalability and sustainability of telehealth services. This phase marks a shift from quantitative management to a fully optimized system that employs advanced analytics, automation, and artificial intelligence (AI) to refine telehealth operations. The primary goal at this stage was to achieve a high level of efficiency, proactive healthcare delivery, and system-wide adaptability, ensuring that telehealth services provide maximum benefits to veterans while maintaining cost-effectiveness and compliance with healthcare regulations (Valenta et al., 2021; Van Dyk & Schutte, 2013).

From an organizational perspective, the optimized stage allows the VA to operate as a data-driven, high-reliability healthcare system. With AI-powered predictive analytics and machine learning models, the VA could proactively identify patient needs, optimize resource allocation, and implement preventative healthcare strategies (Van Cleave et al., 2022). The use of automated workflows enhances clinical decision support, alleviating administrative burdens and ensuring that healthcare providers could concentrate on patient-centered care. Furthermore, real-time monitoring of telehealth performance metrics enables VA leadership to swiftly identify inefficiencies and implement targeted interventions, thereby improving operational effectiveness (Valdez-de-Leon, 2016).

For patients, optimizing telehealth services leads to a more proactive and personalized healthcare experience. AI-driven virtual health assistants, remote patient monitoring, and automated diagnostic support tools allow veterans to receive timely medical interventions and ongoing care management, especially for chronic conditions (Kidholm et al., 2012). The use of AI-powered chatbots and digital health applications enhances communication between patients and providers, improving engagement and adherence to treatment plans. Furthermore, optimized telehealth platforms ensure that veterans in remote or underserved areas receive equitable healthcare access, narrowing disparities in service availability and treatment outcomes (Bashshur et al., 2011).

Veterans Administration (VA) employees, including healthcare providers, administrative personnel, and IT specialists, experience a transformed work environment at the advanced telehealth stage. The integration of AI-driven automation reduces manual administrative tasks, which allows healthcare providers to focus more on patient care (Wilson et al., 2007). Administrative staff benefit from smart scheduling and data management systems that decrease errors and improve workflow efficiency. Information Technology (IT) personnel play a vital role in upholding the integrity of the advanced telehealth infrastructure, ensuring that digital health systems remain secure, interoperable, and adaptable to new healthcare technologies. This stage promotes a culture of continuous learning and innovation, enabling VA employees to develop new skills and adapt to rapidly changing digital healthcare solutions (Urick et al., 2023).

For stakeholders such as policymakers, healthcare administrators, and federal oversight agencies, an optimized telehealth system serves as a model for sustainable, technology-driven healthcare transformation. The ability to track real-time performance indicators enables stakeholders to evaluate the long-term impact of telehealth on patient outcomes, cost savings, and system efficiency (Van Looy et al., 2016). Improved telehealth processes also support

evidence-based policymaking, allowing government agencies to refine regulations, allocate resources effectively, and promote national healthcare innovation. Moreover, enhanced cybersecurity measures and compliance protocols ensure that patient data remains secure, reinforcing trust in the VA's telehealth capabilities (Vargas et al., 2023).

The VA work environment was significantly improved at the optimized stage, as telehealth becomes an integral part of daily healthcare operations. The integration of real-time collaboration tools and AI-driven clinical decision support enhances interdisciplinary teamwork, enabling providers to coordinate care more effectively across various facilities (Türkmen & Sardoğan, 2023). The optimized telehealth model promotes a proactive approach to workforce development, equipping VA personnel with advanced training in digital health tools, cybersecurity best practices, and AI-assisted diagnostics. Furthermore, as telehealth reaches an optimized state, the VA was positioned as a leader in healthcare innovation, influencing wider digital transformation initiatives throughout the national healthcare system (Utz et al., 2019).

The shift to an optimized telehealth system marks the culmination of the VA's telehealth maturity journey. The VA enhances telehealth efficiency, boosts patient outcomes, and ensures sustainable digital healthcare delivery by utilizing AI, predictive analytics, and ongoing process improvement strategies. These advancements establish telehealth as a fundamental aspect of veteran healthcare services, reinforcing the VA's commitment to providing high-quality, accessible, and technology-driven care. The following section would delve into recommendations for practice.

Recommendations for Practice

This study's findings provide valuable insights into implementing a telehealth maturity model within the Veterans Administration (VA). These findings suggest several practice recommendations to enhance telehealth service delivery, standardize processes, and promote

long-term sustainability. These recommendations were grounded in the study's findings and framed within the relevant literature discussed in Chapter 2.

One key recommendation was to establish standardized telehealth workflows to ensure consistency in service delivery across VA facilities. This study's findings indicated that the lack of structured processes in the ad hoc and chaotic stage creates inefficiencies and variations in patient care. The literature supports this, as previous research had shown that standardized procedures improve operational efficiency and healthcare quality (Van Dyk & Schutte, 2013). By adopting well-defined telehealth workflows, the VA could minimize service variability, improve interoperability between facilities, and enhance patient outcomes.

A second recommendation was to invest in telehealth training and competency development for VA healthcare providers and administrative staff. The study findings reveal that employees encounter difficulties in navigating telehealth platforms, especially at lower maturity levels. This aligns with research suggesting that structured training programs enhance provider confidence and telehealth adoption rates (Wilson et al., 2007). A comprehensive training framework should be established to equip VA personnel with the essential skills to manage telehealth technologies effectively and to standardize best practices across all locations.

Another crucial recommendation was integrating data-driven decision-making tools at all stages of telehealth maturity. The study findings demonstrated that transitioning from defined processes to quantitative management significantly improves telehealth efficiency and patient engagement. Supporting literature suggests that incorporating key performance indicators (KPIs) and predictive analytics enhances healthcare decision-making and operational oversight (Valenta et al., 2021). The VA should expand its use of real-time analytics, automated performance tracking, and data visualization tools to optimize telehealth service delivery.

Additionally, prioritizing policy refinement and regulatory alignment was essential to tackle challenges related to compliance and legal issues in telehealth implementation. The study findings indicated that discrepancies in telehealth policies among various VA facilities lead to inequalities in service availability and quality. The literature from Chapter 2 highlights the necessity of clear regulatory frameworks to support telehealth expansion while adhering to federal guidelines (Kidholm et al., 2012). The VA should collaborate with policymakers to create standardized telehealth regulations that encourage uniform service delivery while safeguarding patient privacy and security.

Ultimately, nurturing a continuous improvement and innovation culture was crucial for maintaining telehealth maturity. Evidence suggests that at the optimized stage, telehealth services leverage AI-driven automation, real-time collaboration tools, and ongoing advancements in digital health technologies. Research emphasizes the importance of organizations adopting an iterative improvement model to keep up with healthcare's evolving demands (Van Looy et al., 2016). The VA should prioritize investing in research and development initiatives, pilot emerging technologies, and foster cross-disciplinary collaboration to promote innovation in telehealth service delivery.

In summary, these practice recommendations highlight the importance of standardized workflows, focused training programs, data-driven decision-making, policy refinement, and a culture of continuous improvement. By implementing these strategies, the VA could enhance the efficiency of telehealth services, improve patient care, and ensure the long-term sustainability of its telehealth initiatives.

Research Question 1: Ad-Hoc and Chaotic Processes

At the ad hoc and chaotic stage of telehealth maturity, the Veterans Administration (VA) experiences significant inefficiencies due to the lack of standardized processes. This study's

findings indicated that telehealth services at this stage were inconsistent, leading to operational challenges, disparities in patient care, and barriers to scalability. Addressing these challenges requires structured recommendations to improve telehealth service delivery for the VA, its patients, employees, stakeholders, and the overall work environment.

From an organizational perspective, the VA must establish a structured telehealth governance framework that provided clear policies and standard operating procedures. The study findings highlight that the lack of centralized telehealth guidelines leads to variability in service implementation. Research indicated that healthcare organizations gain from well-defined governance structures that promote standardization and compliance with regulatory requirements (Van Dyk & Schutte, 2013). The VA should implement comprehensive telehealth policies across the system to streamline workflow coordination, enhance interoperability between facilities, and improve service consistency.

For patients, enhancing access to telehealth services at the ad-hoc stage necessitates targeted outreach initiatives and patient education programs. The study findings showed that many veterans, especially those in rural areas, encounter challenges in accessing consistent telehealth services. Previous research indicated that patient engagement increases when healthcare organizations offer clear guidance on telehealth usage and enhance digital literacy among patients (Kidholm et al., 2012). The VA should create patient education programs and digital support initiatives to ensure that veterans were well-informed about telehealth options and could effectively navigate virtual healthcare platforms.

Veterans Administration (VA) employees, including healthcare providers and administrative staff, need structured training and support to effectively manage telehealth services. Study findings reveal that employees frequently encounter inconsistent telehealth platforms and lack clear protocols for remote patient care. The literature emphasizes the

necessity for comprehensive telehealth training programs to boost provider confidence and improve workflow efficiency (Wilson et al., 2007). The VA should invest in training programs that equip employees with telehealth competencies, standardize remote care protocols, and offer ongoing technical assistance to ensure smooth service delivery.

For stakeholders, including policymakers and VA administrators, enhancing telehealth implementation during the ad-hoc stage necessitates data-driven decision-making and investment in telehealth infrastructure. The study findings indicated that the absence of structured telehealth processes obstructs scalability and long-term sustainability. Research emphasizes the significance of utilizing healthcare analytics to pinpoint performance gaps and allocate resources effectively (Valenta et al., 2021). The VA should implement data collection mechanisms to evaluate telehealth utilization rates, monitor service quality, and guide strategic decision-making for telehealth expansion.

The VA's work environment must also be restructured to effectively accommodate telehealth integration. The study findings indicated that the absence of telehealth workflow standardization contributes to employee stress and operational inefficiencies. Previous research underscores the benefits of structured work environments in promoting collaboration and reducing provider burnout (Van Looy et al., 2016). The VA should redesign workflows to ensure smoother telehealth coordination, improve interdepartmental communication, and foster a supportive work environment for telehealth adoption.

Addressing the challenges associated with the ad-hoc and chaotic stage of telehealth maturity requires establishing a governance framework, enhancing patient education, investing in employee training, leveraging data-driven decision-making, and improving the work environment. These recommendations would enable the VA to transition from an unstructured

telehealth system to a more standardized and scalable model, ultimately improving healthcare delivery for veterans.

Research Question 2: Managed and Repeatable Processes

At the managed and repeatable stage of telehealth maturity, the Veterans Administration (VA) starts to implement structured telehealth processes, leading to more predictable and scalable service delivery. This stage represented a shift from the initial, unstructured approach to one where telehealth workflows, policies, and practices become more standardized. However, challenges persist, requiring specific recommendations to enhance telehealth service efficiency, patient outcomes, workforce readiness, stakeholder engagement, and the overall work environment.

From an organizational perspective, the VA must continue to refine and reinforce standardized telehealth procedures to enhance consistency in service delivery across all facilities. The study findings indicated that while a certain level of process standardization had been achieved at this stage, inconsistencies still exist among different VA centers. Research showed that healthcare organizations benefit from clearly defined workflows that improve operational efficiency and care coordination (Van Dyk & Schutte, 2013). To tackle this issue, the VA should develop a centralized telehealth operations framework that ensures uniform implementation of telehealth policies, technological integration, and compliance with regulatory guidelines.

For patients, enhancing the accessibility and reliability of telehealth services was crucial at the managed and repeatable stage. The study findings indicated that while telehealth was becoming more available, challenges related to appointment scheduling, service consistency, and patient engagement persist. Prior research showed that structured telehealth systems reduce disparities in access and improve patient satisfaction (Kidholm et al., 2012). The VA should adopt patient-centered telehealth scheduling systems, broaden remote monitoring capabilities,

and create patient feedback mechanisms to ensure ongoing service improvements and better alignment with veterans' healthcare needs.

Veterans Administration (VA) employees, including healthcare providers, administrative staff, and IT specialists, need ongoing training and resources to effectively manage telehealth operations. The study's findings highlight that while employees begin to adapt to standardized telehealth processes, gaps in training and support still exist. The literature emphasizes the importance of equipping healthcare personnel with the necessary skills and technical expertise to efficiently navigate telehealth systems (Wilson et al., 2007). The VA should implement mandatory telehealth competency programs, provide continuous professional development opportunities, and establish telehealth support teams to help providers overcome technical and procedural challenges.

For stakeholders, including VA administrators, policymakers, and federal agencies, leveraging data analytics to guide telehealth expansion was crucial at this stage. The study findings indicated that while telehealth performance tracking had improved, data-driven decision-making was not yet fully integrated into the management process. Research supports implementing key performance indicators (KPIs) and data analytics tools to assess telehealth effectiveness, identify service gaps, and optimize resource allocation (Valenta et al., 2021). The VA should develop a comprehensive telehealth performance dashboard that allows stakeholders to track service utilization, measure patient outcomes, and make informed policy decisions to further enhance telehealth delivery.

The VA work environment must also evolve to support continued telehealth growth. The study findings showed that the shift to managed and repeatable telehealth processes brings operational changes that affect staff workload, interdepartmental collaboration, and workflow efficiency. Research indicated that organizations with well-structured digital healthcare

environments see improved employee satisfaction and reduced burnout (Van Looy et al., 2016).

The VA should improve workflow coordination across departments, incorporate telehealth operations into standard clinical routines, and offer workplace flexibility initiatives such as hybrid telehealth work models to boost employee productivity and job satisfaction.

The managed and repeatable stage of telehealth maturity demands ongoing process standardization, improved patient accessibility, investments in workforce training, data-driven decision-making, and workplace restructuring to enhance telehealth service delivery. These recommendations would ensure that the VA establishes a solid foundation for telehealth scalability, operational efficiency, and long-term sustainability in veteran healthcare services.

Research Question 3: Defined Processes

At the defined stage of telehealth maturity, the Veterans Administration (VA) implements standardized telehealth processes, ensuring greater consistency and efficiency in service delivery. This stage signifies a transition from repeatable workflows to fully documented procedures that enhance service quality and regulatory compliance. Nevertheless, challenges persist, necessitating recommendations to strengthen telehealth infrastructure, improve patient outcomes, support employees, enhance stakeholder engagement, and optimize the work environment.

From an organizational standpoint, the VA must ensure the comprehensive integration of established telehealth policies across all facilities. The study findings indicated that while telehealth procedures were becoming increasingly standardized, inconsistencies in execution still persist. Research highlights that healthcare organizations benefit from thoroughly documented and enforced procedures that enhance coordination and service delivery (Van Dyk & Schutte, 2013). The VA should establish a telehealth quality assurance framework that guarantees adherence to standardized workflows, improves interoperability, and incorporates best practices across all VA medical centers and outpatient clinics.

At the defined stage, ensuring a consistent and high-quality telehealth experience was critical for patients. The study findings suggest that while telehealth adoption was increasing, disparities in service availability and quality remain. Research indicated that structured telehealth frameworks enhance patient trust and satisfaction by providing clear expectations for virtual care (Kidholm et al., 2012). The VA should expand patient education efforts, develop self-service telehealth portals, and introduce telehealth navigators to guide veterans through virtual healthcare processes, ensuring equitable access to high-quality telehealth services.

Veterans Administration (VA) employees, including healthcare providers, administrative staff, and IT specialists, need further support to adapt to well-defined telehealth workflows. The study findings indicated that while staff members were adjusting to the standardization of telehealth, gaps in compliance training and familiarity with workflows persist. Literature supports implementing structured competency programs to boost provider confidence and enhance workflow efficiency (Wilson et al., 2007). The VA should implement ongoing telehealth training modules, competency assessments, and process improvement initiatives to ensure that employees were well-prepared to manage and deliver high-quality telehealth services.

For stakeholders such as VA administrators, policymakers, and regulatory agencies well-defined telehealth processes presented opportunities for data-driven decision-making and performance monitoring. The study's findings indicated that although telehealth data collection had improved, structured performance evaluations remain underutilized. Research supports utilizing key performance indicators (KPIs) and benchmarking tools to gauge the effectiveness of telehealth initiatives and guide future policy adjustments (Valenta et al., 2021). The VA should establish a national telehealth performance review system, allowing stakeholders to evaluate telehealth outcomes, allocate resources strategically, and pinpoint areas for further process optimization.

The VA work environment also requires improvements to support the transition to established telehealth processes. The study findings reveal that while standardizing telehealth enhances service delivery, it also necessitates adjustments to workflows that affect staff workload and interdepartmental coordination. Research indicated that well-defined processes lead to better collaboration and efficiency when backed by appropriate workplace strategies (Van Looy et al., 2016). The VA should improve communication channels between departments, optimize scheduling systems to balance workloads, and incorporate telehealth into regular clinical workflows to foster a more integrated work environment.

The established stage of telehealth maturity requires ongoing refinement of standardized procedures, improved patient education initiatives, workforce competency programs, stakeholder-driven performance monitoring, and workplace adaptations to fully optimize telehealth service delivery. These recommendations would ensure that the VA maintains efficiency, enhances patient experiences, and achieves regulatory compliance while advancing its telehealth maturity model.

Research Question 4: Quantitative Management

At the quantitative management stage of telehealth maturity, the Veterans Administration (VA) utilizes data-driven decision-making to optimize telehealth operations, improve patient outcomes, and enhance workforce efficiency. This stage represented a transition from established processes to a system where performance metrics, predictive analytics, and key performance indicators (KPIs) were employed to drive continuous improvement. However, challenges persist, requiring recommendations to further strengthen telehealth infrastructure, enhance patient care, support employees, engage stakeholders, and optimize the work environment.

From an organizational standpoint, the VA should integrate advanced data analytics tools to ensure systematic monitoring of telehealth performance. The study findings indicated that

while tracking telehealth performance had improved, data-driven decision-making was not yet fully integrated across all VA facilities. Research suggests that utilizing big data analytics and AI-driven insights enhances healthcare efficiency and resource allocation (Valenta et al., 2021). The VA ought to expand its telehealth analytics framework to include predictive modeling, AI-driven diagnostics, and automated reporting systems that facilitate proactive management of telehealth services.

The quantitative management stage enhances telehealth accessibility and care personalization for patients. The study findings showed that while telehealth services were more structured, data-driven patient engagement strategies were underutilized. Research highlights that predictive analytics and personalized health monitoring improve patient adherence and satisfaction (Kidholm et al., 2012). The VA should implement patient-centered telehealth interventions, including AI-powered chatbots, remote patient monitoring systems, and data-driven wellness plans tailored to individual health needs.

Veterans Administration (VA) employees, including healthcare providers, administrative staff, and IT specialists, need training in data interpretation and technology use to fully benefit from quantitative management. The study findings showed that while structured workflows were in place, employees encounter difficulties in leveraging telehealth data for performance improvement. The literature supports the necessity for workforce training programs that enhance data literacy and analytics skills in healthcare (Wilson et al., 2007). The VA should create telehealth analytics training modules, offer ongoing technical support, and implement data-driven performance review processes to enable employees to make informed decisions.

For stakeholders, including policymakers, healthcare administrators, and federal agencies, quantitative management offered an opportunity for evidence-based policy development and resource allocation. The study findings indicated that while telehealth data

collection had expanded, its role in shaping policy and investment strategies remains limited. Research highlights the importance of comprehensive telehealth performance metrics in informing policy decisions and funding allocations (Van Looy et al., 2016). The VA should establish a telehealth data governance framework that allows stakeholders to evaluate service effectiveness, optimize resource distribution, and refine telehealth policies based on empirical evidence.

The VA work environment must adapt to support a fully data-driven telehealth model. The study findings highlight that while telehealth implementation had matured, workflow adjustments were necessary to integrate real-time analytics and predictive insights into daily operations. Research indicated that organizations with data-driven work environments experience improved coordination and efficiency (Türkmen & Sardoğan, 2023). The VA should enhance work environments by embedding real-time telehealth dashboards into provider workflows, improving cross-departmental data sharing, and fostering a culture of continuous process improvement based on measurable outcomes.

Transitioning to quantitative management necessitates further integration of advanced analytics, data-driven interventions centered around patients, workforce training in data utilization, stakeholder-driven policy refinement, and an environment optimized for data. These recommendations would ensure that the VA maximizes telehealth efficiency, enhances patient care, and maintains a data-driven approach to healthcare management.

Research Question 5: Optimized Processes

At the optimized stage of telehealth maturity, the Veterans Administration (VA) fully integrates continuous improvement methodologies, data-driven innovations, and adaptive strategies to enhance the scalability and sustainability of telehealth services. This phase represented the culmination of telehealth process maturity, where AI-driven automation,

predictive analytics, and real-time performance monitoring refine telehealth operations to deliver high-quality, patient-centered care. Despite achieving this advanced level, ongoing refinements were necessary to maximize efficiency, improve patient outcomes, support employees, engage stakeholders, and optimize the work environment.

From an organizational perspective, the VA should foster a culture of continuous learning and technological advancement. The study findings suggest that while telehealth processes were fully optimized, sustaining peak performance necessitates ongoing adaptation to emerging technologies. Research showed that organizations employing continuous improvement models maintain innovation and long-term efficiency (Valenta et al., 2021). To uphold operational excellence, the VA should adopt a dynamic telehealth strategy that integrates AI-driven diagnostics, automated clinical decision support tools, and a continually evolving set of best practices.

For patients, the optimized stage guarantees the highest level of personalized and proactive telehealth care. The study findings showed that patients benefit from AI-powered health monitoring, customized treatment plans, and real-time access to telehealth consultations. Research indicated that integrating AI-assisted decision-making and automated patient engagement tools significantly enhances healthcare delivery and patient satisfaction (Kidholm et al., 2012). The VA should continue to improve telehealth platforms by expanding predictive health analytics, boosting remote patient monitoring capabilities, and ensuring seamless patient-provider interactions to uphold high levels of engagement and care continuity.

Veterans Administration (VA) employees, including healthcare providers, administrative staff, and IT specialists, must continually adapt to the latest telehealth innovations. The study findings indicated that while employees were skilled in telehealth workflows, ongoing professional development was necessary to keep up with evolving digital healthcare tools.

Literature highlights that continuous education and technical training programs were vital for maintaining high-quality healthcare service delivery (Wilson et al., 2007). The VA should offer advanced telehealth certifications, provide access to AI-assisted clinical training modules, and develop mentorship programs to enhance workforce capabilities in an optimized telehealth environment.

For stakeholders including policymakers, healthcare administrators, and federal agencies an optimized telehealth system offered a model for national healthcare transformation. The study findings suggest that, at this stage, data transparency and real-time analytics enable stakeholders to make well-informed decisions regarding resource allocation and policy development. Research supports the integration of evidence-based policy frameworks to ensure that telehealth remains sustainable and adaptable to future healthcare challenges (Van Looy et al., 2016). The VA should establish a national telehealth policy innovation hub that allows stakeholders to test new regulatory models, refine telehealth reimbursement structures, and guarantee alignment with best practices in digital healthcare governance.

The VA work environment should also evolve to maximize the benefits of an optimized telehealth system. The study findings indicated that telehealth had become a seamless part of daily healthcare operations, but further refinements were necessary to enhance workflow efficiency, cross-departmental collaboration, and employee well-being. Research suggests that organizations with highly integrated telehealth models experience increased job satisfaction, reduced burnout, and improved overall workplace performance (Türkmen & Sardoğan, 2023). The VA should implement AI-enhanced workflow automation, promote interdisciplinary teamwork through digital collaboration tools, and adopt ergonomic workplace solutions to support employees in a high-tech telehealth environment.

The optimized stage of telehealth maturity demands a dedication to continuous innovation, advanced patient-centered care, ongoing workforce training, data-driven stakeholder engagement, and a fully integrated telehealth work environment. These recommendations would ensure that the VA remains at the forefront of digital healthcare excellence while preserving efficiency, accessibility, and sustainability in its telehealth initiatives.

Recommendations for Future Research

Future research should continue to explore the evolving landscape of telehealth maturity and its impact on healthcare delivery within the Veterans Administration (VA). While this study had provided a structured evaluation of telehealth processes, additional research was necessary to address existing gaps, validate findings, and enhance the applicability of telehealth maturity models across diverse healthcare environments. Investigating the barriers and facilitators of telehealth adoption at different stages of maturity would further contribute to the development of best practices and policies that enhance telehealth effectiveness and scalability. The following research areas were recommended to advance the understanding and optimization of telehealth services within the VA.

Research Question 1: Ad-Hoc and Chaotic Processes

Future research should explore the factors leading to the lack of structure during the initial stages of telehealth adoption at the VA. The study's findings indicated that the lack of standardized workflows and policies leads to inefficiencies, service inconsistencies, and challenges in scaling telehealth services. Further research could examine the impacts of unstructured telehealth processes on healthcare access, patient outcomes, and provider workload (Kidholm et al., 2012).

From an organizational perspective, future research should explore strategies for developing an effective governance framework that supports the early structuring of telehealth

services. Research could examine the role of leadership, change management strategies, and regulatory policies in transitioning from an ad hoc model to a more structured system.

Comparative studies among VA facilities with different levels of telehealth maturity could offer insights into best practices for addressing early-stage implementation challenges (Van Dyk & Schutte, 2013).

Future studies should explore how unstructured telehealth processes affect patient engagement, appointment accessibility, and continuity of care. Research could evaluate patient experiences during the early stages of telehealth adoption and identify key factors that contribute to hesitancy or dissatisfaction with telehealth. Additionally, studies could examine digital literacy initiatives and patient education strategies designed to enhance telehealth adoption and usability among veterans (Wilson et al., 2007).

Veterans Administration (VA) employees, including healthcare providers and administrative staff, encounter significant operational challenges when telehealth processes lack structure. Future research should investigate how disorganized telehealth workflows impact provider efficiency, job satisfaction, and burnout. Studies could also assess the effectiveness of training programs designed to help employees navigate the early stages of telehealth environments and adapt to changing digital healthcare practices (Valenta et al., 2021).

Future research should examine the long-term effects of delaying structured telehealth implementation for stakeholders, such as policymakers and VA administrators. Analyzing cost impacts, resource allocation strategies, and regulatory compliance challenges could yield valuable insights into the risks associated with adopting unstructured telehealth. Additionally, future research could evaluate the effectiveness of policy interventions aimed at accelerating the transition from an unstructured to a structured telehealth model (Van Looy et al., 2016).

Future studies should also examine the VA work environment to understand how unstructured telehealth processes affect interdepartmental coordination, workflow efficiency, and the integration of digital health technologies. Research could investigate workplace redesign strategies, such as AI-driven scheduling, electronic health record (EHR) optimizations, and improvements to telehealth infrastructure to enhance overall operational effectiveness (Türkmen & Sardoğan, 2023).

Future studies at the VA or similar healthcare organizations should use longitudinal designs to monitor the development of telehealth over time and evaluate the long-term benefits of structured implementation strategies. Additionally, comparative analyses of telehealth adoption in other government healthcare systems or private sector organizations could offer broader insights into best practices for establishing structured telehealth processes.

Research Question 2: Managed and Repeatable Processes.

Future research should investigate how the VA could effectively implement and sustain managed and repeatable telehealth processes to enhance service quality and scalability. The study's findings indicated that while initial process standardization efforts had improved efficiency, there remain gaps in workflow consistency and operational execution. Further research was crucial to identify best practices for bolstering and expanding structured telehealth processes across all VA facilities (Butzner & Cuffee, 2021).

From an organizational standpoint, future research should investigate how the VA could integrate managed telehealth workflows with broader healthcare policies and operational frameworks. Studies could explore how standardized telehealth protocols enhance interoperability among VA medical centers, ensuring consistent service delivery. Furthermore, research should analyzed how leadership engagement and regulatory compliance influence the effectiveness of repeatable telehealth processes (Cartwright et al., 2013).

Future studies should also explore how structured and consistent telehealth processes influence patient access, care continuity, and satisfaction. Research could investigate whether well-managed telehealth workflows improve appointment scheduling, reduce wait times, and enhance adherence to treatment plans. Additionally, studies should examine how consistent telehealth processes impact healthcare disparities, particularly among rural and underserved veteran populations, by ensuring reliable access to virtual healthcare services (Greenhalgh et al., 2012).

Veterans Administration (VA) employees, including healthcare providers, administrative staff, and IT specialists, play a crucial role in implementing managed and repeatable telehealth workflows. Future research should explore how standardizing workflows affects provider efficiency, job satisfaction, and administrative workload. Studies could assess whether telehealth training programs effectively prepare employees to function within structured systems and identify areas where additional training or support was needed (Hilty et al., 2018).

Future research should examine how data-driven insights from managed telehealth processes influence policy formulation and resource allocation for stakeholders such as VA administrators, policymakers, and federal agencies. Studies could evaluate whether performance tracking and key performance indicators (KPIs) enhance the scalability and long-term viability of telehealth programs. Additionally, further investigation could explore the role of funding strategies in expanding managed telehealth workflows across VA facilities while ensuring cost efficiency (Der-Martirosian et al., 2021).

Future research should also investigate how managed telehealth processes affect the VA work environment, especially regarding workflow integration, technology adoption, and cross-departmental collaboration. Studies might explore whether the incorporation of AI-driven automation and advanced telehealth platforms improves workflow efficiency and decreases

operational redundancies. Furthermore, research should assess whether workplace restructuring initiatives, such as optimizing telehealth workflows, lead to better employee collaboration and service delivery (Kho et al., 2020).

Future studies at the VA or similar healthcare institutions should include longitudinal research to assess the long-term effects of managed telehealth workflows on service quality, efficiency, and patient outcomes. Comparative studies between VA facilities and private-sector healthcare organizations could offer insights into best practices for maintaining efficient and repeatable telehealth processes. Moreover, research should examine how emerging technologies, such as AI-driven clinical decision support systems and predictive analytics, could further enhance the effectiveness of structured telehealth workflows.

Research Question 3: Defined Processes

Future research should investigate how the VA could strengthen and refine established telehealth processes to improve service quality, efficiency, and patient outcomes. The study's findings showed that while telehealth processes had achieved a structured and standardized stage, challenges persist in maintaining consistency across VA facilities. Research was necessary to identify best practices for sustaining well-defined workflows that ensure reliable telehealth service delivery (Buntin et al., 2011).

From an organizational standpoint, future studies should assess how the VA could incorporate defined telehealth processes into broader healthcare frameworks while ensuring they remain adaptable to new technologies. Research could examine how regulatory policies, leadership involvement, and healthcare accreditation standards affect the implementation and sustainability of structured telehealth workflows. Additionally, studies could explore the effects of standardization on efficiency, cost-effectiveness, and patient safety across various VA medical centers (Kruk et al., 2018).

Future research should also investigate how well-defined telehealth processes impact patient experiences and healthcare accessibility. Studies could assess whether structured workflows enhance care coordination, minimize appointment delays, and improve patient-provider communication. Additionally, research should examine how digital health literacy initiatives could assist veterans in navigating telehealth platforms and fostering positive patient engagement (Mehrotra et al., 2017).

Veterans Administration (VA) employees, including healthcare providers, administrative personnel, and IT specialists, require continuous training and adaptation strategies to sustain defined telehealth processes. Future studies should assess the effectiveness of ongoing competency-based training programs, workflow optimization strategies, and technology integration support to ensure that employees could efficiently operate within structured telehealth systems. Research could also examine how the shift toward defined telehealth processes affects provider satisfaction and burnout rates (Nilsen et al., 2020).

Future research should evaluate how defined telehealth processes contribute to long-term strategic planning, performance measurement, and resource allocation for stakeholders such as VA administrators, policymakers, and federal agencies. Studies could explore how data-driven policy adjustments based on structured workflows improve service delivery, compliance with federal healthcare mandates, and the sustainability of VA telehealth programs. Additionally, the research could examine whether defined processes facilitate more effective telehealth reimbursement models and funding mechanisms (Ammenwerth et al., 2019).

Future studies should also examine how structured telehealth workflows affect the VA work environment, especially in terms of interdepartmental collaboration, workflow efficiency, and technology adoption. Research could explore how telehealth workflow standardization improves interdisciplinary teamwork and enhances coordination between various departments.

Additionally, studies could investigate how advancements in automation, AI-driven clinical decision support, and remote patient monitoring help to optimize established telehealth processes (Lyles et al., 2020).

Future research at the VA or similar healthcare organizations should focus on longitudinal studies to evaluate the sustainability and effectiveness of structured telehealth workflows over time. Comparative studies between VA facilities and private-sector healthcare organizations could offer insights into best practices for maintaining structured telehealth operations. Additionally, further research should investigate how emerging digital health innovations, such as blockchain for telehealth data security and machine learning algorithms for process optimization, could improve established telehealth workflows in large healthcare systems.

Research Question 4: Quantitative Management.

Future research should examine how the VA could enhance its quantitative management strategies to optimize telehealth service delivery, improve patient outcomes, and boost operational efficiency. The study's findings suggest that although telehealth data collection and analytics had progressed, there remains significant underutilization of performance tracking tools. Research should investigate the most effective methods for incorporating real-time data analysis, predictive modeling, and performance measurement frameworks into the VA's telehealth infrastructure (Bashshur et al., 2011).

From an organizational standpoint, future studies should evaluate how the VA could create a comprehensive telehealth performance measurement system that integrates with electronic health records (EHRs) and other digital health platforms. Research could investigate how predictive analytics and machine learning tools enhance decision-making, enabling VA administrators to allocate resources more effectively, improve service scalability, and decrease

operational inefficiencies (Der-Martirosian et al., 2021). Furthermore, studies could examine how AI-driven automation could streamline administrative processes and clinical workflows within telehealth operations (Kho et al., 2020).

Future research should also explore how enhanced quantitative management affects the quality and accessibility of patient care. Studies could assess whether utilizing performance tracking tools results in better appointment scheduling, fewer delays in care, and improved patient adherence to treatment plans. Additional research should examine how personalized, data-driven care models improve telehealth experiences for veterans, especially those managing chronic conditions or living in rural and underserved areas (Kidholm et al., 2012).

For VA employees, including healthcare providers, administrative staff, and IT specialists, future research should examine how data-driven performance management influences job efficiency and workforce satisfaction. Studies could investigate the effects of real-time analytics dashboards, telehealth performance benchmarks, and automated reporting systems on reducing administrative burdens and enhancing clinical decision-making. Additionally, research should assess the effectiveness of telehealth training programs aimed at improving employee proficiency in utilizing data analytics tools for performance tracking and service optimization (Hilty et al., 2018).

Future research should investigate how quantitative performance tracking informs the development of telehealth policy, allocation of funding, and regulatory compliance for stakeholders, including VA administrators, policymakers, and federal agencies. Studies could examine how data-driven insights affect telehealth reimbursement structures, ensuring that financial resources align with patient outcomes and service utilization rates. Furthermore, research could evaluate how the implementation of a national telehealth performance

benchmarking system fosters continuous quality improvement and accountability in VA telehealth programs (Butzner & Cuffee, 2021).

Future studies should also explore the impact of quantitative management on the VA work environment, particularly in fostering interdepartmental collaboration and workflow optimization. Research could examine how integrating AI-powered decision support tools and blockchain technology for data security enhances the reliability and accuracy of telehealth performance tracking. Additionally, studies could investigate whether real-time performance monitoring strengthens communication among healthcare teams, improving coordination and service efficiency (Valenta et al., 2021).

Future research at the VA and similar healthcare institutions should incorporate longitudinal studies to evaluate the long-term impact of quantitative performance tracking on telehealth service efficiency, provider satisfaction, and patient outcomes. Comparative studies between VA facilities and high-performing private-sector healthcare organizations could provide insights into best practices for optimizing telehealth efficiency through data-driven performance monitoring. Moreover, additional research should examine how emerging technologies, such as AI-assisted analytics and blockchain, could improve the accuracy, security, and transparency of telehealth performance data (Türkmen & Sardoğan, 2023).

Research Question 5: Optimized Processes.

Future research should examine how the VA could sustain and enhance optimized telehealth processes to ensure long-term efficiency, innovation, and high-quality patient care. The study's findings suggest that while telehealth operations had achieved a high level of process maturity, ongoing refinements were essential for maintaining effectiveness and adaptability. Research should explore best practices for leveraging artificial intelligence (AI), automation, and real-time analytics to further optimize telehealth service delivery (Valdez-de-Leon, 2016).

From an organizational perspective, future studies should evaluate how the VA could incorporate emerging technologies into optimized telehealth workflows to improve service efficiency. Research could investigate the effectiveness of AI-driven clinical decision support tools, automated scheduling systems, and predictive analytics in enhancing telehealth operations. Additionally, studies could assess the impact of telehealth optimization on cost savings, resource allocation, and the overall sustainability of digital healthcare programs within the VA (Buntin et al., 2011).

Future research should also investigate how optimized telehealth processes influence patient engagement, satisfaction, and healthcare accessibility. Studies could assess whether real-time monitoring, AI-powered patient interactions, and remote diagnostic tools improve treatment adherence and health outcomes. Further research should also evaluate how enhanced digital health platforms reduce disparities in telehealth access for veterans in rural and underserved areas, ensuring equitable healthcare services (Kruk et al., 2018).

Future research should explore how optimized telehealth processes impact job satisfaction, workload efficiency, and professional development for VA employees, including healthcare providers, administrative staff, and IT specialists. Studies could examine how AI-assisted automation reduces administrative burdens, allowing providers to focus more on direct patient care. Additionally, research should assess how continuous professional development programs prepare employees to adapt to evolving digital health tools and maintain proficiency in optimized telehealth environments (Mehrotra et al., 2017).

Future research should also examine how optimized telehealth operations influence policy development, funding strategies, and regulatory compliance for stakeholders including VA administrators, policymakers, and federal agencies. Studies could investigate how telehealth optimization impacts decision-making processes regarding reimbursement models, technology

investments, and long-term healthcare planning. Furthermore, research should evaluate the effectiveness of incorporating data-driven insights into federal telehealth policies to enhance sustainability and scalability (Ammenwerth et al., 2019).

Future studies should also investigate how optimized telehealth processes transform the VA work environment, particularly in terms of interdepartmental collaboration, workflow efficiency, and technology adoption. Research could examine how AI-powered interoperability solutions improve communication among various healthcare departments, resulting in more cohesive and coordinated care. Additionally, studies should explore whether integrating blockchain technology for secure data management enhances patient privacy and adherence to regulatory standards (Lyles et al., 2020).

Future research at the VA or similar healthcare institutions should include longitudinal studies to assess the long-term impact of telehealth optimization on healthcare delivery, system efficiency, and patient satisfaction. Comparative studies between VA facilities and private-sector healthcare organizations could provide insights into best practices for sustaining optimized telehealth operations. Additionally, research should explore how emerging technologies, such as machine learning-based diagnostics and virtual reality (VR) for remote consultations, could further enhance telehealth service quality and accessibility. By continuing to refine and advance optimized telehealth processes, the VA could position itself as a leader in digital healthcare transformation and ensure that telehealth remains a sustainable and high-performing component of veteran healthcare services.

Conclusion

The findings of this study emphasize the importance of structured telehealth maturity models within the Veterans Administration (VA). This research examined the applicability of the Capability Maturity Model (CMM) framework in the VA's telehealth environment, assessing

whether structured processes improved service efficiency, standardization, and scalability. The study addressed the lack of a defined telehealth maturity model suited to the VA's unique operational and regulatory context. Given the increasing reliance on digital healthcare solutions, ensuring the scalability and sustainability of telehealth services was vital.

The study's findings had significant implications for the VA, its employees, patients, and stakeholders. The results showed that well-defined telehealth workflows enhance patient access, improve provider efficiency, and standardize service delivery across VA facilities. For patients, structured telehealth processes decrease wait times, improve care coordination, and increase accessibility, especially for veterans in rural and underserved areas (Kidholm et al., 2012). Employees gain from streamlined workflows, reduced administrative burdens, and better integration of telehealth technologies into routine healthcare delivery (Nilsen et al., 2020). For stakeholders, including policymakers and VA administrators, these findings provide a framework for informed decision-making regarding telehealth investments, regulatory compliance, and infrastructure development (Buntin et al., 2011). The study emphasizes the need for ongoing evaluation of telehealth processes to ensure continuous improvement and alignment with emerging healthcare trends.

The findings also provide a deeper understanding of the CMM framework's applicability in telehealth maturity. While CMM principles offer a structured approach to process optimization, this study revealed that direct application within the VA setting was complex. The hierarchical nature of CMM maturity levels did not always align seamlessly with the dynamic and evolving landscape of telehealth services. Adjustments to the framework were necessary to accommodate regulatory requirements, technological advancements, and patient-centered care initiatives (Mehrotra et al., 2017). Despite these limitations, the study confirms that telehealth services could achieve increased efficiency, interoperability, and sustainability when guided by a

structured maturity model. Future adaptations of the CMM framework should incorporate telehealth-specific performance metrics and considerations for scalability to enhance its applicability in the VA and similar healthcare institutions (Lyles et al., 2020).

The study contributes to the existing literature by building on prior research regarding telehealth maturity models and digital healthcare transformation. Previous studies had highlighted the importance of structured workflows in telehealth adoption; however, limited research had investigated their impact within large government healthcare systems (Ammenwerth et al., 2019). This study bridges that gap by providing empirical evidence on optimizing telehealth processes within the VA. The findings align with research that advocates for data-driven decision-making, continuous quality improvement, and interoperability as essential components of telehealth scalability (Kruk et al., 2018). Additionally, this study extends the research lineage by stressing the role of structured telehealth maturity models in enhancing service delivery efficiency and patient outcomes. It serves as a critical "brick in the wall" by demonstrating the feasibility of applying process maturity frameworks to digital health transformation in complex healthcare environments (Hilty et al., 2018). Future research should continue to explore telehealth maturity frameworks in diverse healthcare settings to validate their adaptability and long-term benefits further.

In conclusion, this study highlights the importance of structured telehealth processes in improving service delivery, patient care, and operational efficiency within the VA. While the CMM framework offered a solid foundation, adjustments were necessary to tackle the distinct challenges of telehealth implementation in large-scale healthcare systems. The findings of this study guide future telehealth development initiatives, ensuring that telehealth services remain accessible, efficient, and sustainable for veterans. By incorporating continuous improvement

strategies and evidence-based decision-making, the VA could enhance telehealth maturity and create a scalable digital healthcare model that sets a standard for other organizations.

References

- Abd Halim, M., Mohd Foozy, C. F., Brahmi, I., & Mustapha, A. (2018). A review of live survey application: SurveyMonkey and SurveyGizmo. *JOIV: International Journal on Informatics Visualization*, 2(4–2), 309–312.
- Abdel Nasser, A., Mohammed Alzahrani, R., Al-Falah, C. A., Muwafak Jerash, D., Talea Almouled, N., Salem Bakula, D., & Abed Ra'oof, R. (2021). Measuring the patients' satisfaction about telemedicine used in Saudi Arabia during the COVID-19 pandemic. <https://doi.org/10.7759/cureus.13382>. *Cureus*, 13(6), e13382.
- Abdel-Hamid, A. N., & Hamouda, A. E. (2015). Lean CMMI: An iterative and incremental approach to CMMI-based process improvement. *2015 Agile Conference*. <https://doi.org/10.1109/agile.2015.20>
- Aceto, G., Persico, V., & Pescapé, A. (2020). Industry 4.0 and health: Internet of things, big data, and cloud computing for Healthcare 4.0. *Journal of Industrial Information Integration*, 18, 100129. <https://doi.org/10.1016/j.jii.2020.100129>
- Administration, V. H. (2008, April 29), April 29)
- Agboola, S., Hale, T. M., Masters, C., Kvedar, J., & Jethwani, K. (2014). “Real-World” practical evaluation strategies: A review of telehealth evaluation. *JMIR Research Protocols*, 3(4)
- Ahmad, S., Miskon, S., Alabdan, R., & Tlili, I. (2020). Towards sustainable textile and apparel industry: Exploring the role of business intelligence systems in the ERA of Industry 4.0. *Sustainability*, 12(7), 2632
- Akbar, M. A., Leiva, V., Rafi, S., Qadri, S. F., Mahmood, S., & Alsanad, A. (2022). Towards roadmap to implement blockchain in healthcare systems based on a maturity model. *Journal of Software: Evolution and Process*, 34(12).

- Akhlaghpour, S., & Lapointe, L. (2018). From placebo to panacea: Studying the diffusion of its management techniques with ambiguous efficiencies: The case of Capability Maturity Model. *Journal of the Association for Information Systems*, 19(06), 441–502.
- Al-Osail, A. M., al-sheikh, M. H., Al-Osail, E. M., Al-Ghamdi, M. A., Al-Hawas, A. M., Al-Bahussain, A. S., & Al-Dajani, A. A. (2015). *was Cronbach's alpha sufficient for assessing the reliability of the OSCE for an internal medicine course? BMC Research Notes*, 8(1)
- Al-Sai, Z. A., Abdullah, R., & Husin, M. H. (2019). A review on Big Data Maturity models. *2019 IEEE Jordan International Joint Conference on Electrical Engineering and Information Technology (JEEIT)*. <https://doi.org/10.1109/jeeit.2019.8717398>
- Alami, H., Gagnon, M. P., & Fortin, J.-P. (2019). Some multidimensional unintended consequences of Telehealth Utilization: A multi-project evaluation synthesis. *International Journal of Health Policy and Management*, 8(6), 337–352.
- Amoako, R., Jiang, Y., Adu-Yeboah, S. S., Frempong, M. F., & Tetteh, S. (2023). Factors influencing electronic human resource management implementation in public organizations in an emerging economy: An empirical study. *South African Journal of Business Management*, 54(1).
- Andrade, C. (2020). Sample size and its importance in research. *Indian Journal of Psychological Medicine*, 42(1), 102–103.
- Andrews, D., Nonnecke, B., & Preece, J. (2003). Electronic survey methodology: A case study in reaching hard-to-involve internet users. *International Journal of Human-Computer Interaction*, 16(2), 185–210.
- ANOVA as regression. (2009). *Modern regression techniques using R* (pp. 29–47). <https://doi.org/10.4135/9780857024497.d4>

- Anthopoulos, L. (2017). Smart utopia vs smart reality: Learning by experience from 10 Smart City cases. *Cities*, 63, 128–148, 63, 128–148
- Arkwright, B. T., Leslie, M., & Light, M. (2019). Telehealth finance variables and successful business models. *Telehealth and Medicine Today*, 3, 3
- Asah-Kissiedu, M., Manu, P., Booth, C. A., Abdul Majeed Mahamadu, A. M., & Agyekum, K. (2021). An Integrated Safety, Health and Environmental Management Capability Maturity Model for Construction Organizations: A Case Study in Ghana. *Buildings*, 11(2), 1–28
- Bahauddin, K., & Uddin, M. (2012). Prospect of solid waste situation and an approach of environmental management measure (EMM) model for Sustainable Solid Waste Management: Case study of Dhaka City. *Journal of Environmental Science and Natural Resources*, 5(1), 99–111.
- Bai, X., Nath, I., Capon, A., Hasan, N., & Jaron, D. (2012). Health and well-being in the Changing Urban Environment: Complex Challenges, Scientific Responses, and the Way Forward. *Current Opinion in Environmental Sustainability*, 4(4), 465–472
- Baird, S. (2023, February 14). What was the process? Process Model.
<https://www.processmodel.com/blog/what-was-a-process/>
- Baker, D. C., & Bufka, L. F. (2011). Preparing for the Telehealth World: Navigating legal, regulatory, reimbursement, and ethical issues in an electronic age. *Professional Psychology: Research and Practice*, 42(6), 405–411
- Balachandran, B. M., & Prasad, S. (2017). Challenges and benefits of deploying Big Data Analytics in the cloud for business intelligence. *Procedia Computer Science*, 112, 1112–1122, 112, 1112–1122

- Bartlett, J. E., Kotrlik, J. W., & Higgins, C. C. (2001). Organizational Research: Determining Appropriate Sample Size in Survey Research. *Information Technology, Learning, and Performance Journal*, 19(1), 43–50.
- Bashshur, R., Shannon, G., Krupinski, E., & Grigsby, J. (2011a). The taxonomy of Telemedicine. *Telemedicine and E-Health*, 17(6), 484–494
- Bashshur, R., Shannon, G., Krupinski, E., & Grigsby, J. (2011b). The taxonomy of Telemedicine. *Telemedicine and E-Health*, 17(6), 484–494
- Basile, L. J., Carbonara, N., Pellegrino, R., & Panniello, U. (2023). Business intelligence in the healthcare industry: The utilization of a data-driven approach to support clinical decision making. *Technovation*, 120, 102482, 120, 102482
- Bauer, M., Bräuer, C., Schuldt, J., & Krömker, H. (2018). Adaptive e-learning technologies for sustained learning motivation in engineering science - acquisition of motivation through self-reports and wearable technology. *Proceedings of the 10th International Conference on Computer Supported Education*. <https://doi.org/10.5220/0006787104180425>.
- Becker, J., Knackstedt, R., & Pöppelbuß, J. (2009). Developing maturity models for IT management. *Business & Information Systems Engineering*, 1(3), 213–222
- Bellido-Contreras, S. I., Diaz Romani, M. C., Unchupaico-Huaringa, J. C., Guevara-Jimenez, J. A., & Moggiano, N. (2022). *Development of a web application to improve price quotation of pledge items in San Martín de Pangoa City, Peru*. Universidad Continental. <https://cris.continental.edu.pe/en/publications/development-of-a-web-application-to-improve-price-quotation-of-pl>
- Bhandari, P. (2023, June 22). Statistical Power and why it matters: A simple introduction. Scribbr. <https://www.scribbr.com/statistics/statistical-power/>

- Bhardwaj, R. K. (2017). Academic Social Networking Sites. *Information and Learning Science*, 118(5/6), 298–316
- Bhaskar, S., Tan, J., Bogers, M. L., Minssen, T., Badaruddin, H., Israeli-Korn, S., & Chesbrough, H. (2020). At the epicenter of COVID-19, the tragic failure of the global supply chain for medical supplies, the tragic failure of the global supply chain for medical supplies. *Health and Technology*, 10(3), 813–823.
<https://doi.org/10.1007/s12553-020-00429-2>
- Bifulco, P., Narducci, F., Vertucci, R., Ambruosi, P., Cesarelli, M., & Romano, M. (2014). Telemedicine supported by augmented reality: An interactive guide for untrained people in performing an ECG test. *BioMedical Engineering OnLine*, 13(1), 153
- Blondiau, A., Mettler, T., & Winter, R. (2016). Designing and implementing maturity models in hospitals: An experience report from 5 years of research. *Health Informatics Journal*, 22(3), 758–767.
- Boell, S. K., & Cecez-Kecmanovic, D. (2014). A hermeneutic approach for conducting literature reviews and literature searches. *Communications of the Association for Information Systems*, 34, 34
- Bollinger, T., & McGowan, C. (2009). A critical look at software capability evaluations: An update. *IEEE Software*, 26(5), 80–83.
- Bonett, D. G., & wright, T. A. (2014). Cronbach's alpha reliability: Interval estimation, hypothesis testing, and sample size planning. *Journal of Organizational Behavior*, 36(1), 3–15.
- Bowen, H. P., & Wiersema, M. F. (1999). Matching method to paradigm in strategy research: Limitations of cross-sectional analysis and some methodological alternatives. *Strategic Management Journal*, 20(7), 625–636.

- Braithwaite, J., Westbrook, M. T., Robinson, M., Michael, S., Pirone, C., & Robinson, P. (2011). Improving patient safety: The comparative views of patient-safety specialists, workforce staff and managers. *BMJ Quality & Safety*, 20(5), 424–431.
- Brinkman, L. N., Saeed, M. S., Beck, A. F., Ponti-Zins, M. C., Unaka, N. I., Burkhardt, M. C., Meinen-Derr, J., & Hanke, S. P. (2021). Sociodemographic characteristics and patient and family experience survey response biases. *Patient Experience Journal*, 8(2), 18–25.
- Brooks, P., El-Gayar, O., & Sarnikar, S. (2013). Towards a business intelligence maturity model for Healthcare. *2013 46th Hawaii International Conference on System Sciences*.
<https://doi.org/10.1109/hicss.2013.560>.
- Brooks, P., El-Gayar, O., & Sarnikar, S. (2015). A framework for developing a domain specific business intelligence maturity model: Application to healthcare. *International Journal of Information Management*, 35(3), 337–345.
- Bujang, M. A., Sa'at, N., Tg Abu Bakar Sidik, T. M., & Chien Joo, L. (2018). Sample size guidelines for logistic regression from observational studies with large population: Emphasis on the accuracy between statistics and parameters based on real-life clinical data. *Malaysian Journal of Medical Sciences*, 25(4), 122–130.
- Bunker, G., & Thomson, D. (2005). *Delivering Utility Computing*.
<https://doi.org/10.1002/0470017880>
- Butzner, M., & Cuffee, Y. (2021). Telehealth interventions and outcomes across rural communities in the United States: Narrative review. *Journal of Medical Internet Research*, 23(8).
- Bak, S., & Jedynak, P. (2022). Development of a new risk management maturity assessment model. *Risk Management Maturity*, 45–56, 45–56

- Cao, J., Chow, J. C., Lee, F. S. C., & Watson, J. G. (2013). Evolution of PM_{2.5} measurements and standards in the U.S. and future perspectives for China. *Aerosol and Air Quality Research*, 13(4), 1197–1211
- Cao, Y., Jia, F., & Manogaran, G. (2020). Efficient traceability systems of steel products using blockchain-based industrial internet of things. *IEEE Transactions on Industrial Informatics*, 16(9), 6004–6012.
- Caralli, R., Knight, M., & Montgomery, A. (2012). *Maturity Models 101: A Primer for Applying Maturity Models to Smart Grid Security, Resilience, and Interoperability, Resilience, and Interoperability*
- Cartwright, M., Hirani, S. P., Rixon, L., Beynon, M., Doll, H., Bower, P., Bardsley, M., Steventon, A., Knapp, M., Henderson, C., Rogers, A., Sanders, C., Fitzpatrick, R., Barlow, J., & Newman, S. P. (2013). Effect of telehealth on quality of life and psychological outcomes over 12 months (Whole Systems Demonstrator telehealth questionnaire study): *NESTED study of patient-reported outcomes in a pragmatic, cluster randomized controlled trial*. *BMJ Open*, 3(6), e003019. <https://doi.org/10.1136/bmjopen-2013-003019>
- Carvalho, J. V., Rocha, A., & Abreu, A. (2016). Information systems and technologies maturity models for Healthcare: A systematic literature review. *New Advances in Information Systems and Technologies*, 83–94, 83–94
- Carvalho, J. V., Rocha, A., Van de Wetering, R., & Abreu, A. (2019). A maturity model for hospital information systems. *Journal of Business Research*, 94, 388–399, 94, 388–399.
- Carvalho, J. V., Rocha, A., Vasconcelos, J., & Abreu, A. (2019). A health data analytics maturity model for hospitals information systems. *International Journal of Information Management*, 46, 278–285, 46, 278–285.

- Chang, G., Guida, W. C., & still, W. C. (1989). An internal-coordinate Monte Carlo method for searching conformational space. *Journal of the American Chemical Society*, 111(12), 4379–4386.
- Chang, H. (2015). Evaluation framework for telemedicine using the Logical Framework Approach and a fishbone diagram. *Healthcare Informatics Research*, 21(4), 230
- Chike-Harris, K. E., Durham, C., Logan, A., Smith, G., & Dubose-Morris, R. (2021). Integration of telehealth education into the Health Care Provider Curriculum: A Review. *Telemedicine and E-Health*, 27(2), 137–149
- Chuah, M.-H., & Wong, K.-L. (2012). Construct an enterprise business intelligence maturity model (EBI2M) using an integration approach: A conceptual framework. *Business Intelligence - Solution for Business Development*. <https://doi.org/10.5772/35457>
- CMMI. (2023). CMMi Institute. CMMI Institute - What was CMMI? <https://cmmiinstitute.com/cmmi/intro>
- Coker, D. C. (2023). A thematic analysis of the structure of delimitations in the dissertation. *InSite Conference*. <https://doi.org/10.28945/5093>.
- Constantinescu, R., & Iacob, I. M. (2007). Capability Maturity Model Integration. *Journal of Applied Quantitative Methods*, 2(1), 114–121. https://www.jaqm.ro/issues/volume-2/issue-1/pdfs/constantinescu_iacob.pdf
- Corsi, P., & Neau, E. (2015). A recap of the five innovation capability maturity levels. *Innovation Capability Maturity Model*, 205–209, 205–209
- Couckuyt, D., & Van Looy, A. (2021). An exploration of green business process maturity based on Ecolabels. *Business Process Management Journal*, 27(7), 1999–2020.

- Craighead, C. W., Ketchen, D. J., & Darby, J. L. (2019). Taking aim at theoretical and pragmatic impact: A call for actionable, insightful, and measurable research. *Journal of Business Logistics*, 40(4), 289–298.
- Croteau, A.-M., & Vieru, D. (2002). Telemedicine adoption by different groups of Physicians. *Proceedings of the 35th Annual Hawaii International Conference on System Sciences*. <https://doi.org/10.1109/hicss.2002.994121>.
- Cummins, F. A. (2017). Next-generation Business Process Management (BPM). *Building the Agile Enterprise*, 115–154, 115–154
- Cunningham, C. T., Quan, H., Hemmelgarn, B., Noseworthy, T., Beck, C. A., Dixon, E., Samuel, S., Ghali, W. A., Sykes, L. L., & Jetté, N. (2015). Exploring physician specialist response rates to web-based surveys. *BMC Medical Research Methodology*, 15(1)
- Curtis, B., Hefley, B., & Miller, S. (2009). *People Capability Maturity Model (P-CMM) version 2.0* (2nd ed.). Software Engineering Institute, Carnegie Mellon University.
- Curtis, B., Hefley, W. E., & Miller, S. A. (2001). *People Capability Maturity Model (P-CMM) version 2.0* (CMU/SEI-2001-MM-001). Software Engineering Institute, Carnegie Mellon University. <https://doi.org/10.21236/ADA388676>
- Dahlin, G. (2020). What could we learn from process maturity models – a literature review of models addressing process maturity. *International Journal of Process Management and Benchmarking*, 10(4), 495.
- Dangmei, J. (2017). People capability maturity model (P-CMM) facing the key HR challenges of MSME in India: A theoretical approach. *Asian Journal of Management*, 8(2), 132.
- Daniel, J. (2012). *Sampling Essentials: Practical guidelines for making sampling choices*. <https://doi.org/10.4135/9781452272047>

- Das, S., Rohilla, V., Vijarana, Dr. M., & Arora, Dr. Y. (2020). Two-step verification and data security through the network. *International Journal of Innovative Research in Computer Science & Technology*, 8(3).
- David Patón-Romero, J., Baldassarre, M. T., Rodríguez, M., & Piattini, M. (2019). Maturity model based on CMMI for governance and management of green it. *IET Software*, 13(6), 555–563
- Davis, S. M., Jones, A., Jaynes, M. E., Woodrum, K. N., Canaday, M., Allen, L., & Mallow, J. A. (2020). Designing a multifaceted telehealth intervention for a rural population using a model for developing complex interventions in nursing. *BMC Nursing*, 19(1)
- De Keyser, A., Köcher, S., Alkire (Née Nasr), L., Verbeeck, C., & Kandampully, J. (2019). Frontline Service Technology Infusion: Conceptual Archetypes and future research directions. *Journal of Service Management*, 30(1), 156–183.
- de Sousa Pereira, R., & da Silva, M. M. (2010). A maturity model for implementing ITIL V3. *2010 6th World Congress on Services*. <https://doi.org/10.1109/services.2010.80>
- Deakin University. (2023). Libguides: Quantitative Study Designs: *Cross-sectional studies*. *Cross-sectional studies – Quantitative study designs*. LibGuides. <https://deakin.libguides.com/quantitative-study-designs/cross-sectional>
- Der-Martirosian, C., Wyte-Lake, T., Balut, M., Chu, K., Heyworth, L., Leung, L., Ziaecian, B., Tubbesing, S., Mullur, R., & Dobalian, A. (2021a). Implementation of telehealth services at the US Department of Veterans Affairs during the COVID-19 pandemic: Mixed Methods Study. *JMIR Formative Research*, 5(9)
- Der-Martirosian, C., Wyte-Lake, T., Balut, M., Chu, K., Heyworth, L., Leung, L., Ziaecian, B., Tubbesing, S., Mullur, R., & Dobalian, A. (2021b). Implementation of telehealth services

- at the US Department of Veterans Affairs during the COVID-19 pandemic: Mixed Methods Study. *JMIR Formative Research*, 5(9)
- Derby-Davis, M. J. (2014). Predictors of nursing faculty's job satisfaction and intent to stay in academia. *Journal of Professional Nursing*, 30(1), 19–25.
- DeVon, H. A., Block, M. E., Moyle-Wright, P., Ernst, D. M., Hayden, S. J., Lazzara, D. J., Savoy, S. M., & Kostas-Polston, E. (2007). A Psychometric Toolbox for Testing Validity and Reliability. *Journal of Nursing Scholarship*, 39(2), 155–164.
- Dimitrov, D. V. (2016). Medical internet of things and Big Data in Healthcare. *Healthcare Informatics Research*, 22(3), 156
- Donati, M., Celli, A., Ruiiu, A., Saponara, S., & Fanucci, L. (2019). A telemedicine service system exploiting BT/BLE wireless sensors for remote management of chronic patients. *Technologies*, 7(1), 13
- Dooley, K., Subra, A., & Anderson, J. (2001). Maturity and its impact on New Product Development Project Performance. *Research in Engineering Design*, 13(1), 23–29
- Doss, B. D., Feinberg, L. K., Rothman, K., Roddy, M. K., & Comer, J. S. (2017). Using technology to enhance and expand interventions for couples and families: Conceptual and methodological considerations. *Journal of Family Psychology*, 31(8), 983–993.
- Doss, D. A. (2004). *An investigation of adapting the software capability maturity model architecture and framework within traditional industrial environments* (dissertation). ProQuest Dissertation Publishing, Prescott, AZ.
- Doss, D. A. (2014). *The capability maturity model as a criminal justice process improvement paradigm* [Doctoral dissertation, Northcentral University]. ProQuest Dissertations Publishing.

- Doss, D. A., & Kamery, R. (2005). The capability maturity model: A valid architecture to support a baseline environmental management maturity model. *In Proceedings of the Academy of Information and Management Sciences*, 9(2) (pp. 1-10).
- Doss, D. A., Chen, I. C., & Holland, L. D. (2008). A proposed variation of the capability maturity model framework among financial management settings. *Proceedings of the Academy of Accounting and Financial Studies* 13(1)
- Doss, D. A., Goza, R., Tesiero, R., Gokaraju, B., & McElreath, D. H. (2017). The capability maturity model as an industrial process improvement model. *Manufacturing Science and Technology*, 4(2), 17–24
- Doss, D. A., Henley, R., Hong, Q., & Pickett, T. (2019). Investigating process maturity modeling as an advertising process improvement paradigm. *Marketing of Scientific and Research Organizations*, 32(2), 1–26
- Doss, D. A., Henley, R., McElreath, D. H., Mallory, S. L., Gokaraju, B., Tesiero, R., Hong, Q., & Taylor, L. N. (2021a). The capability maturity model integrated as a market engineering maturity model. *International Journal of Service Science, Management, Engineering, and Technology*, 12(3), 175–196.
- Doss, D. A., Henley, R., McElreath, D. H., Mallory, S. L., Gokaraju, B., Tesiero, R., Hong, Q., & Taylor, L. N. (2021b). The capability maturity model integrated as a market engineering maturity model. *International Journal of Service Science, Management, Engineering, and Technology*, 12(3), 175–196.
- Doss, D. A., Henley, R., McElreath, D., Gokaraju, B., Lin, S., Lusk, G., & Goza, R. (2017, March). Managerial perspectives of the integrated capability maturity model among law enforcement personnel. *In Proceedings of the Southwest Academy of Management* (Vol. 2017, pp. 24-33).

- Doss, D. A., & Kamery, R. H. (2006). A study of the benefits of adapting software process improvement frameworks and architectures in traditional settings. In *Allied Academies International Conference. Academy of Educational Leadership. Proceedings* (Vol. 11, No. 1, p. 167). Jordan Whitney Enterprises, Inc.
- Doss, D. A., McElreath, D., Henley, R., Goza, R., Lusk, G., & Gokaraju, B. (2017). Process improvement and Total Quality Management for justice system organizations. In Southwest Academy of Management. *CONGRATULATIONS!*, 34.
- Doss, D. A., Tesiero, R., Gokaraju, B., & McElreath, D. (2018). Urban versus rural perspectives of the integrated Capability Maturity Model (CMMi) as a maturity framework for police organizations. In *Joint Conference of the Academic Business World International Conference and the International Conference on Learning and Academic Administration*. Nashville, TN. May 24-25..
- Drake, C., Lian, T., Cameron, B., Medynskaya, K., Bosworth, H. B., & Shah, K. (2022). Understanding telemedicine's "New normal": Variations in telemedicine use by Specialty Line and patient demographics. *Telemedicine and E-Health*, 28(1), 51–59
- Duggineni, S. (2023). Impact of Controls on Data Integrity and Information Systems. *Science and Technology*, 13(2), 29–35.
- Dunn, C. S., & Austin, E. W. (1998). Protecting confidentiality in archival data resources. *IASSIST Quarterly*, 22(2), 16
- Durach, C. F., Blesik, T., Düring, M., & Bick, M. (2020). Blockchain applications in supply chain transactions. *Journal of Business Logistics*, 42(1), 7–24.
- Dávalos, M. E., French, M. T., Burdick, A. E., & Simmons, S. C. (2009). Economic evaluation of telemedicine: Review of the literature and research guidelines for benefit–cost analysis. *Telemedicine and E-Health*, 15(10), 933–948

- Edmonds, W. A., & Kennedy, T. D. (2016). *An Applied Guide to Research Designs Quantitative, Qualitative, and Mixed Methods*. Sage Publications.
- Elbashir, M. Z., Collier, P. A., & Davern, M. J. (2008). Measuring the effects of business intelligence systems: The relationship between business process and organizational performance. *International Journal of Accounting Information Systems*, 9(3), 135–153.
- Ellimoottil, C., An, L., Moyer, M., Sossong, S., & Hollander, J. E. (2018). Challenges and opportunities faced by large health systems implementing Telehealth. *Health Affairs*, 37(12), 1955–1959
- Enago Academy. (2020, November 17). *Consider publishing in Taylor & Francis Journals – here's why*. <https://www.enago.com/academy/why-consider-publishing-in-taylor-francis-journals/>
- Englander, L. (2018). Telehealth and digital technology. *Telehealth and Medicine Today*, 1(1)
- Epstein, R. M., & Peters, E. (2009). Beyond information. *JAMA*, 302(2), 195
- Esdar, M. (2022). The relationship between innovation capabilities, information management, and HIT Quality in healthcare organizations: *Development of a Sociotechnical Model*. (Dr. Rer. Medic.) from the School of Human Sciences at Osnabrück University.
- Eysenbach, G., & Wyatt, J. (2002). Using the internet for surveys and Health Research. *Journal of Medical Internet Research*, 4(2).
- Fabbro, E., & Tonchia, S. (2021). Project management maturity models: Literature review and new developments. *The Journal of Modern Project*, 8(3), 31–45.
- Facchini, F., Oleśków-Szłapka, J., Ranieri, L., & Urbinati, A. (2019). A maturity model for Logistics 4.0: An empirical analysis and a roadmap for future research. *Sustainability*, 12(1), 86

- Faiola, A., Papautsky, E. L., & Isola, M. (2019). Empowering the aging with Mobile Health: A mHealth Framework for supporting Sustainable Healthy Lifestyle Behavior. *Current Problems in Cardiology*, 44(8), 232–266
- Fan, Y., Liu, S., Liu, J., Javed, S. A., & Fang, Z. (2020). Habit or utility: A key choice point in promoting the adoption of telehealth in China. *Complexity*, 2020, 1–11, 2020, 1–11
- Far. Subpart 16.5 - Indefinite-delivery contracts | acquisition.gov. (2023, December 4), December 4)
- Farkas, K. J., & Romaniuk, J. R. (2020). *Social Work*, ethics and vulnerable groups in the time of coronavirus and covid-19, ethics and vulnerable groups in the time of coronavirus and covid-19
- Farrell, M., & Gallagher, R. (2014). The valuation implications of enterprise risk management maturity. *Journal of Risk and Insurance*, 82(3), 625–657.
- Fatehi, F., Smith, A. C., Maeder, A., Wade, V., & Gray, L. C. (2016). How to formulate research questions and Design Studies for Telehealth Assessment and Evaluation. *Journal of Telemedicine and Telecare*, 23(9), 759–763.
- Felden, C., Koschtial, C., & Chamoni, P. (2015). Towards an evaluation framework to structure business intelligence project patterns as enhancement of Business Intelligence Maturity models. *Business Intelligence for New-Generation Managers*, 67–76, 67–76
- Ferguson, S. M., Olewnik, A. T., & Cormier, P. (2013). A review of mass customization across marketing, engineering and distribution domains toward development of a process framework. *Research in Engineering Design*, 25(1), 11–30
- Ferraiolo, D., & Barkley, J. (1997). Specifying and managing role-based access control within a corporate intranet. *Proceedings of the Second ACM Workshop on Role-Based Access Control - RBAC '97*. <https://doi.org/10.1145/266741.266761>

- Fitterer, R., & Rohner, P. (2009). Towards assessing the Networkability of health care providers: A maturity model approach. *Information Systems and E-Business Management*, 8(3), 309–333
- Forza, C. (2002). Survey research in operations management: A process-based perspective. *International Journal of Operations & Production Management*, 22(2), 152–194.
- Foster, P., & Hoult, S. (2013). The safety journey: Using a safety maturity model for safety planning and assurance in the UK coal mining industry. *Minerals*, 3(1), 59–72
- Franke, T. M., Ho, T., & Christie, C. A. (2011). The chi-square test. *American Journal of Evaluation*, 33(3), 448–458.
- Gagnon, M.-P., Duplantie, J., Fortin, J.-P., & Landry, R. (2006). *Implementing telehealth to support medical practice in rural/remote regions: What were the conditions for success?* *Implementation Science*, 1(1)
- Gajarawala, S. N., & Pelkowski, J. N. (2021). Telehealth benefits and barriers. *The Journal for Nurse Practitioners*, 17(2), 218–221.
- Gajdzik, B. (2022). Frameworks of the maturity model for Industry 4.0 with an assessment of maturity levels on the example of the segment of Steel Enterprises in Poland. *Journal of Open Innovation: Technology, Market, and Complexity*, 8(2), 77.
- Gallagher, B. P. (2002). Interpreting Capability Maturity Model Integration (CMMI) for *Operational Organizations*. <https://doi.org/10.21236/ada401709>
- Galup, S. D., Dattero, R., Quan, J. J., & Conger, S. (2009). An overview of IT service management. *Communications of the ACM*, 52(5), 124–127
- Gama, N., Nunes da Silva, R., & Mira da Silva, M. (2013). Using people-CMM for diminishing resistance to Ital. *Enhancing the Modern Organization through Information Technology Professionals*, 83–98, 83–98

- Gammon, D., Berntsen, G. K., Koricho, A. T., Sygna, K., & Ruland, C. (2015). The chronic care model and technological research and innovation: A scoping review at the Crossroads. *Journal of Medical Internet Research*, 17(2).
- Gan, R. C., & Chin, C. M. (2019). Project Management Guide and Project Management Maturity Models as generic tools capable for diverse applications. *Diverse Applications and Transferability of Maturity Models*, 269–315, 269–315
- Ganzarain, J., & Errasti, N. (2016). Three stage maturity model in SMEs toward Industry 4.0. *Journal of Industrial Engineering and Management*, 9(5), 1119.
- Gao, H., Yous, M. L., Connelly, D., Hung, L., Garnett, A., Hay, M., & Snobelen, N. (2023). Implementation and impacts of virtual team-based care planning for older persons in formal care settings: A scoping review. *Digital health*, 9, 20552076231151567.
- Gessl, A. S., Schlögl, S., & Mevenkamp, N. (2019). On the perceptions and acceptance of artificially intelligent robotics and the psychology of the future elderly. *Behavior & Information Technology*, 38(11), 1068–1087
- Ghazal, T. M., Soomro, T. R., & Shaalan, K. (2013). Integration of project management maturity (PMM) based on Capability Maturity Model Integration (CMMI). *ResearchGate*.
https://www.researchgate.net/publication/236201657_Integration_of_Project_Management_Maturity_PMM_based_on_Capability_Maturity_Model_Integration_CMMI
- Gibson, D. L., Goldenson, D. R., & Kost, K. (2006). Performance Results of CMMI-Based Process Improvement. *Software Engineering Institute*, 8.
<https://doi.org/10.21236/ada454687>
- Gibson, M. C., Arnott, D. R., & Jagielska, I. (1970). Evaluating the intangible benefits of business intelligence: Review & research agenda. *Monash University*.

<https://research.monash.edu/en/publications/evaluating-the-intangible-benefits-of-business-intelligence-revie>

- Gill, F. J., Leslie, G. D., Grech, C., & Latour, J. M. (2013). Using a web-based survey tool to undertake a Delphi Study: Application for Nurse Education Research. *Nurse Education Today*, 33(11), 1322–1328
- Gitlin, A., & Russell, R. (2014). Alternative methodologies and the research context. *Power and method* (pp. 181–202). Routledge.
- Goldenson, D. R., & Gibson, D. L. (2003). Demonstrating the Impact and Benefits of CMMI (Trademark): An Update and Preliminary Results. <https://doi.org/10.21236/ada418481>
- Gomes, J., & Romão, M. (2018). Information system maturity models in healthcare. *Journal of Medical Systems*, 42(12).
- Gonçalves, R. L., Pagano, A. S., Reis, Z. S., Brackstone, K., Lopes, T. C., Cordeiro, S. A., Nunes, J. M., Afagbedzi, S. K., Head, M., Meira Jr, W., Batchelor, J., & Ribeiro, A. L. (2023). Usability of Telehealth Systems for noncommunicable diseases in primary care from the COVID-19 pandemic onward: Systematic review. *Journal of Medical Internet Research*, 25, 25.
- Grant, K. P., & Pennypacker, J. S. (2006). Project Management Maturity: An Assessment of project management capabilities among and between selected industries. *IEEE Transactions on Engineering Management*, 53(1), 59–68.
- Green, L. A., Lowery, J. C., Kowalski, C. P., & Wyszewianski, L. (2005). Impact of Institutional Review Board practice variation on observational health services research. *Health Services Research*, 41(1), 214–230
- Greenhalgh, T., Procter, R., Wherton, J., Sugarhood, P., & Shaw, S. (2012). The organizing vision for telehealth and telecare: Discourse analysis: Table 1. *BMJ Open*, 2(4)

- Gregoski, M. J., Mueller, M., Vertegel, A., Shaporev, A., Jackson, B. B., Frenzel, R. M., Sprehn, S. M., & Treiber, F. A. (2012). Development and validation of a smartphone heart rate acquisition application for Health Promotion and Wellness Telehealth Applications. *International Journal of Telemedicine and Applications*, 2012, 1–7, 2012, 1–7.
- Gresse von Wangenheim, C., von Wangenheim, A., McCaffery, F., Hauck, J. C., & Buglione, L. (2013). Tailoring software process capability/maturity models for the Health Domain. *Health and Technology*, 3(1), 11–28
- Grewal, A., Kataria, H., & Dhawan, I. (2016). Literature Search for Research Planning and identification of research problem. *Indian Journal of Anaesthesia*, 60(9), 635.
- Gupta, B., Goul, M., & Dinter, B. (2015). Business intelligence and big data in higher education: Status of a multi-year model curriculum development effort for Business School undergraduates, MS graduates, and MBAs. *Communications of the Association for Information Systems*, 36, 36
- Gusenbauer, M., & Haddaway, N. R. (2020). Which academic search systems were suitable for systematic reviews or meta-analyses? Evaluating retrieval qualities of Google Scholar, PubMed, and 26 other resources. *Research Synthesis Methods*, 11(2), 181–217.
<https://doi.org/10.1002/jrsm.1378>
- Guzik, A. K., Martin-Schild, S., Tadi, P., Chapman, S. N., Al Kasab, S., Martini, S. R., Meyer, B. C., Demaerschalk, B. M., Wozniak, M. A., & Southerland, A. M. (2021). Telestroke across the continuum of care: Lessons from the COVID-19 pandemic. *Journal of Stroke and Cerebrovascular Diseases*, 30(7), 105802.
- Gökalp, M. O., Gökalp, E., Kayabay, K., Koçyiğit, A., & Eren, P. E. (2021a). Data-driven manufacturing: An assessment model for Data Science Maturity. *Journal of Manufacturing Systems*, 60, 527–546, 60, 527–546.

- Gökalp, M. O., Gökalp, E., Kayabay, K., Koçyiğit, A., & Eren, P. E. (2021b). The development of the Data Science Capability Maturity Model: Survey-based research. *Online Information Review*, 46(3), 547–567
- Hafeez, K., Zhang, Y., & Malak, N. (2002). Determining key capabilities of a firm using analytic hierarchy process. *International Journal of Production Economics*, 76(1), 39–51.
- Haleem, A., Javaid, M., Pratap Singh, R., & Suman, R. (2022). Medical 4.0 technologies for healthcare: Features, capabilities, and applications, capabilities, and applications. *Internet of Things and Cyber-Physical Systems*, 2, 12-30.
- Hall, J. L., & McGraw, D. (2014). For telehealth to succeed, privacy and security risks must be identified and addressed. *Health Affairs*, 33(2), 216–221
- Hans, E. W., van Houdenhoven, M., & Hulshof, P. J. (2011). A framework for healthcare planning and Control. *International Series in Operations Research & Management Science*, 303–320, 303–320
- Harahap, N. C., Handayani, P. W., & Hidayanto, A. N. (2020). Functionalities and issues in the implementation of personal health record: A systematic review (preprint). *Journal of Medical Internet Research*. <https://doi.org/10.2196/26236>.
- Hare, N., Bansal, P., Bajowala, S. S., Abramson, S. L., Chervinskiy, S., Corriel, R., Hauswirth, D. W., Kakumanu, S., Mehta, R., Rashid, Q., Rupp, M. R., Shih, J., & Mosnaim, G. S. (2020). Workgroup report: COVID-19: Unmasking telemedicine. *The Journal of Allergy and Clinical Immunology: In Practice*, 8(8).
- Hareskov Jensen, N., Vallentin-Holbech, L., Dash, G. F., Feldstein Ewing, S. W., & Rømer Thomsen, K. (2023). Validity of an online, self-administered timeline follow back for alcohol use with adolescents. *Frontiers in Psychiatry*, 14, 14
- Haroon, S., Voo, T. C., Chua, H., Tan, G. L., & Lau, T. (2022, April 29), April 29)

- Harrell, H. (1996). Differences in perception of employee empowerment between managers and non-managers in the hotel industry. *Tourism Recreation Research*, 21(2), 69–75
- Hartono, B., Sulisty, S. R., Chai, K. H., & Indarti, N. (2019). Knowledge management maturity and performance in a project environment: Moderating roles of firm size and Project Complexity. *Journal of Management in Engineering*, 35(6).
- Hasan, A. S., & Trianni, A. (2020). A review of Energy Management Assessment Models for Industrial Energy Efficiency. *Energies*, 13(21), 5713
- Hauck, R. (2021). Blockchain, smart contracts and intellectual property using distributed ledger technology to protect, license and enforce Intellectual Property Rights. *Legal Issues in the Digital Age*, 1(1), 17–41
- Hayes, A. (2023). How stratified random sampling works, with examples. *Investopedia*.
https://www.investopedia.com/terms/stratified_random_sampling.asp
- Hebert, M. A., Jansen, J. J., Brant, R., Hailey, D., & Van Der Pol, M. (2004). Successes and challenges in a field-based, multi-method study of Home Telehealth. *Journal of Telemedicine and Telecare*, 10(1_suppl), 41–44.
- Hedman, J., & Borell, A. (2002). The impact of Enterprise Resource Planning Systems on organizational effectiveness. *Enterprise Resource Planning*, 125–142, 125–142
- Heravi, G., & Gholami, A. (2018). The influence of project risk management maturity and organizational learning on the success of Power Plant Construction Projects. *Project Management Journal*, 49(5), 22–37.
- Herbsleb, J., Zubrow, D., Goldenson, D., Hayes, W., & Paulk, M. (1997). Software quality and the Capability Maturity Model. *Communications of the ACM*, 40(6), 30–40
- Hidayati, A., Purwandari, B., Budiardjo, E. K., & Solichah, I. (2018). Global software development and Capability Maturity Model Integration: A systematic literature review.

2018 Third International Conference on Informatics and Computing (ICIC).

<https://doi.org/10.1109/iac.2018.8780489>.

Higdon, M. L. (2018). Trading your health: Assessing the need for domestic regulation of telemedicine and ability to conform to U.S. trade agreements. *LAW eCommons*.

<https://lawecommons.luc.edu/lclr/vol30/iss3/4>

Hilty, D. M., Maheu, M., Drude, K. P., & Hertlein, K. M. (2018). The need to implement and evaluate telehealth competency frameworks to ensure quality care across Behavioral Health Professions. *Academic Psychiatry*, 42(6), 818–824

Howard, I. M., & Kaufman, M. S. (2018). Telehealth applications for outpatients with neuromuscular or musculoskeletal disorders. *Muscle & Nerve*, 58(4), 475–485

Hua, J., Gu, G., Meng, W., & Wu, Z. (2013). Age band 1 of the Movement Assessment Battery for children-second edition: Exploring its usefulness in Mainland China. *Research in Developmental Disabilities*, 34(2), 801–808

Hung, H. M., O'Neill, R. T., Bauer, P., & Kohne, K. (1997). The behavior of the P-value when the alternative hypothesis was true. *Biometrics*, 53(1), 11

Iacobucci, D., Popovich, D. L., Moon, S., & Román, S. (2022). How to calculate, use, and report variance explained effect size indices and not die trying. *Journal of Consumer Psychology*, 33(1), 45–61.

Iden, J., & Eikebrokk, T. R. (2013). Implementing IT Service Management: A systematic literature review. *International Journal of Information Management*, 33(3), 512–523.

Islam, M. R. (2018). Sample size and its role in Central Limit Theorem (CLT). *Computational and Applied Mathematics Journal*, 4(1), 1–7.

Iyanna, S., Kaur, P., Ractham, P., Talwar, S., & Najmul Islam, A. K. M. (2022). Digital transformation of the healthcare sector. *What was impeding the adoption and continued*

- usage of technology-driven innovations by end-users? Journal of Business Research*, 153, 150–161, 153, 150–161.
- Jack, S. M., Munro Kramer, M. L., Williams, J. R., Schminkey, D., Tomlinson, E., Jennings Mayo-Wilson, L., Bradbury-Jones, C., & Campbell, J. C. (2020). Recognizing and responding to intimate partner violence using telehealth: Practical guidance for nurses and Midwives. *Journal of Clinical Nursing*, 30(3–4), 588–602.
- Jackson, G. L., Cutrona, S. L., White, B. S., Reardon, C. M., Orvek, E., Nevedal, A. L., Lindquist, J., Gifford, A. L., White, L., King, H. A., DeLaughter, K., Houston, T. K., Henderson, B., Vega, R., Kilbourne, A. M., & Damschroder, L. J. (2021). Merging implementation practice and science to scale up promising practices: The Veterans Health Administration (VHA) diffusion of excellence (DOE) program. *The Joint Commission Journal on Quality and Patient Safety*, 47(4), 217–227.
- Jacsó, P. (2005). Google Scholar: The Pros and the cons. *Online Information Review*, 29(2), 208–214
- Jakobsen, C. R., & Johnson, K. A. (2008). Mature agile with a twist of CMMI. *Agile 2008 Conference*. <https://doi.org/10.1109/agile.2008.10>.
- Jami Pour, M., & Jafari, S. M. (2019). Toward a maturity model for the application of social media in Healthcare. *Online Information Review*, 43(3), 404–425
- Jan, S., & Javed, A. (2013a). SCXTREME framework: A customized approach of process improvements in agile blend with CMMI practices in Pakistan. *International Journal of Information Technology and Computer Science*, 5(3), 69–78.
- Jan, S., & Javed, A. (2013b). SCXTREME framework: A customized approach of process improvements in agile blend with CMMI practices in Pakistan. *International Journal of Information Technology and Computer Science*, 5(3), 69–78.

- Jarvis-Selinger, S., Chan, E., Payne, R., Plohman, K., & Ho, K. (2008). Clinical telehealth across the disciplines: Lessons learned. *Telemedicine and E-Health*, 14(7), 720–725
- Javaid, M., Haleem, A., Pratap Singh, R., Khan, S., & Suman, R. (2021). Blockchain technology applications for Industry 4.0: A literature-based review. *Blockchain: Research and Applications*, 2(4), 100027
- Jayakrishnan, M., Mohamad, A. K., Azmi, F. R., & Abdullah, A. (2018). Implementing a Business Intelligence Framework for the Malaysian halal food manufacturing industry towards initiating strategic financial performance management. *Management Science Letters*, 1059–1076, 1059–1076
- Johansen, N., Miserez, M., de Beaux, A., Montgomery, A., Faylona, J. M., Carbonell, A., & Bisgaard, T. (2020). Surgical strategy for contralateral groin management in patients scheduled for unilateral inguinal hernia repair: An international web-based SurveyMonkey questionnaire: Strategy for contralateral groin management during inguinal hernia repair. *Scandinavian Journal of Surgery*, 110(3), 368–372.
- Johnson, D. E., & Benedek, D. M. (2023). Defining the limits of medical privacy within the US Departments of Defense and Veterans Affairs. *Veteran and Military Mental Health*, 127–142, 127–142
- Johnston, M. P. (2014). Secondary data analysis: A method of which the time had come. *Qualitative and Quantitative Methods in Libraries*, 3(3), 619–626.
- Jokela, T., & Lalli, T. (2003). Usability and CMMI. CHI '03 Extended Abstracts on Human Factors in Computing Systems - CHI '03. <https://doi.org/10.1145/765891.766121>
- Joly, Y., Dalpé, G., So, D., & Birko, S. (2015). Fair shares and Sharing fairly: A survey of public views on open science, informed consent and participatory research in Biobanking. *PLOS ONE*, 10(7)

- Jones, S. R., Carley, S., & Harrison, M. (2003). An introduction to power and sample size estimation. *Emergency Medicine Journal*, 20(5), 453–458.
- Jonnagaddala, J., Godinho, M. A., & Liaw, S. T. (2021). From telehealth to Virtual Primary Care in Australia? A rapid scoping review. *International Journal of Medical Informatics*, 151, 104470, 151, 104470.
- Jovanović, B., & Filipović, J. (2016). ISO 50001 standard-based Energy Management Maturity Model – proposal and validation in industry. *Journal of Cleaner Production*, 112, 2744–2755, 112, 2744–2755.
- Junaid, S. B., Imam, A. A., Balogun, A. O., De Silva, L. C., Surakat, Y. A., Kumar, G., Abdulkarim, M., Shuaibu, A. N., Garba, A., Sahalu, Y., Mohammed, A., Mohammed, T. Y., Abdulkadir, B. A., Abba, A. A., Kakumi, N. A., & Mahamad, S. (2022). Recent advancements in emerging technologies for Healthcare Management Systems: A survey. *Healthcare*, 10(10), 1940
- Jury, S. C., & Kornberg, A. J. (2016). Integrating telehealth into ‘business as usual’: was it really possible? *Journal of Telemedicine and Telecare*, 22(8), 499–503.
- Kahn, E. N., la Marca, F., & Mazzola, C. A. (2016). Neurosurgery and telemedicine in the United States: Assessment of the risks and opportunities. *World Neurosurgery*, 89, 133–138, 89, 133–138
- Kalid, N., Zaidan, A. A., Zaidan, B. B., Salman, O. H., Hashim, M., & Muzammil, H. (2017). Based real-time remote health monitoring systems: A review on patients’ prioritization and related “big data” using body sensors information and Communication Technology. *Journal of Medical Systems*, 42(2).

- Kamal, S. A., Shafiq, M., & Kakria, P. (2020). Investigating acceptance of telemedicine services through an extended technology acceptance model (TAM). *Technology in Society*, 60, 101212, 60, 101212
- Kamsu-Foguem, B. (2014). Systemic modeling in Telemedicine. *European Research in Telemedicine / La Recherche Européenne En Télémédecine*, 3(2), 57–65
- Kang, H. G., Mahoney, D. F., Hoenig, H., Hirth, V. A., Bonato, P., Hajjar, I., & Lipsitz, L. A. (2010). In situ monitoring of health in older adults: Technologies and issues. *Journal of the American Geriatrics Society*, 58(8), 1579–1586.
- Kangwantrakool, T., Theeramunkong, T., Usanavasin, S., & Piyabundit, C. (2022). R3P2: A Performance Model For Readiness Review Process Improvement In Capability Maturity Model Integration Level 3. *Journal of Positive School Psychology*, 6(8), 8074–8084..
- Karsh, B.-T. (2004). Beyond usability: Designing effective technology implementation systems to promote patient safety. *Quality and Safety in Health Care*, 13(5), 388–394
- Karwacka, M., Cieurzyńska, A., Lenart, A., & Janowicz, M. (2020). Sustainable development in the Agri-food sector in terms of the carbon footprint: A Review. *Sustainability*, 12(16), 6463
- Kastania, A. N., & Kossida, S. (2011). Statistics in telemedicine. *Advances in Telemedicine: Technologies, Enabling Factors and Scenarios*. <https://doi.org/10.5772/14356>
- Kaundinya, T., & Agrawal, R. (2021). Unpacking a telemedical takeover: Recommendations for improving the sustainability and usage of telemedicine post-COVID-19. *Quality Management in Health Care*, 31(2), 68–73
- Kayes, D. C. (2005). Internal validity and reliability of Kolb’s Learning Style Inventory Version 3 (1999). *Journal of Business and Psychology*, 20(2), 249–257.

- Keles, B., McCrae, N., & Grealish, A. (2019). A systematic review: The influence of social media on depression, anxiety and psychological distress in adolescents. *International Journal of Adolescence and Youth*, 25(1), 79–93.
- Kelley, K. (2003). Good practice in the conduct and Reporting of Survey Research. *International Journal for Quality in Health Care*, 15(3), 261–266.
- Kho, J., Gillespie, N., & Martin-Khan, M. (2020). A systematic scoping review of change management practices used for telemedicine service implementations. *BMC Health Services Research*, 20(1)
- Kidholm, K., Ekeland, A. G., Jensen, L. K., Rasmussen, J., Pedersen, C. D., Bowes, A., Flottorp, S. A., & Bech, M. (2012). A model for assessment of telemedicine applications: Mast. *International Journal of Technology Assessment in Health Care*, 28(1), 44–51.
- Kishore, R., Swinarski, M. E., Jackson, E., & Rao, H. R. (2012a). A quality-distinction model of IT capabilities: Conceptualization and two-stage empirical validation using CMMI processes. *IEEE Transactions on Engineering Management*, 59(3), 457–469.
- Kishore, R., Swinarski, M., Jackson, E., & Rao, H. (2012b). A quality-distinction model of IT capabilities: Conceptualization and two-stage empirical validation using CMMI processes. *IEEE Transactions on Engineering Management*, 59(3), 457–469.
- Kluger, A. N., & Tikoichinsky, A. (2001). The error of accepting the “theoretical” null hypothesis: The rise, fall, and resurrection of commonsense hypotheses in psychology. *Psychological Bulletin*, 127(3), 408–423
- Klun, M., & Trkman, P. (2018). Business process management – at the Crossroads. *Business Process Management Journal*, 24(3), 786–813.

- Knörr, V., Dini, L., Gunkel, S., Hoffmann, J., Mause, L., Ohnhäuser, T., Stöcker, A., & Scholten, N. (2022). Use of telemedicine in the outpatient sector during the COVID-19 pandemic: A cross-sectional survey of German physicians. *BMC Primary Care*, 23(1)
- Kolker, A. (2012). Efficient managerial decision-making in healthcare settings. *Management Engineering for Effective Healthcare Delivery*, 1–45, 1–45
- Kolukısa Tarhan, A., Garousi, V., Turetken, O., Söylemez, M., & Garossi, S. (2020). Maturity assessment and maturity models in health care: A multivocal literature review. *Digital health*, 6, 2055207620914772.
- Kootahi, K., Kwan Leung, A., & Wang, Y. (2023). An algorithm based on distribution similarity to determine the representative sample size for particle morphology characterization. *Computers and Geotechnics*, 162, 105696, 162, 105696
- Korachi, Z., & Bounabat, B. (2018). Data-Driven Maturity Model for Assessing Smart Cities. *Proceedings of the 2nd International Conference on Smart Digital Environment*.
<https://doi.org/10.1145/3289100.3289123>.
- Kosieradzka, A., & Rostek, K. (2021). Models of process maturity in organizations. *Process Management and Organizational Process Maturity*, 191–235, 191–235
- Kostalova, J., & Tetreva, L. (2018). Proposal of project management methods and tools-oriented maturity model. *Revista de Gestão e Projetos*, 9(1), 01–23
- Kouhizadeh, M., Sarkis, J., & Zhu, Q. (2019). At the nexus of blockchain technology, the circular economy, and product deletion. *Applied Sciences*, 9(8), 1712
- Kozak, J., Kania, K., Juszczuk, P., & Mitreęa, M. (2021). Swarm intelligence goal-oriented approach to data-driven innovation in Customer Churn Management. *International Journal of Information Management*, 60, 102357, 60, 102357.

- Krantz, D. H. (1999). The null hypothesis testing controversy in psychology. *Journal of the American Statistical Association*, 94(448), 1372–1381.
- Krause, N. (2002). A comprehensive strategy for developing closed-ended survey items for use in studies of older adults. *The Journals of Gerontology Series B: Psychological Sciences and Social Sciences*, 57(5).
- Kroes, A. D., & Finley, J. R. (2023). Demystifying omega squared: Practical guidance for effect size in common analysis of variance designs. *Psychological Methods*.
<https://doi.org/10.1037/met0000581>
- Krosnick, J. A. (1999). Survey research. *Annual Review of Psychology*, 50(1), 537–567
- Krupinski, E., & Bernard, J. (2014). Standards and guidelines in telemedicine and Telehealth. *Healthcare*, 2(1), 74–93
- Kukull, W. A., & Ganguli, M. (2012). Generalizability: The trees, the forest, and the low-hanging fruit. *Neurology*, 78(23), 1886–1891
- Kulpa, M. K., & Johnson, K. A. (2003). A high-maturity perspective. *Interpreting the CMMI (R)*, 257–274, 257–274
- Kumar, A., Mani, V., Jain, V., Gupta, H., & Venkatesh, V. G. (2023). Managing Healthcare Supply Chain through Artificial Intelligence (AI): A study of critical success factors. *Computers & Industrial Engineering*, 175, 108815, 175, 108815
- Kumar, P., Chauhan, S., & Awasthi, L. K. (2023). Artificial Intelligence in Healthcare: Review, Ethics, Trust Challenges & Future Research Directions. *Engineering Applications of Artificial Intelligence*, 120, 105894, 120, 105894
- Kurihara, H., Maeno, T., & Maeno, T. (2014). Importance of physicians' attire: Factors influencing the impression it makes on patients, a cross-sectional study. *Asia Pacific Family Medicine*, 13(1), 2

Kwak, Y. H., & Ibbs, C. W. (2002). Project Management Process Maturity (PM)2 model.

Journal of Management in Engineering, 18(3), 150–155.

Lakens, D. (2013). Calculating and reporting effect sizes to facilitate cumulative science: A practical primer for t-tests and ANOVAs. *Frontiers in Psychology*, 4, 863, 4, 863

Lamash, L., Little, L., & Hen-Herbst, L. (2022, January 21). Telehealth interventions to promote health and behavior-related outcomes in adolescents with autism spectrum disorder - journal of autism and developmental disorders. *SpringerLink*.
<https://link.springer.com/article/10.1007/s10803-022-05440-2>

Lamoureux, E. L., Chou, S. L., Larizza, M. F., & Keeffe, J. E. (2006). The reliability of data collection periods of personal costs associated with vision impairment. *Ophthalmic Epidemiology*, 13(2), 121–126

Landes, S. J., Pitcock, J. A., Harned, M. S., Connolly, S. L., Meyers, L. L., & Oliver, C. M. (2022). Provider perspectives on delivering dialectical behavior therapy via telehealth during COVID-19 in the Department of Veterans Affairs. *Psychological Services*, 19(3), 562–572

Langbecker, D., Caffery, L. J., Gillespie, N., & smith, A. C. (2017). Using survey methods in Telehealth Research: A practical guide. *Journal of Telemedicine and Telecare*, 23(9), 770–779.

Lau, J., & Knudsen, J. (2022). Reducing disparities in telemedicine: An equity-focused, public health approach. *Health Affairs*, 41(5), 647–650

Laupland, K. B. (2013). Incidence of bloodstream infection: A review of population-based studies. *Clinical Microbiology and Infection*, 19(6), 492–500

Lee, D., Gu, J., & Jung, H. (2019). Process maturity models: Classification by application sectors and validities studies. *Journal of Software: Evolution and Process*, 31(4).

- Lee, J., Lee, D., & Kang, S. (N.D.). An overview of the Business Process Maturity Model (BPMM). *Lecture Notes in Computer Science*, 384–395, 384–395
- Lenert, L., & McSwain, B. Y. (2020). Balancing Health Privacy, Health Information Exchange, and research in the context of the COVID-19 pandemic. *Journal of the American Medical Informatics Association*, 27(6), 963-966.
- Leppink, J., O’Sullivan, P., & Winston, K. (2016). Effect size – large, medium, and small. *Perspectives on Medical Education*, 5(6), 347–349
- Levesque, J. F., Pineault, R., Hamel, M., Roberge, D., Kapetanakis, C., Simard, B., & Prud’Homme, A. (2012). Emerging organizational models of primary healthcare and unmet needs for care: Insights from a population-based survey in Quebec Province. *BMC Family Practice*, 13(1)
- Li, J., & Wilson, L. S. (2013). Telehealth trends and the challenge for Infrastructure. *Telemedicine and E-Health*, 19(10), 772–779
- Li, J.-P. O., Liu, H., Ting, D. S. J., Jeon, S., Chan, R. V. P., Kim, J. E., Sim, D. A., Thomas, P. B. M., Lin, H., Chen, Y., Sakomoto, T., Loewenstein, A., Lam, D. S. C., Pasquale, L. R., Wong, T. Y., Lam, L. A., & Ting, D. S. W. (2021). Digital Technology, Tele-Medicine and Artificial Intelligence in Ophthalmology: A global perspective. *Progress in Retinal and Eye Research*, 82, 100900, 82, 100900
- Lianying, Z., Jing, H., & Xinxing, Z. (2012). The project management maturity model and application based on PRINCE2. *Procedia Engineering*, 29, 3691–3697, 29, 3691–3697
- Linders, B. (2010). Can the P-CMM Be Used to Implement Agile in Organizations? *Software Quality Professional*, 13(1), 13
- Loane, M., & Wootton, R. (2002). A review of guidelines and standards for Telemedicine. *Journal of Telemedicine and Telecare*, 8(2), 63–71.

- Long, H. (2014). An empirical review of research methodologies and methods in creativity studies (2003–2012). *Creativity Research Journal*, 26(4), 427–438.
- Lowry, P. B., D’Arcy, J., hammer, B., & moody, G. D. (2016). “Cargo Cult” science in Traditional Organization and Information Systems Survey Research: A case for using nontraditional methods of data collection, including mechanical turk and online panels. *The Journal of Strategic Information Systems*, 25(3), 232–240.
- Lundon, D. (2023). Regulations and requirements for telehealth provision. *Telehealth in Urology*, 3–13, 3–13
- Lytvyn, V., Vysotska, V., Uhryn, D., Hrendus, M., & Naum, O. (2018). Analysis of statistical methods for stable combinations determination of keywords identification. *Eastern-European Journal of Enterprise Technologies*, 2(2 (92)), 23–37.
- MacDonald-Rencz, S., Craddock, T., & Parker-Taillon, D. (2004). The National Initiative for Telehealth guidelines. *Telemedicine Journal and E-Health*, 10(1), 113–114.
- Machado, C. G., Winroth, M., Carlsson, D., Almström, P., Centerholt, V., & Hallin, M. (2019). Industry 4.0 readiness in manufacturing companies: Challenges and enablers towards increased digitalization. *Procedia CIRP*, 81, 1113–1118.
- Maeder, A., & Poultney, N. (2016). Evaluation strategies for telehealth implementations. 2016 *IEEE International Conference on Healthcare Informatics (ICHI)*.
<https://doi.org/10.1109/ichi.2016.65>.
- Mahafza, S. R. (2005). A performance-based methodology to assess Department of Defense technologies. The University of Alabama in Huntsville.
- Maisiri, W., & van Dyk, L. (2020). Industry 4.0 competence maturity model design requirements: A systematic mapping review. *FEES World Engineering Education Forum-Global Engineering Deans Council (WEEF-GEDC)*, IEEE., 1–6.

- Makridis, C., Hurley, S., Klote, M., & Alterovitz, G. (2021, April 29). Ethical applications of Artificial Intelligence: Evidence from Veterans' Health Research. by Christos Makridis, Seth Hurley, Molly Klote, Gil Alterovitz: *SSRN*. <https://dx.doi.org/10.2139/ssrn.3828046>
- Mamlin, B. W., & Tierney, W. M. (2016). The Promise of Information and communication technology in Healthcare: Extracting value from the Chaos. *The American Journal of the Medical Sciences*, 351(1), 59–68.
- Martín-Martín, A., Costas, R., van Leeuwen, T., & Delgado López-Cózar, E. (2018). Evidence of open access of scientific publications in google scholar: A large-scale analysis. *Journal of Informetrics*, 12(3), 819–841..
- Maxwell, J. A. (2013). *Qualitative research design: An interactive approach* (3rd ed.). SAGE Publications.
- McHugh, M. L. (2013). The chi-square test of Independence. *Biochemia Medica*, 143–149, 143–149
- Meade, A. W., & craig, S. B. (2012). Identifying careless responses in survey data. *Psychological Methods*, 17(3), 437–455
- Mellinger, C. D., & Hanson, T. A. (2021). Methodological considerations for survey research: Validity, reliability, and Quantitative Analysis. *Linguistica Antverpiensia, New Series – Themes in Translation Studies*, 19.
- Melnyk, S. A., Sroufe, R. P., & Calantone, R. (2002). Assessing the impact of environmental management systems on corporate and environmental performance. *Journal of Operations Management*, 21(3), 329–351.
- Memon, M. A., ting, H., Cheah, J. H., Thurasamy, R., Chuah, F., & Cham, T. H. (2020). Sample Size For Survey Research: Review and recommendations. *Journal of Applied Structural Equation Modeling*, 4(2), i–xx.

- Meyer, D., Slone, S. E., Ogungbe, O., Duroseau, B., & Farley, J. E. (2022). Impact of the COVID-19 pandemic on HIV Healthcare Service engagement, treatment adherence, and viral suppression in the United States: A systematic literature review, treatment adherence, and viral suppression in the United States: A systematic literature review. *AIDS and Behavior*, 27(1), 344-357.
- Miranda, R., Oliveira, M. D., Nicola, P., Baptista, F. M., & Albuquerque, I. (2023). Towards a framework for implementing remote patient monitoring from an Integrated Care Perspective: A scoping review. *International Journal of Health Policy and Management*. <https://doi.org/10.34172/ijhpm.2023.7299>.
- Mishra, V., & Sharma, M. G. (2022). Digital Transformation Evaluation of telehealth using convergence, maturity, and adoption. *Health Policy and Technology*, 11(4), 100684
- Mittal, S., Khan, M. A., Romero, D., & Wuest, T. (2018). A critical review of smart manufacturing & industry 4.0 maturity models: Implications for small and medium-sized enterprises (SMEs). *Journal of Manufacturing Systems*, 49, 194–214, 49, 194–214.
- Moates, K. N., Armenakis, A. A., Gregory, B. T., Albritton, M. D., & Field, H. S. (2005). Achieving content representativeness in organizational diagnosis: Use of action groups for successful organizational change. *Action Research*, 3(4), 403–416
- Moehr, J. R., Schaafsma, J., Anglin, C., Pantazi, S. V., Grimm, N. A., & Anglin, S. (2006). Success factors for telehealth a case study. *International Journal of Medical Informatics*, 75(10–11), 755–763.
- Moghadam, M. M., Safari, H., & Maleki, M. (2013). A novel model for business process maturity assessment through combining maturity models with EFQM and ISO 9004:2009. *International Journal of Business Process Integration and Management*, 6(2), 167.

- Moran, M. (2021, July 27). Take the guesswork out of sample size: Why and how to conduct a power analysis. *Statistics Solutions*. <https://www.statisticssolutions.com/take-the-guesswork-out-of-sample-size-why-and-how-to-conduct-a-power-analysis/>
- Morris, M., & Venkatesh, V. (2010). Job characteristics and job satisfaction: Understanding the role of Enterprise Resource Planning System implementation. *MIS Quarterly*, 34(1), 143–161
- Moser, S. M. (2007). Some expectations of a non-central chi-square distribution with an even number of degrees of freedom. *TENCON 2007 - 2007 IEEE Region 10 Conference*. <https://doi.org/10.1109/tencon.2007.4429039>.
- Moskovkin, V. M. (2009). The potential of using the Google Scholar search engine for estimating the publication activities of Universities. *Scientific and Technical Information Processing*, 36(4), 198–202
- Mott, D. A., Pederson, C. A., Doucette, W. R., Gaither, C. A., & Schommer, J. C. (2001). A national survey of U.S. pharmacists in 2000: Assessing nonresponse bias of a survey methodology. *AAPS PharmSci*, 3(4), 76–86
- Mrugalska, B., & Stasiuk-Piekarska, A. (2020). Readiness and maturity of Manufacturing Enterprises for Industry 4.0. *Advances in Manufacturing, Production Management and Process Control*, 263–270.
- Mullaly, M. (2006). Longitudinal analysis of Project Management Maturity. *Project Management Journal*, 37(3), 62–73.
- Mullen-Fortino, M., Rising, K. L., Duckworth, J., Gwynn, V., sites, F. D., & Hollander, J. E. (2019). Presurgical assessment using telemedicine technology: Impact on efficiency, effectiveness, and patient experience of care. *Telemedicine and E-Health*, 25(2), 137–142

- Murphy, L. (2023). The Questionnaire Surveying Research Method: *Pros, Cons and Best Practices*. <https://doi.org/10.14293/s2199-1006.1.sor-.pp3wys8.v1>
- Musah, S., Medeni, T. D., & Soylu, D. (2019). Assessment of role of innovative technology through blockchain technology in Ghana's cocoa beans food supply chains. *2019 3rd International Symposium on Multidisciplinary Studies and Innovative Technologies (ISMSIT)*. <https://doi.org/10.1109/ismsit.2019.8932936>
- Musco, C., Musco, C., & Tsourakakis, C. E. (2018). Minimizing polarization and disagreement in social networks. *Proceedings of the 2018 World Wide Web Conference on World Wide Web - WWW '18*. <https://doi.org/10.1145/3178876.3186103>.
- Mustapha, M. (2019). The Art of Effective Research: *A study utilizing multiple regression analysis (Hayes process macros for SPSS)*. SAGE Publications Ltd.
<https://doi.org/10.4135/9781526492401>
- Mutafelija, B., & Stromberg, H. (2008). Capability Maturity Model Integration (CMMI). *Process Improvement with CMMI® v1.2 and ISO Standards*, 25–64, 25–64
- Myers, C. R. (2018). Using telehealth to remediate rural mental health and healthcare disparities. *Issues in Mental Health Nursing*, 40(3), 233–239
- Naito, A., Wills, A.-M., Tropea, T. F., Ramirez Zamora, A., Hauser, R. A., Martino, D., Turner, T. H., Rafferty, M. R., Afshari, M., Williams, K. L., Vaou, O., McKeown, M. J., Ginsburg, L., Ezra, A., Iansek, R., Wallock, K., Evers, C., Schroeder, K., DeLeon, R., ... Beck, J. C. (2021). Expediting telehealth use in clinical research studies: *Recommendations for overcoming barriers in North America*. *Npj Parkinson's Disease*, 7(1)
- Najmi, M., Rigas, J., & Fan, I. (2005). A framework to review performance measurement systems. *Business Process Management Journal*, 11(2), 109–122.

- Najmi, M., Sepehri, M., & Hashemi, S. (2010). The evaluation of business intelligence maturity level in Iranian banking industry. *2010 IEEE 17Th International Conference on Industrial Engineering and Engineering Management*.
<https://doi.org/10.1109/icieem.2010.5646571>.
- Nunnally, J. C., & Bernstein, I. H. (1994). *Psychometric theory* (3rd ed.). McGraw-Hill.
- Nelson, E.-L., & Palsbo, S. (2006). Challenges in telemedicine equivalence studies. *Evaluation and Program Planning*, 29(4), 419–425
- Nematovich, V. A. Z., & Karimberdiyevich, S. S. (2022). Teaching perspective based on innovative technologies. *Web of Scientist: International Scientific Research Journal*, 3(1), 678–687..
- Neri, A. J., Whitfield, G. P., Umeakunne, E. T., Hall, J. E., DeFrances, C. J., Shah, A. B., Sandhu, P. K., Demeke, H. B., Board, A. R., Iqbal, N. J., Martinez, K., Harris, A. M., & Strona, F. V. (2022). Telehealth and public health practice in the United States before, during, and after the COVID-19 pandemic. *Preventing Chronic Disease*, 19, E90.
<https://doi.org/10.5888/pcd19.220109>
- Ngai, E. W. T., Chau, D. C. K., Poon, J. K. L., & To, C. K. M. (2013). Energy and utility management maturity model for sustainable manufacturing process. *International Journal of Production Economics*, 146(2), 453–464.
- Nguyen, T. M., Schiefer, J., & Tjoa, A. M. (2005). Sense & Response Service Architecture (SARESA). *Proceedings of the 8th ACM International Workshop on Data Warehousing and OLAP*. <https://doi.org/10.1145/1097002.1097015>
- Ni Riordain, R., Hodgson, T., Porter, S., & Fedele, S. (2016). Validity and reliability of the chronic oral mucosal disease questionnaire in a UK population. *Journal of Oral Pathology & Medicine*, 45(8), 613–616.

- Ning, L., Niu, J., Bi, X., Yang, C., Liu, Z., Wu, Q., Ning, N., Liang, L., Liu, A., Hao, Y., Gao, L., & Liu, C. (2020). The impacts of knowledge, risk perception, emotion and information on citizens' protective behaviors during the outbreak of COVID-19: A cross-sectional study in China. *BMC Public Health*, 20, 1751. <https://doi.org/10.1186/s12889-020-09892-y>
- Obstfelder, A., Engeseth, K. H., & wynn, R. (2007). Characteristics of successfully implemented Telemedical Applications. *Implementation Science*, 2(1)
- Office of Research and Development. (2018). *Draft electronic mailtext - Office of Research & Development*. Veterans' Health Administration. <https://www.research.va.gov/resources/policies/guidance/draft-electronic-mailtext.pdf>
- Ogbeifun, E., Agwa-Ejon, J., Mbohwa, C., & Pretorius, J. H. (2016). The Delphi technique: A credible research methodology. *Proceedings of the International Conference on Industrial Engineering and Operations Management*, 572–578.
- Olejnik, S., & Algina, J. (2003). Generalized eta and omega squared statistics: Measures of effect size for some common research designs. *Psychological Methods*, 8(4), 434–447
- Oliva, F. L. (2016). A maturity model for enterprise risk management. *International Journal of Production Economics*, 173, 66–79, 173, 66–79.
- Oliveira, M. A., & lopes, I. (2019). Evaluation and improvement of maintenance management performance using a maturity model. *International Journal of Productivity and Performance Management*, 69(3), 559–581.
- Omboni, S., Padwal, R. S., Alessa, T., Benczúr, B., Green, B. B., Hubbard, I., Kario, K., Khan, N. A., Konradi, A., Logan, A. G., Lu, Y., Mars, M., McManus, R. J., Melville, S., Neumann, C. L., Parati, G., Renna, N. F., Ryvlin, P., Saner, H., ... Wang, J. (2022). The worldwide impact of telemedicine during COVID-19: Current evidence and

- recommendations for the future. *Connected Health*, 1(1), 7–35.
<https://doi.org/10.20517/ch.2021.03>
- Ormazabal, M., Sarriegi, J. M., & Viles, E. (2017, August 14), August 14)
- Otto, L., Schlieter, H., Harst, L., Whitehouse, D., & Maeder, A. (2023). The Telemedicine Community Readiness Model successful telemedicine implementation and scale-up. *Frontiers in Digital Health*, 5, 5
- Otto, L., Whitehouse, D., & Schlieter, H. (2019). On the road to telemedicine maturity: A systematic review and classification of Telemedicine Maturity Models. *Humanizing Technology for a Sustainable Society*. <https://doi.org/10.18690/978-961-286-280-0.26>
- Overby, E., Slaughter, S. A., & Konsynski, B. (2010). The design, use, and consequences of virtual processes. *Information Systems Research*, 21(4), 700–710
- O’Brien, H. L., & Toms, E. G. (2009). The development and evaluation of a survey to measure user engagement. *Journal of the American Society for Information Science and Technology*, 61(1), 50–69.
- O’Connell, M. E., Crossley, M., Cammer, A., Morgan, D., Allingham, W., Cheavins, B., Dalziel, D., Lemire, M., Mitchell, S., & Morgan, E. (2013). Development and evaluation of a telehealth video conferenced support group for rural spouses of individuals diagnosed with atypical early-onset dementias. *Dementia*, 13(3), 382–395
- O’Connor, A., Jackson, L., Goldsmith, L., & Skirton, H. (2013). could I get a retweet please? Health Research Recruitment and the Twittersphere. *Journal of Advanced Nursing*, 70(3), 599–609.
- O’Donovan, P., Bruton, K., & T.J. O’Sullivan, D. (2020). IAMM: A maturity model for measuring industrial analytics capabilities in large-scale manufacturing facilities. *International Journal of Prognostics and Health Management*, 7(4).

- Palacholla, R. S., Fischer, N., Coleman, A., Agboola, S., Kirley, K., Felsted, J., Katz, C., Lloyd, S., & Jethwani, K. (2019). Provider- and patient-related barriers to and facilitators of Digital Health Technology Adoption for Hypertension Management: Scoping Review. *JMIR Cardio*, 3(1)
- Pan, B., Litvin, S. W., & O'Donnell, T. E. (2007). Understanding accommodation search query formulation: The first step in putting heads in beds. *Journal of Vacation Marketing*, 13(4), 371–381.
- Papoutsis, C., Wherton, J., Shaw, S., Morrison, C., & Greenhalgh, T. (2020). Putting the social back into sociotechnical: Case studies of co-design in Digital Health. *Journal of the American Medical Informatics Association*, 28(2), 284–293.
- Park, H. M. (2009). *Comparing Group Means: T-tests and One-way ANOVA Using Stata, SAS, R, and SPSS*. University Information Technology Services.
- Parker, C. (2021). Choosing secure & private communications tools: EMAIL, Messaging & File Transfer. *Maximum Privacy with End-to-End Encryption*. https://doi.org/10.1007/978-1-4842-7034-9_4
- Parker, S., Amy Prince, A., Louise Thomas, L., Hyun Song, H., Diana Milosevic, D., & Harris, M. F. (2018). Electronic, mobile and telehealth tools for vulnerable patients with chronic disease: a systematic review and realist synthesis. *BMJ Open*, (8), 1–23.
- Parmanto, B., Lewis, Jr., A. N., graham, K. M., & Bertolet, M. H. (2016). Development of the Telehealth Usability Questionnaire (TUG). *International Journal of Telerehabilitation*, 8(1), 3–10.
- Paul, D. L. (2006). Collaborative activities in virtual settings: A knowledge management perspective of telemedicine. *Journal of Management Information Systems*, 22(4), 143–176.

- Paulk, M. (2002). Capability maturity model for software. *Encyclopedia of Software Engineering*. <https://doi.org/10.1002/0471028959.sof589>
- Peracca, S. B., Fonseca, A. S., Lachica, O., Jackson, G. L., Morris, I. J., King, H. A., Misitzis, A., Whited, J. D., Mohr, D. C., Lamkin, R. P., Gifford, A. L., Weinstock, M. A., & Oh, D. H. (2023). Organizational readiness for patient-facing mobile tele dermatology to care for established veteran patients in the United States. *Telemedicine and E-Health*, 29(1), 72–80
- Pereira, R., & da Silva, M. M. (2014, July 6), July 6)
- Perimal-Lewis, L., Williams, P. A. H., Mudd, G., & Gunasekara, G. (2021). Virtual care: *The future for Telehealth. Studies in Health Technology and Informatics*.
<https://doi.org/10.3233/shti210033>
- Picket, T. (2016). *The Capability Maturity Model as an advertising process maturity paradigm*. University of West Alabama.
- Poepelbuss, J., Niehaves, B., Simons, A., & Becker, J. (2011). Maturity models in information systems research: Literature search and analysis. *Communications of the Association for Information Systems*, 29, 29
- Polit, D. F. (2015). Assessing measurement in Health: Beyond Reliability and validity. *International Journal of Nursing Studies*, 52(11), 1746–1753.
- Pollard, J. S., Karimi, K. A., & Ficcaglia, M. B. (2017). Ethical considerations in the design and implementation of a telehealth service delivery model. *Behavior Analysis: Research and Practice*, 17(4), 298–311
- Pool, J., Akhlaghpour, S., Fatehi, F., & Gray, L. C. (2022). Data Privacy concerns and use of telehealth in the aged care context: An integrative review and research agenda. *International Journal of Medical Informatics*, 160, 104707, 160, 104707.

- Porter, L. W. (1962). Job attitudes in management: I. Perceived deficiencies in need fulfillment as a function of job level. *Journal of Applied Psychology*, 46(6), 375–384.
- Pour, M. J., Abbasi, F., & Sohrabi, B. (2022). Toward a maturity model for Big Data Analytics: A roadmap for complex data processing. *International Journal of Information Technology & Decision Making*, 22(01), 377–419.
- Priescu, I., & Oncioiu, I. (2022). Measuring the impact of virtual communities on the intention to use telemedicine services. *Healthcare*, 10(9), 1685
- Prieto-Sandoval, V., Jaca, C., Santos, J., Baumgartner, R. J., & Ormazabal, M. (2019). Key strategies, resources, and capabilities for implementing circular economy in industrial small and Medium Enterprises. *Corporate Social Responsibility and Environmental Management*, 26(6), 1473–1484
- PSU.EDU. (2023). *S.5 power analysis: Stat online*. Statistics Online Courses. Penn State. <https://online.stat.psu.edu/statprogram/reviews/statistical-concepts/power-analysis>
- Pustokhina, I. V., Pustokhin, D. A., RH, A., Jayasankar, T., Jeyalakshmi, C., Díaz, V. G., & Shankar, K. (2021). Dynamic customer churn prediction strategy for business intelligence using text analytics with evolutionary optimization algorithms. *Information Processing & Management*, 58(6), 102706
- Queirós, A., Faria, D., & Almeida, F. (2017). Strengths and limitations of qualitative and quantitative research methods. *European Journal of Education Studies*. <https://oapub.org/edu/index.php/ejes/article/view/1017>.
- Quick, J., & Hall, S. (2015). Part Three: The quantitative approach. *Journal of Perioperative Practice*, 25(10), 192–196.
- Raghupathi, W., & Raghupathi, V. (2021). Contemporary Business Analytics: An overview. *Data*, 6(8), 86

- Rahi, S. (2017). Research design and methods: A systematic review of research paradigms, sampling issues and instruments development. *International Journal of Economics & Management Sciences*, 06(02).
- Rahi, S. (2021). Assessing individual behavior towards adoption of telemedicine application during COVID-19 pandemic: Evidence from emerging market. *Library Hi Tech*, 40(2), 394–420
- Rajic, M., Banic, M., Maksimovic, R., Mancic, M., & Milosavljevic, P. (2022). Energy and utility management maturity model for sustainable industry. *Machine and Industrial Design in Mechanical Engineering*, 353–361, 353–361
- Recker, J., & Mendling, J. (2015). The State of the Art of Business Process Management Research as published in the BPM conference. *Business & Information Systems Engineering*, 58(1), 55–72
- Regmi, P. R., Waithaka, E., Paudyal, A., Simkhada, P., & van Teijlingen, E. (2017). Guide to the design and application of online questionnaire surveys. *Nepal Journal of Epidemiology*, 6(4), 640–644.
- Rhon, D. I., fritz, J. M., Kerns, R. D., McGeary, D. D., Coleman, B. C., Farrokhi, S., Burgess, D. J., Goertz, C. M., Taylor, S. L., & Hoffmann, T. (2022). Tidier-telehealth: Precision in reporting of telehealth interventions used in clinical trials - unique considerations for the template for the intervention description and replication (Tidier) checklist. *BMC Medical Research Methodology*, 22(1)
- Richter, B., Mengelkamp, E., & Weinhardt, C. (2018). Maturity of blockchain technology in local electricity markets. *2018 15th International Conference on the European Energy Market (EEM)*. <https://doi.org/10.1109/eem.2018.8469955>.

- Rolke, W., & Gongora, C. G. (2020). A chi-square goodness-of-fit test for continuous distributions against a known alternative. *Computational Statistics*, 36(3), 1885–1900
- Romão, M., & Carvalho, H. (2022). Organizational maturity and project success in Healthcare: The mediating effect of project management. *International Journal of Healthcare Technology and Management*, 19(2), 91.
- Ronaghi, M. H. (2021). A blockchain maturity model in agricultural supply chain. *Information Processing in Agriculture*, 8(3), 398–408
- Rosnow, R. L., Rotheram-Borus, M. J., Ceci, S. J., Blanck, P. D., & Koocher, G. P. (1993). The Institutional Review Board as a mirror of scientific and ethical standards. *American Psychologist*, 48(7), 821–826
- Rossi, J. S. (1988). Onaway: A basic program for computing Anova from group summary statistics. *Behavior Research Methods, Instruments, & Computers*, 20(3), 347–348
- Royce, W. (2002). CMM vs. CMMi: *From conventional to modern software management*. http://www.imsi-pm.com/library/CMM_vs_CMMI_Software_Management.pdf
- Ruiz-Cosignani, D., Chen, Y., Cheung, G., Lawrence, M., Lyndon, M. P., Ma'u, E., & Ramalho, R. (2022). Adaptation models, barriers, and facilitators for cultural safety in Tele-psychiatry: A systematic scoping review. *Journal of Telemedicine and Telecare*. <https://doi.org/10.1177/1357633x211069664>.
- Röglinger, M., Pöppelbuß, J., & Becker, J. (2012). Maturity models in business process management. *Business Process Management Journal*, 18(2), 328–346.
- Saigi-Rubió, F., Borges did Nascimento, I. J., robles, N., Ivanovska, K., Katz, C., Azzopardi-Muscat, N., & Novillo Ortiz, D. (2022). The current status of telemedicine technology use across the World Health Organization European Region: An overview of Systematic Reviews. *Journal of Medical Internet Research*, 24(10).

- Santos, R. C., & Martinho, J. L. (2019). An industry 4.0 maturity model proposal. *Journal of Manufacturing Technology Management*, 31(5), 1023–1043.
- Sarmiento, K. F., Folmer, R. L., Stepnowsky, C. J., Whooley, M. A., Boudreau, E. A., Kuna, S. T., Atwood, C. W., Smith, C. J., & Yarbrough, W. C. (2019). National expansion of sleep telemedicine for veterans: The Telesleep Program. *Journal of Clinical Sleep Medicine*, 15(09), 1355–1364.
- Sauers-Ford, H. S., Hamline, M. Y., Gosdin, M. M., Kair, L. R., Weinberg, G. M., Marcin, J. P., & Rosenthal, J. L. (2019). Acceptability, usability, and effectiveness: A qualitative study evaluating a pediatric telemedicine program. *Academic Emergency Medicine*, 26(9), 1022–1033
- Scala, N. M., Schiavone, V., Olivieri, H., Seppänen, O., Alves, T. D. C., Liu, M., & Granja, A. D. (2023). Comparative analysis of planning with the critical path method, last planner system, and location-based techniques in brazil, finland, and the united states. *Engineering management journal*, 35(3), 237-256.
- Scanlon, D. P., Harvey, J. B., Wolf, L. J., Vanderbrink, J. M., Shaw, B., Shi, Y., Mahmud, Y., Ridgely, M. S., & Damberg, C. L. (2020). *Were health Systems redesigning how health care was delivered?* *Health Services Research*, 55(S3), 1129–1143
- Scweb. (2001). Concepts: Process Quality. Concepts: *Process quality*.
https://scweb.uhcl.edu/helm/RationalUnifiedProcess/process/workflow/environm/co_sqlty.htm
- Scharkow, M. (2016). The accuracy of self-reported internet use a validation study using client log data. *Communication Methods and Measures*, 10(1), 13–27
- Schieltz, K. M., & Wacker, D. P. (2020, July), July)

- Schleyer, T. K., & Forrest, J. L. (2000). Methods for the design and administration of web-based surveys. *Journal of the American Medical Informatics Association*, 7(4), 416–425.
- Schlieter, H., Marsch, L. A., Whitehouse, D., Otto, L., Londral, A. R., Teepe, G. W., Benedict, M., Ollier, J., Ulmer, T., Gasser, N., Ultsch, S., Wollschlaeger, B., & Kowatsch, T. (2022). Scale-up of Digital Innovations in Health Care: Expert commentary on enablers and barriers. *Journal of Medical Internet Research*, 24(3).
- Schmiedel, T., vom Brocke, J., & Recker, J. (2013). *Which cultural values matter to business process management?* *Business Process Management Journal*, 19(2), 292–317.
- Schoefer, K., & Ennew, C. (2005). The impact of perceived justice on consumers' emotional responses to service complaint experiences. *Journal of Services Marketing*, 19(5), 261–270.
- Schoville, R. R., & Titler, M. G. (2015). Guiding Healthcare Technology Implementation. *CIN: Computers, Informatics, Nursing*, 33(3), 99–107
- Schumacher, A., Erol, S., & Sih, W. (2016). A maturity model for assessing Industry 4.0 readiness and maturity of manufacturing enterprises. *Procedia CIRP*, 52, 161–166, 52, 161–166
- Schumacher, A., Nemeth, T., & Sih, W. (2019). Road mapping towards industrial digitalization based on an Industry 4.0 maturity model for manufacturing enterprises. *Procedia CIRP*, 79, 409–414, 79, 409–414
- Schwamm, L. H. (2014, February 14), February 14)
- Schwamm, L. H., Audebert, H. J., Amarenco, P., Chumbler, N. R., Frankel, M. R., George, M. G., Gorelick, P. B., Horton, K. B., Kaste, M., Lackland, D. T., Levine, S. R., Meyer, B. C., Meyers, P. M., Patterson, V., Stranne, S. K., & White, C. J. (2009).

- Recommendations for the implementation of telemedicine within stroke systems of care. *Stroke*, 40(7), 2635–2660
- Schönwälder, J., & Weber, A. (2022). Maturity Levels of sustainable corporate entrepreneurship: The role of collaboration between a firm's corporate venture and Corporate Sustainability Departments. *Business Strategy and the Environment*, 32(2), 976–990
- Shah, H. (2018). Internet of things (IoT) rollouts in the telehealth industry. *Telehealth and Medicine Today*, 2(3)
- Shah, L., Siadat, A., & Vernadat, F. (2009). Maturity assessment in risk management in manufacturing engineering. *2009 3rd Annual IEEE Systems Conference* (pp. 296-301). IEEE. <https://doi.org/10.1109/systems.2009.4815815>.
- Shah, M. N., Morris, D., Jones, C. M., Gillespie, S. M., Nelson, D. L., McConnochie, K. M., & Dozier, A. (2013). A qualitative evaluation of a telemedicine-enhanced emergency care program for older adults. *Journal of the American Geriatrics Society*, 61(4), 571–576.
- Shaikh, T. A., Dar, T. R., & Sofi, S. (2022). A data-centric artificial intelligent and extended reality technology in Smart Healthcare Systems. *Social Network Analysis and Mining*, 12(1)
- Shang, S. S. C., & Lin, S.-F. (2009). Understanding the effectiveness of Capability Maturity Model integration by examining the knowledge management of software development processes. *Total Quality Management & Business Excellence*, 20(5), 509–521
- Sharma, B. (2016). A focus on reliability in developmental research through Cronbach's alpha among medical, dental and paramedical professionals. *Asian Pacific Journal of Health Sciences*, 3(4), 271–278.

- Sharma, H. K., Kumar, A., Pant, S., & ram, M. (2022). 3 Role of artificial intelligence, IoT and Blockchain in smart healthcare. *Artificial Intelligence, Blockchain and IoT for Smart Healthcare*, 25–36, *Blockchain and IoT for Smart Healthcare*, 25–36
- Shaygan, A., & Daim, T. (2019). Technology management maturity assessment model: an exploratory multi-criteria approach for healthcare organizations. *In 2019 Portland International Conference on Management of Engineering and Technology (PICMET)*, 1–9., 1–9..
- Shaygan, Amir, & Daim, T. (2023). Technology management maturity assessment model in Healthcare Research Centers. *Technovation*, 120, 102444, 120, 102444
- Sheikhshoei, F., Naghshineh, N., Alidousti, S., & Nakhoda, M. (2018). Design of a Digital Library Maturity Model (DLMM). *The Electronic Library*, 36(4), 607–619
- Shiferaw, F., & Zolfo, M. (2012). The role of Information Communication Technology (ICT) towards Universal Health Coverage: The first steps of a telemedicine project in Ethiopia. *Global Health Action*, 5(1), 15638
- Shulver, W., Killington, M., & Crotty, M. (2016). ‘Massive potential’ or ‘safety risk’? Health worker views on telehealth in the care of older people and implications for successful normalization. *BMC Medical Informatics and Decision Making*, 16(1)
- Singh, H., & Samalia, H. V. (2014). A business intelligence perspective for churn management. *Procedia - Social and Behavioral Sciences*, 109, 51–56, 109, 51–56
- Sinn, C.-L. J., Pasat, Z., Klea, L., Hogeveen, S., Holditch, C., Beltzner, C., & Costa, A. (2022). A maturity model framework for integrated virtual care. *Journal of Integrated Care*, 30(4), 351–362.
- IT Governance. (2013). *Software Capability Maturity Model (CMM)*.
<https://www.itgovernance.asia/capability-maturity-model>

- Soydan, G. H., & Kokar, M. (2006). An OWL ontology for representing the CMMI-SW model. Workshop on Semantic Web Enabled Software Engineering (SWESE), Department of Electrical and Computer Engineering, 1–14., Department of Electrical and Computer Engineering, 1–14.
- Springer, P. R., Bischoff, R. J., Taylor, N. C., Neuhaus, V., & Leow, C. (2021). Competency-based training in the supervision of Relational Telemental Supervision. *Journal of Marital and Family Therapy*, 47(2), 375–391.
- Spruit, M. (2018). Improving operational risk management using Business Performance Management Technologies. *Sustainability*, 10(3), 640
- Staron, M., Meding, W., Karlsson, G., & Nilsson, C. (2011). Developing measurement systems: An industrial case study. *Journal of Software Maintenance and Evolution: Research and Practice*, 23(2), 89–107.
- Stasolla, F., Perilli, V., & Boccasini, A. (2016). Assistive technologies for persons with severe-profound intellectual and developmental disabilities. *Computer-Assisted and Web-Based Innovations in Psychology, Special Education, and Health*, 287–310.
- Statistics Solutions. (2022, October 21). *Sample Size & Power Analysis*. *Statistics Solutions - Sample Size & Power Analysis*. <https://www.statisticssolutions.com/free-resources/sample-size-power-analysis/>
- Stauss, M., Floyd, L., Becker, S., Ponnusamy, A., & Woywodt, A. (2020). Opportunities in the cloud or PIE in the Sky? current status and future perspectives of telemedicine in nephrology. *Clinical Kidney Journal*, 14(2), 492–506.
- Stelzer, D., & Mellis, W. (1998). Success factors of organizational change in software process improvement. *Software Process: Improvement and Practice*, 4(4), 227–250

- Storey, M.-A. D., Wong, K., & Müller, H. A. (2000). *How did program understanding tools affect how programmers understand programs? Science of Computer Programming*, 36(2–3), 183–207
- Su, D., Zhang, L., Peng, H., Saeidi, P., & Tirkolaei, E. B. (2023). Technical challenges of blockchain technology for Sustainable Manufacturing Paradigm in industry 4.0 era using a fuzzy decision support system. *Technological Forecasting and Social Change*, 188, 122275, 188, 122275
- Subramanyam, V., Deb, S., Krishnaswamy, P., & Ghosh, R. (2004). An Integrated Approach to Software Process Improvement at Wipro Technologies: *Veloci-Q*.
<https://doi.org/10.21236/ada421628>
- Succar, B. (2010). Building Information Modelling Maturity Matrix. *Handbook of Research on Building Information Modeling and Construction Informatics*, 65–103, 65–103
- Sullivan, G. M., & Feinn, R. (2012). Using effect size or why the p value was not enough. *Journal of Graduate Medical Education*, 4(3), 279–282.
- Sullivan, J. L., & Hughes, J. M. (2021). The Veterans Health Administration: Opportunities and considerations for implementing innovations in a national, Integrated Health Care System. *Public Policy & Aging Report*, 32(1), 19–24
- Surega, N. (2019, February). *Application of people capability maturity model in business process*.
https://www.researchgate.net/publication/330166131_Application_of_People_Capability_Maturity_Model_in_Business_Process_Outourcing_Enterprises-_A_Study_With_Reference_to_Tamil_Nadu
- SurveyMonkey. (2023). *Demographic survey questions: Why & how to ask*.
<https://www.surveymonkey.com/mp/gathering-demographic-information-from-surveys/>

- Swanson, J. (2015). Mine or theirs, where did users go? A comparison of e-journal usage at the OhioLINK Electronic Journal Center Platform Versus the Elsevier ScienceDirect platform. *Journal of Electronic Resources Librarianship*, 27(1), 1–9.
- Swedan, S., Khabour, O. F., Alzoubi, K. H., & Aljabali, A. A. A. (2020). Graduate students reported practices regarding the issue of informed consent and maintaining of data confidentiality in a developing country. *Heliyon*, 6(9)
- Sydle. (2022, February). *What's process optimization? Why was it so important for your business?* <https://www.sydle.com/blog/what-was-process-optimization-6126ac39b060f57604039a57/>
- Symonds, E. (2011). A practical application of SurveyMonkey as a remote usability-testing tool. *Library Hi Tech*, 29(3), 436–445
- Taber, K. S. (2017). The use of Cronbach's alpha when developing and Reporting Research Instruments in science education. *Research in Science Education*, 48(6), 1273–1296
- Taherdoost, H. (2019). What was the best response scale for survey and questionnaire design; review of different lengths of rating scale/attitude scale/Likert scale. *International Journal of Academic Research in Management (IJARM)*, 8(1), 1–10.
- Taherdoost, H. (2022). What were different research approaches? Comprehensive Review of Qualitative, quantitative, and mixed method research, their applications, types, and limitations. *Journal of Management Science & Engineering Research*, 5(1), 53–63.
- Talab, H., & Flayyih, H. (2023). An empirical study to measure the impact of information technology governance under the control objectives for information and related technologies on financial performance. *International Journal of Professional Business Review*, 8(4).

- Talal, A. H., Sofikitou, E. M., Jaanimägi, U., Zeremski, M., Tobin, J. N., & Markatou, M. (2020). A framework for patient-centered telemedicine: Application and lessons learned from vulnerable populations. *Journal of Biomedical Informatics*, 112, 103622, 112, 103622.
- Tarhan, A., Turetken, O., & Reijers, H. A. (2016). Business process maturity models: A systematic literature review. *Information and Software Technology*, 75, 122–134, 75, 122–134
- Tarka, P. (2015). Likert scale and change in range of response categories vs. the factor's extraction in EFA model. *Acta Universitatis Lodzianis. Folia Oeconomica*, 1(311)
- Tashakkori, A., & Teddlie, C. (2009). Integrating qualitative and quantitative approaches to research. *The SAGE Handbook of Applied Social Research Methods*, 283–317, 283–317
- Tavakol, M., & Dennick, R. (2011). Making sense of Cronbach's alpha. *International Journal of Medical Education*, 2, 53–55, 2, 53–55.
- Tay, K.-M., & Butler, C. (1999). Methodologies for experimental design: A survey, comparison, and future predictions. *Quality Engineering*, 11(3), 343–356
- Terrasse, M., Gorin, M., & Sisti, D. (2019). Social Media, e-health, and medical ethics. *Hastings Center Report*, 49(1), 24–33
- Thabit, W., & Sharekh, N. A. (2023). The Impact of Dynamic Capabilities on Institutional Maturity via CMMI. *International Journal of Research in Education Humanities and Commerce*, 04(01), 11–24.
- Thomas, L., & Woodside, J. M. (2016). Social Media Maturity Model. *International Journal of Healthcare Management*, 9(1), 67–73.

- Toews, I., Glenton, C., Lewin, S., Berg, R. C., Noyes, J., Booth, A., Marusic, A., Malicki, M., Munthe-Kaas, H. M., & Meerpohl, J. J. (2016). Extent, awareness and perception of dissemination bias in qualitative research: An explorative survey. *PLOS ONE*, 11(8)
- Togatorop, P. R. (2023). Auditing learning management system using information technology infrastructure library (ITIL). *Jurnal Mantik*, 6(4), 3808–3817.
- Trafimow, D. (2019). A frequentist alternative to significance testing, p-values, and confidence intervals. *Econometrics*, 7(2), 26
- Traore, B. B., Kamsu-Foguem, B., & Tangara, F. (2016). Integrating MDA and SOA for improving telemedicine services. *Telematics and Informatics*, 33(3), 733–741
- Tsikriktsis, N., & Heineke, J. (2004). The impact of process variation on customer dissatisfaction: Evidence from the U.S. domestic airline industry. *Decision Sciences*, 35(1), 129–141
- Tukey, J. W. (1949). Comparing individual means in the analysis of variance. *Biometrics*, 5(2), 99
- Tuli, F. (2011). The basis of distinction between qualitative and quantitative research in social science: reflection on ontological, epistemological and methodological perspectives. *Ethiopian Journal of Education and Sciences*, 6(1), 97–108.
- Twycross, A. (2004). Research design: Qualitative, quantitative and mixed methods approaches *Nurse Researcher*, 12(1), 82-84.
- Tyagi, A. (2021). Impact of blockchain on financial technology – challenges and strategies. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.3832937>.
- Türetken, O., & Demirörs, O. (2004). *People capability maturity model and human resource management systems: did they benefit each other?* *Human Systems Management*, 23(3), 179–190

- Türkmen, İ., & Sardoğan, B. (2023). A holistic view of remote health care. *Advances in Healthcare Information Systems and Administration*, 208–222.
- Urick, B. Y., Peters, A., Pathak, S., Vest, M. H., Colmenares, E., Blanchard, C., Easter, J., Foushee, L., & DeFalco, P. (2023). Telehealth medication management and health care spending in a Medicare accountable care organization. *Journal of Managed Care & Specialty Pharmacy*, 29(4), 357–364.
- US Census Bureau. (2021, October 8). Additional instructions for respondents. *Census.gov*. <https://www.census.gov/programs-surveys/decennial-census/technical-documentation/questionnaires/2020/response-guidance.html>
- Uttley, J. (2019). Power analysis, Sample Size, and assessment of statistical assumptions improving the evidential value of Lighting Research. *LEUKOS*, 15(2–3), 143–162
- Utz, C., Degeling, M., Fahl, S., Schaub, F., & Holz, T. (2019). (un)informed consent. *Proceedings of the 2019 ACM SIGSAC Conference on Computer and Communications Security*. <https://doi.org/10.1145/3319535.3354212>.
- VA Telehealth National Planning Strategy. (2021). MAHSO *National Planning Strategy – Telehealth*. <https://www.va.gov/AIRCOMMISSIONREPORT/docs/Telehealth-National-Planning-Strategy.pdf>
- Valdez-de-Leon, O. (2016). A digital maturity model for telecommunications service providers. *Technology Innovation Management Review*, 6(8), 19–32
- Valenta, S., Harvey, J., Sederstrom, E., Glanville, M., Walsh, T., & Ford, D. (2021). Enterprise adoption of Telehealth: An academic medical center’s experience utilizing the Telehealth Service Implementation Model. *Telemedicine Reports*, 2(1), 163–170
- Vallerand, J., Lapalme, J., & Moïse, A. (2015). Analyzing enterprise architecture maturity models: A learning perspective. *Enterprise Information Systems*, 11(6), 859–883

- Van Cleave, J., Stille, C., & Hall, D. E. (2022). Child health, vulnerability, and complexity: Use of telehealth to enhance care for children and youth with special health care needs. *Academic Pediatrics, 22*(2).
- van Dyk, L., & Schutte, C. (2013). The Telemedicine Service Maturity Model: A Framework for the measurement and improvement of telemedicine services. *Telemedicine*.
<https://doi.org/10.5772/56116>
- van Kessel, R., Hrzic, R., O’Nuallain, E., Weir, E., Wong, B. L., Anderson, M., Baron-Cohen, S., & Mossialos, E. (2022). Digital Health Paradox: International policy perspectives to address increased health inequalities for people living with disabilities. *Journal of Medical Internet Research, 24*(2).
- Van Looy, A. (2014). Business process maturity. *SpringerBriefs in Business Process Management*. <https://doi.org/10.1007/978-3-319-04202-2>
- Van Looy, A., Poels, G., & Snoeck, M. (2016). *Evaluating business process maturity models - KU Leuven*. <https://lirias.kuleuven.be/retrieve/415158>
- Van Velsen, L., Wildevuur, S., Flierman, I., Van Schooten, B., Tabak, M., & Hermens, H. (2015). Trust in telemedicine portals for rehabilitation care: An exploratory focus group study with patients and healthcare professionals. *BMC Medical Informatics and Decision Making, 16*(1)
- Vargas, V. B., Gomes, J. de, Fernandes, P. C., Vallejos, R. V., & Carvalho, J. V. (2023). Influential factors for hospital management maturity models in a post-covid-19 scenario - systematic literature review. *Journal of Information Systems Engineering and Management, 8*(1), 19556.

- Vaudreuil, R., Langston, D. G., Magee, W. L., Betts, D., Kass, S., & Levy, C. (2020). Implementing music therapy through telehealth: Considerations for military populations. *Disability and Rehabilitation: Assistive Technology*, 17(2), 201–210.
- Verdugo, M. A., Vicente, E., Guillén, V. M., Sánchez, S., Ibáñez, A., & Gómez, L. E. (2021). A measurement of self-determination for people with intellectual disability: Description of the AUTODDIS scale and evidence of reliability and external validity. *International Journal of Developmental Disabilities*, 69(2), 317–326.
- Verduzco-Gutierrez, M., Estores, I. M., Graf, M. J., Barshikar, S., Cabrera, J. A., Chang, L. E., Eapen, B. C., & Bell, K. R. (2021). Models of care for Post acute COVID-19 clinics. *American Journal of Physical Medicine & Rehabilitation*, 100(12), 1133–1139.
- Vergutz, A., Noubir, G., & Nogueira, M. (2020). Reliability for Smart Healthcare: A network slicing perspective. *IEEE Network*, 34(4), 91–97.
- Vidal Carvalho, J., Rocha, Á., & Abreu, A. (2017). Maturity of hospital information systems: Most important influencing factors. *Health Informatics Journal*, 25(3), 617–631.
- vom Brocke, J., & Rosemann, M. (2015). *Handbook on business process management 1: and information systems*. Springer. <https://doi.org/10.1007/978-3-642-45100-3>
- Vranas, K. C., Slatore, C. G., & Kerlin, M. P. (2018). Telemedicine coverage of Intensive Care Units: A narrative review. *Annals of the American Thoracic Society*, 15(11), 1256–1264.
- Vukšić, V. B., Bach, M. P., & Popović, A. (2013). Supporting performance management with business process management and business intelligence: A case analysis of integration and orchestration. *International Journal of Information Management*, 33(4), 613–619.
- Waclawski, E. (2012). How I use it: Survey monkey. *Occupational Medicine*, 62(6), 477–477.

- Wade, V. A., Taylor, A. D., Kidd, M. R., & Carati, C. (2016). Transitioning a home telehealth project into a sustainable, large-scale service: A qualitative study. *BMC Health Services Research, 16*(1)
- Wademan, M. R., Spuches, C. M., & Doughty, P. L. (2008). The People Capability Maturity Model. *Performance Improvement Quarterly, 20*(1), 97–123
- Wan, P. K., Huang, L., & Holtskog, H. (2020). Blockchain-enabled information sharing within a supply chain: A Systematic Literature Review. *IEEE Access, 8*, 49645–49656, 8, 49645–49656.
- Wang, H., Chen, K., & Xu, D. (2016). A maturity model for blockchain adoption. *Financial Innovation, 2*(1)
- Waseh, S., & Dicker, A. P. (2019). Telemedicine training in Undergraduate Medical Education: Mixed-methods review. *JMIR Medical Education, 5*(1)
- Wason, R., Bhardwaj, B. R., & Jain, V. (2020). Enabling digital twin through blockchain: A strategic perspective. *Blockchain Technology and Applications, 185–212, 185–212*
- Watts, S. (2020, February 19). CMMI: An introduction to capability maturity model integration. *BMC Blogs*. <https://www.bmc.com/blogs/cmmi-capability-maturity-model-integration/>
- Watts, S. (2020). *Process improvement using the Capability Maturity Model Integration (CMMI): A practical guide*. Productivity Press.
- Watzlaf, V. J. M., Zhou, L., DeAlmeida, D. R., & Hartman, L. M. (2017). A systematic review of research studies examining telehealth privacy and security practices used by healthcare providers. *International Journal of Telerehabilitation, 9*(2), 39–58.
- Weber, C., Königsberger, J., Kassner, L., & Mitschang, B. (2017). M2DDM – a maturity model for data-driven manufacturing. *Procedia CIRP, 63, 173–178, 63, 173–178*

- Whelan, R. (2008). Effective analysis of Reaction Time Data. *The Psychological Record*, 58(3), 475–482.
- White, M. (2021). A comprehensive review of the Capability Maturity Model Integration: Application in healthcare and government. *Journal of Technology and Innovation Management*, 5(2), 20–32..
- White, S. K. (2019, August 27). *What was process improvement? A business methodology for efficiency and Productivity*. CIO. <https://www.cio.com/article/220557/what-was-process-improvement-a-business-methodology-for-efficiency-and-productivity.html>
- White, S. K. (2021, June 1). *What was CMMi? A model for optimizing development processes*. CIO. <https://www.cio.com/article/274530/process-improvement-capability-maturity-model-integration-cmmi-definition-and-solutions.html>
- World Health Organization. (2022). *Emerging trends and technologies: a horizon scan for global public health*. World Health Organization.
- Williams, T. (2021, June 14). *Why was quantitative research important?* Doctoral Journey. <https://www.gcu.edu/blog/doctoral-journey/why-quantitative-research-important>
- Wilson, J. A., & Schild, S. (2014). Provision of mental health care services to deaf individuals using telehealth. *Professional Psychology: Research and Practice*, 45(5), 324–331
- Wilson, M., Smith, N. C., & Holmes, P. S. (2007). The role of effort in influencing the effect of anxiety on performance: Testing the conflicting predictions of processing efficiency theory and the conscious processing hypothesis. *British Journal of Psychology*, 98(3), 411–428.
- Wisniewski, H., Gorrindo , T., Rauseo-Ricupero, N., Hilty, D., & Torous, J. (2020). The role of digital navigators in promoting clinical care and technology integration into practice. *Digital Biomarkers*, 4(Suppl. 1), 119–135. <https://doi.org/10.1159/000510144>

Wootton, A. R., McCuistian, C., Legnitto Packard, D. A., Gruber, V. A., & Saberi, P. (2020).

Overcoming technological challenges: Lessons learned from a telehealth counseling study. *Telemedicine and E-Health*, 26(10), 1278–1283

Workflows of Telemedicine. (2021). *Field Guide to Telehealth and Telemedicine for Nurse Practitioners and Other Healthcare Providers*.

<https://doi.org/10.1891/9780826172921.0003>

Wosik, J., Fudim, M., Cameron, B., Gellad, Z. F., Cho, A., Phinney, D., Curtis, S., Roman, M.,

Poon, E. G., Ferranti, J., Katz, J. N., & Tchong, J. (2020). Telehealth Transformation:

Covid-19 and the rise of virtual care. *Journal of the American Medical Informatics Association*, 27(6), 957–962.

Wu, D. D., Chen, S. H., & Olson, D. L. (2014). Business intelligence in risk management: Some recent progresses. *Information Sciences*, 256, 1–7, 256, 1–7

Wulf, J., Winkler, T. J., & Brenner, W. (2015, April 3). Measuring IT service management capability: Scale Development and empirical validation. *AIS Electronic Library (AISeL)*.

<http://aisel.aisnet.org/wi2015/43>

Yang, C., Lan, S., Wang, L., Shen, W., & Huang, G. G. (2020). Big Data Driven Edge-Cloud

Collaboration Architecture for Cloud Manufacturing: A Software Defined Perspective.

IEEE Access, 8, 45938–45950, 8, 45938–45950.

Yang, Z., Zhu, R., & Zhang, L. (2016, might 1). Research on the capability maturity model of

Digital Library Knowledge Management. *Research on the Capability Maturity Model of*

Digital Library Knowledge Management | Atlantis Press. [https://www.atlantis-](https://www.atlantis-press.com/proceedings/itoec-16/25853131)

[press.com/proceedings/itoec-16/25853131](https://www.atlantis-press.com/proceedings/itoec-16/25853131)

Yangyin, Z., & Changbin, J. (2016). Investigating the impact factors forming users' intention in

utilizing live online platforms: Case study douyu TV. *2016 International Conference on*

- Industrial Informatics - Computing Technology, Intelligent Technology, Industrial Information Integration (ICIICII)*. <https://doi.org/10.1109/iciicii.2016.0041>
- Yaël, O. (2022). Optimized patient engagement and the consumer experience – how digital health improves timely patient engagement, effectiveness and experience. *Telehealth and Medicine Today*. <https://doi.org/10.30953/tmt.v7.344>
- Young, T. J. (2015). Questionnaires and Surveys. *Research Methods in Intercultural Communication*, 163–180. <https://doi.org/10.1002/9781119166283.ch11>
- Zahrullaili, M. H., & Noordin, M. F. (2018). Towards developing a comprehensive business intelligence maturity model for Malaysian Public Sector: Application of Mixed Methodology. *2018 Fourth International Conference on Information Retrieval and Knowledge Management (CAMP)*. <https://doi.org/10.1109/infrkm.2018.8464776>
- Zhang, H., Arens, E., & Zhai, Y. (2015). A review of the corrective power of Personal Comfort Systems in non-neutral ambient environments. *Building and Environment*, 91, 15–41. <https://doi.org/10.1016/J.buildenv.2015.03.013>
- Zhang, X., & Saltman, R. (2022). Impact of electronic health record interoperability on Telehealth Service Outcomes. *JMIR Medical Informatics*, 10(1). <https://doi.org/10.2196/31837>
- Zhou, L., Varadharajan, V., & Gopinath, K. (2016). A secure role-based cloud storage system for encrypted patient-centric health records. *The Computer Journal*, 59(11), 1593–1611. <https://doi.org/10.1093/comjnl/bxw019>
- Zhu, L., Gai, K., & Li, M. (2019). Blockchain and internet of things. *Blockchain Technology in Internet of Things*, 9–28. https://doi.org/10.1007/978-3-030-21766-2_2

- Zhu, S., Song, M., Lim, M. K., Wang, J., & Zhao, J. (2020). *The development of energy blockchain and its implications for China's energy sector*. *Resources Policy*, 66, 101595. <https://doi.org/10.1016/J.resourpol.2020.101595>
- Zulman, D. M., Wong, E. P., Slightam, C., Gregory, A., Jacobs, J. C., Kimerling, R., Blonigen, D. M., Peters, J., & Heyworth, L. (2019). *Making connections: Nationwide implementation of video telehealth tablets to address access barriers in veterans*. *JAMIA Open*, 2(3), 323–329. <https://doi.org/10.1093/jamiaopen/ooz024>
- Škrinjar, R., & Štemberger, M. I. (2008). *Improving organizational performance by raising the level of business process orientation maturity: Empirical test and case study*. *Information Systems Development*, 723–740. https://doi.org/10.1007/978-0-387-78578-3_11

Appendix A

Significant Findings for the Individual Questions 1 – 20

Table A. 1

Significant Findings for the Individual Questions 1 – 20

Question	Topic	F Value	F-Critical	p-Value	Effect Size (ω^2)	Significance Level
1	Telehealth processes may be defined as ad hoc, chaotic, or random.	41.75	3.87	0.00	0.13	0.05
2	Telehealth processes are managed.	73.25	3.87	0.00	0.20	0.05
3	Telehealth processes are defined/specific.	53.35	3.87	0.00	0.16	0.05
4	Telehealth processes are quantitatively managed.	65.02	3.87	0.00	0.18	0.05

5	Telehealth processes are optimized.	22.57	3.87	0.00	0.07	0.05
7	Telehealth processes are reactive.	48.38	3.87	0.00	0.14	0.05
8	Telehealth processes are uncoordinated.	52.55	3.87	0.00	0.15	0.05
9	Telehealth processes are planned.	32.79	3.87	0.00	0.10	0.05
10	Telehealth processes are controlled.	81.70	3.87	0.00	0.22	0.05
11	Telehealth processes are well-defined.	45.09	3.87	0.00	0.14	0.05
12	Telehealth processes are consistent.	66.39	3.87	0.00	0.19	0.05
13	Telehealth processes are followed.	80.14	3.87	0.00	0.22	0.05

14	Telehealth	66.50	3.87	0.00	0.19	0.05
	processes					
	involve					
	quantitative or					
	measurable					
	objectives.					
15	Telehealth	69.90	3.87	0.00	0.20	0.05
	processes					
	involve metrics					
	analysis.					
16	Telehealth	61.87	3.87	0.00	0.18	0.05
	processes					
	involve					
	statistical					
	analysis.					
17	Telehealth	48.16	3.87	0.00	0.14	0.05
	processes are					
	improved					
	incrementally.					
18	Telehealth	33.67	3.87	0.00	0.10	0.05
	processes are					
	efficient.					

20	Process	106.92	3.87	0.00	0.27	0.05
	maturity is not					
	addressed by					
	our current					
	process					
	improvement					
	initiatives.					

Appendix B

Survey Instrument

Note: Adapted from Dr. Daniel Adrian Doss's surveys for process maturity and improvement.

Dear Sir or Ma'am,

You were invited to anonymously share your valuable insights on the maturity levels of process architecture in relation to a process improvement framework for telehealth. Your feedback would systematically enhance telehealth processes over time.

Why Participate?

- Contribute to advancing telehealth practices.
- Help improve process architecture for better healthcare outcomes.
- Influence future frameworks for systematic process improvements.

This survey would gather your opinions on the maturity levels of process architecture in relation to a process improvement framework that systematically improves telehealth processes over time. The desired telehealth framework proposes that the process architecture begin at Level 1 and progress sequentially through the model's subsequent stages until it reaches an optimized state of process improvement at Level 5. Telehealth structural processes were assumed to be chaotic and random in the initial stage (Level 1) and optimized and refined in the final stage (Level 5).

Please share your thoughts on your current work environment and respond according to your perceptions.

Please keep the following concepts in mind while completing this survey.

Level 1: Ad hoc, unstructured, and chaotic processes often exceed budgetary expectations.

Level 2: Processes might be reactive and managed, as well as planned, performed, measured, and controlled.

Level 3: Processes might be characterized and understood and were described in standards, procedures, tools, and methods.

Level 4: Processes were managed quantitatively via measurement and control, and

Level 5: Processes exhibit optimization, and continuous improvement was emphasized.

Questions 1-5 reflect the foundational concepts represented by levels 1 – 5.

Questions 6-8 were affiliated with Level 1

Questions 9-11 were affiliated with Level 2.

Questions 12-14 were affiliated with Level 3.

Questions 15-17 were affiliated with Level 4.

Questions 18-20 were affiliated with Level 5.

Questions 21-33 query your perceptions of your work environment.

The remaining questions query the demographic attributes of your work setting.

How to Participate:

Click [\[here\]](#)(survey link) to start the survey. It shouldn't take more than 15 minutes to complete.

Your input was crucial in shaping the future of telehealth. Thank you for your participation!

Thank you for your time and attention.

Best regards,

James Jamison

A Questionnaire for Telehealth Process Maturity Assessment.

Please check the box that best reflects your perception of the query statement.

Item	Query	Strongly Disagree	Disagree	Neither Agree nor Disagree	Agree	Strongly Agree
1	Telehealth processes may be defined as ad hoc, chaotic, or random.	1	2	3	4	5
2	Telehealth processes are managed.	1	2	3	4	5
3	Telehealth processes are defined/specific.	1	2	3	4	5
4	Telehealth processes are quantitatively managed.	1	2	3	4	5

5	Telehealth processes are optimized.	1	2	3	4	5
6	Telehealth processes are unpredictable.	1	2	3	4	5
7	Telehealth processes are reactive.	1	2	3	4	5
8	Telehealth processes are uncoordinated.	1	2	3	4	5
9	Telehealth processes are planned.	1	2	3	4	5
10	Telehealth processes are managed.	1	2	3	4	5
11	Telehealth processes are controlled.	1	2	3	4	5
12	Telehealth processes are well-defined.	1	2	3	4	5
13	Telehealth processes are consistent.	1	2	3	4	5
14	Telehealth processes are followed.	1	2	3	4	5
15	Telehealth processes involve quantitative or measurable objectives.	1	2	3	4	5

16	Telehealth processes involve metrics analysis.	1	2	3	4	5
17	Telehealth processes involve statistical analysis.	1	2	3	4	5
18	Telehealth processes are improved incrementally.	1	2	3	4	5
19	Telehealth processes are efficient.	1	2	3	4	5
20	Telehealth processes are effective.	1	2	3	4	5
21	Process maturity is not addressed by our current process improvement initiatives.	1	2	3	4	5
22	Process improvement is advocated within my telehealth setting.	1	2	3	4	5
23	Process initiatives are tracked to examine process performance.	1	2	3	4	5

24	Grouping of processes, according to maturity level, would improve the outcomes of our processes.	1	2	3	4	5
25	Categorical process grouping is advocated within my telehealth setting.	1	2	3	4	5
26	Process maturity is a contributor to successful process output within my telehealth setting.	1	2	3	4	5
27	Processes are informal within my telehealth setting.	1	2	3	4	5
28	Telehealth policies influence processes.	1	2	3	4	5
29	Methods of managing processes vary within my telehealth setting.	1	2	3	4	5
30	Telehealth processes are inefficient.	1	2	3	4	5

31	Telehealth processes are ineffective.	1	2	3	4	5
32	Telehealth processes change frequently.	1	2	3	4	5
33	My telehealth setting advocates process training.	1	2	3	4	5

34. My job was classified as ____ Non-Managerial ____ Managerial

Non-Managerial roles focused on gathering and recording patient data through digital tools, while Management roles analyzed this data and made informed medical decisions during virtual consultations.

35. Our previous improvement initiatives consisted of (check all that apply):

No previous improvement initiative or unknown

CompStat

Total Quality Management (TQM)

Business Process Re-Engineering (BPR)

Business Process Improvement (BPI)

Business Process Management (BPM)

Benchmarking

Six-Sigma

Regulation (government or corporate)

ISO Standards

Process Maturity Modeling

Proprietary Initiative

36. Our current improvement initiatives consisted of (check all that apply)

No previous improvement initiative or unknown

CompStat

Total Quality Management (TQM)

Business Process Re-Engineering (BPR)

Business Process Improvement (BPI)

Business Process Management (BPM)

Benchmarking

Six-Sigma

Regulation (government or corporate)

ISO Standards

Process Maturity Modeling

Proprietary Initiative

37. How many employees does your telehealth service have?

☐ 1-100 ☐ 101-200 ☐ 201-300 ☐ 301- 400 ☐ 401-500 ☐ 501-600

☐ 601 – 700 ☐ 701 – 800 ☐ 801 – 900 ☐ 901 – 1,000 ☐ Over 1,000

38. My job description was best described as: (Check all that might apply)

☐ Anesthesiology

☐ Nursing

☐ Audiology

☐ Occupational Therapist

☐ Behavioral Health Specialist

☐ Optometry

☐ Cardiology

☐ Orthopedic

☐ Case Management

☐ Palliative Care

☐ Chronic Disease Management

☐ Pathology

☐ Dental

☐ Patient Monitoring

☐ Dermatology

☐ Patient Navigator (Virtual Care)

___ Dietitian/Nutritionist	___ Pharmacy
___ Endocrinology	___ Physical Therapy
___ Gastroenterologist	___ Provider
___ Geriatric	___ Psychiatric
___ Health Coach	___ Radiation Oncology
___ Medical Services	___ Respiratory
___ Medical Social Work	___ Rheumatology
___ Mental Health	___ Social Work
___ Neurology	___ Telehealth
___ Nuclear Medicine	___ Vocational Rehabilitation

Key Words:

Ad hoc Processes, Benchmarking, Business Process Improvement (BPI), Business Process Management (BPM), Business Process Re-Engineering (BPR), Categorical Process Grouping, Chaotic Processes, CompStat, Consistent Processes, Continuous Improvement, Controlled Processes, Defined Processes, Effective Processes, Efficient Processes, Informal Processes, ISO Standards, Managed Processes, Metrics Analysis, Optimized Processes, Planned Processes, Process Architecture, Process Improvement Framework, Process Maturity, Process Maturity Modeling, Proprietary Initiative, Quantitative Management, Reactive Processes, Six Sigma,

Structured Processes, Telehealth, Total Quality Management (TQM), Uncoordinated Processes, Unpredictable Processes

Glossary of Terms:

Ad hoc Processes: Activities that were unstructured, spontaneous, and often unpredictable, typically leading to inefficiencies and exceeding budgetary constraints.

Benchmarking: Comparing business processes and performance metrics to industry bests and best practices from other companies.

Business Process Improvement (BPI): A systematic approach to help an organization optimize its underlying processes to achieve more efficient results.

Business Process Management (BPM): A disciplined approach to identify, design, execute, document, measure, monitor, and control automated and non-automated business processes to achieve consistent, targeted results aligned with an organization's strategic goals.

Business Process Re-Engineering (BPR): A business management strategy that analyzed and designed organizational workflows and processes, aiming for dramatic improvements in critical performance measures.

Categorical Process Grouping: Organizing processes according to maturity level to improve the outcomes of processes.

Chaotic Processes: Initial stage of process maturity (Level 1) where activities were random, unstructured, and unmanaged.

CompStat: A performance management system used to reduce crime and achieve other police department goals.

Consistent Processes: Processes performed the same way each time, ensuring reliability and predictability.

Continuous Improvement: Ongoing effort to improve products, services, or processes through incremental and breakthrough improvements, typically found at Level 5 of process maturity.

Controlled Processes were monitored and measured to ensure they operate within defined parameters. They were often found at Level 2 of process maturity.

Defined Processes: Well-understood and documented processes, typically found at Level 3 of process maturity.

Effective Processes: Processes that achieve their intended outcomes.

Efficient Processes: Processes that achieve their outcomes with minimal waste of resources.

Informal Processes: Unstructured and loosely organized processes that might lack consistency.

ISO Standards ensure product, service, and system quality, safety, and efficiency.

Managed Processes: Processes that were systematically controlled and planned to ensure they meet predefined criteria at Levels 2 and 4 of process maturity.

Metrics Analysis: The process of evaluating data to understand the performance and effectiveness of a process.

Optimized Processes: The highest maturity level was where processes were continually refined and improved for efficiency and effectiveness.

Planned Processes: Processes that were intentionally designed and scheduled, often found at Level 2 of process maturity.

Process Architecture: Structured arrangement of processes within an organization that guides how tasks were completed and improvements were implemented.

Process Improvement Framework: Structured approach to enhance processes systematically over time, moving through defined maturity levels.

Process Maturity: The level of sophistication and optimization of processes, ranging from chaotic (Level 1) to optimized (Level 5).

Process Maturity Modeling: A framework for assessing and improving process maturity.

Proprietary Initiative: Custom-developed process improvement initiatives specific to an organization.

Quantitative Management: The use of data and statistical analysis to measure, control, and improve processes was typically found at Level 4 of process maturity.

Reactive Processes were activities that responded to issues as they arise rather than being planned and proactive. They were often seen at Level 2 of process maturity.

Six Sigma: A set of techniques and tools for process improvement aiming to reduce defects and variability.

Structured Processes: Well-organized and planned processes that follow a defined sequence and method, found at Levels 3, 4, and 5 of process maturity.

Telehealth: Delivery of healthcare services through telecommunications technology, such as video conferencing and remote monitoring.

Total Quality Management (TQM): An organization-wide approach to continuous improvement, focusing on long-term success through customer satisfaction.

Uncoordinated Processes: These lack synchronization and integration, often resulting in inefficiencies.

Unpredictable Processes: Processes that produce varying and unreliable outcomes, typically found at Level 1 of process maturity.

Appendix C

Permission From the Veterans Administration to Conduct Research

From: Christina Bennett <no-reply@gov.irbnet.org>

To: James Jamison

Please note that VANEOMS Research Administration has published the following Board

Document on IRBNet:

Project Title: [1806721-1] A Maturity Model for Implementing Innovative Telehealth Technology

Principal Investigator: James Jamison, MSTIM

Submission Type: New Project

Date Submitted: April 22, 2024

Document Type: Administrative Prereview Letter

Document Description: Administrative Prereview Letter 2024-04-22

Publish Date: April 22, 2024

Should you have any questions you might contact Christina Bennett at christina.bennett2@va.gov.

Thank you,

Christina Bennett, The IRBNet Support Team

gov.irbnet.org

You have Full access to this project.	
Research Institution	VA Northeast Ohio Healthcare System, Cleveland, OH
Title	A Maturity Model for Implementing Innovative Telehealth Technology
Principal Investigator	Jamison, James, MSTIM
Keywords	Maturity Model, CMMi, Telehealth, Innovative Technology
Sponsor	National University
Internal Reference Number	1806721-1

Appendix D

Consent Letter

Dear Sir or Ma'am,

My name was James Jamison. I am a National University (NU) doctoral student and a senior IT specialist at the Veterans Administration Medical Center (VAMC) in Cleveland, OH. I had submitted an IRB application to conduct technical research and distribute this survey. You were invited to anonymously share your valuable insights on the maturity levels of process architecture to a process improvement framework for telehealth. Your feedback would systematically enhance telehealth processes over time.

Purpose of the Study

The desired telehealth framework proposes that the process architecture begin at Level 1 and progress sequentially through Level 5 as the model's subsequent stages until it reaches an optimized state of process improvement. The title of this research was "A Maturity Model for Implementing Innovative Telehealth Technology."

- Level 1: Ad hoc, unstructured, and chaotic processes often exceed budgetary expectations.
- Level 2: Processes might be reactive and managed, as well as planned, performed, measured, and controlled.
- Level 3: Processes might be characterized and understood and were described in standards, procedures, tools, and methods.
- Level 4: Processes had been managed quantitatively via measurement and control.
- Level 5: Processes exhibit optimization, and continuous improvement was emphasized.

Why Participate?

- Contribute to advancing telehealth practices.
- Help improve process architecture for better healthcare outcomes.
- Influence future frameworks for systematic process improvements.

Eligibility Criteria

You might participate in this research if you meet the following criteria:

1. A Veterans Administration employee who was involved in patient care or administration and had been exposed to telehealth technology.
2. Participants had been willing to answer 37 questions and share your views on levels of telehealth structural processes.
3. Share your thoughts on your current work environment and respond according to your perceptions.

I aim to include 500 people in this study.

Research Activities

If you agree to be in this study, you would be asked to complete an online survey for 20 minutes to provide your opinions on the maturity levels of process architecture in relation to a process improvement framework that systematically enhances telehealth processes over time.

During the survey, the questions would focus on the following:

- Questions 1-5 reflect the foundational concepts represented by levels 1 – 5.
- Questions 6-8 were affiliated with Level 1.
- Questions 9-11 were affiliated with Level 2.
- Questions 12-14 were affiliated with Level 3.
- Questions 15-17 were affiliated with Level 4.
- Questions 18-20 were affiliated with Level 5.
- Questions 21-33 query your perceptions of your work environment.

The remaining questions query the demographic attributes of your work setting.

How to Participate

Click [\[here\]](#)(survey link) to start the survey. It shouldn't take more than 15 minutes to complete.

Your input was crucial in shaping the future of telehealth. Thank you for your participation!

Confidentiality

I would keep the records of this study private and take reasonable measures to protect the security of all your personal information. In any report I make public, I would not include any information that would make it possible to identify you.

Voluntary Participation

Participation in this study was completely voluntary. You might quit at any time. This was the link for the Survey:

Thank you for your time and attention.

Best regards,

James C. Jamison II, MSTIM, BSIT, MCSE

Senior Information Technology Specialist

Office of Information and Technology (OIT),

Development, Security, and Operations (DevSecOps),

End User Operations (EUO)

Area 541, Territory 3 (T3), Midwest District (D3), VAMC Cleveland

Office: (216) 791-2300 x41222

Gov Cell # 216.785.5698

M-F 6 A.M. – 2:30 P.M. EST



[IRB00000686 VA Northeast Ohio Healthcare System: 1806721-1]

Appendix E**Permission to Use Survey Instrument from the Original Author**

Daniel Doss<ddoss@nu.edu>

To:James Jamison

Cc:Daniel Doss <ddoss@nu.edu>

Fri 3/29/2024 8:01 PM

Mr. Jamison,

Thank you for your inquiry regarding the survey instrument used within my preceding research.

Per our telephone conversation, I granted you permission to the use survey instrument for your dissertation research endeavors. Please ensure that you appropriately credit the survey and its variations within your dissertation and any other outlet(s) where your results might be presented or published.

Good luck with your research.

Best.

Dr. Daniel Adrian Doss



Daniel Adrian Doss, Ph.D.

Professor of Technology Management
National University | www.nu.edu
ddoss@nu.edu
855-355-6288

Appendix F

Curriculum Vitae

James C. Jamison II

3261 Alderwood Way, Cuyahoga Falls, OH 44223

Phone: 6464094135 | Email: jjamison@msn.com

Professional Summary

Senior IT Specialist with extensive experience in technical support, asset management, network security, and project management.

Education

Doctoral Candidate (Ph.D. In Technology & Innovation Management)

National University, Scottsdale, AZ | GPA: 3.8

Master of Science (Information Technology & Innovation Management)

Northcentral University, Scottsdale, AZ | GPA: 3.6

Bachelor of Science (Information Technology)

Sanford Brown, IADT Online, Tampa, FL | GPA: 3.5

Professional Experience

Senior IT Specialist (Customer Support) U.S. Department of Veterans Affairs – Cleveland, OH
(2018 – presented)

IT Specialist (Customer Support & Asset Manager) U.S. Department of Veterans Affairs –
Montrose, NY (2014 – 2018)

Director of IT Graff Diamonds, Inc. USA – New York, NY (2001 – 2005)

Military Service

United States Navy (1971 – 1979) – Honorable Discharge

Instructor of Aviation Electronics (8309 – AT / 6685 - AT)

Avionics Electronics Reconnaissance

Certifications & Training

Microsoft MCSE / MCP+I ID = 1412909 / Web Development Program E-Commerce CIW

Information Systems Project Management (AMA) / AT&T Telecommunications Systems

Administration / CNE

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